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Quaternionic decorations of hyperbolic 3-manifolds

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Introduction

A hyperbolic structure on a hyperbolic n-manifold M, if any, is uniquely determined up to isometry by the conjugacy class of its holonomy representation

$$\rho\colon \pi_1(M) \to \mathrm{Isom}(\mathbb{H}^n),$$

together with a developing map $D\colon \widetilde{M} \to \mathbb{H}^n$. The pair (D, ρ) is sometimes referred to as a *developing pair* for M. By Mostow-Prasad rigidity [Mos68], [Pra73], if $n \geq 3$ and M is complete with finite volume, the hyperbolic structure is unique.

Assume M is a complete, oriented, non-compact hyperbolic 3-manifold with finite volume, and let (D_0, ρ_0) be a developing pair associated to its unique complete structure. From M, we can construct a complete, 5-dimensional hyperbolic manifold M' (with infinite volume) which is diffeomorphic to $M \times \mathbb{R}^2$ and in which M lies geodesically as $M \times \{0\}$. The holonomy representation ρ_0' of M' is simply the composition

$$\rho_0'\colon \pi_1(M) \xrightarrow{\rho_0} \mathrm{Isom}^+(\mathbb{H}^3) \hookrightarrow \mathrm{Isom}^+(\mathbb{H}^5).$$

Since M' has infinite volume, Mostow-Prasad rigidity does not apply and we may look for new complete, hyperbolic structures on it. We want to study the possible deformations of the hyperbolic structure of M', and a natural way to achieve this is by studying the deformations of its holonomy representation. Strictly speaking, we would be ultimately interested in finding a continuous path of non-conjugate representations

$$\rho_t'\colon \pi_1(M) \to \mathrm{Isom}^+(\mathbb{H}^5)$$

satisfying suitable peripheral properties (e.g., mapping the peripheral subgroups to parabolic isometries), such that $\rho_t' \to \rho_0'$ as $t \to 0$.

Our approach is combinatorial and inspired by the works of Thurston [Thu02] and Neumann and Zagier [NZ85]. We assume that M admits an ideal triangulation $\mathcal{T}$, which we lift to a triangulation $\widetilde{\mathcal{T}}$ of the universal cover $\widetilde{M}$. Denoting by $|\widetilde{\mathcal{T}}|$ the geometric realization of the lifted triangulation, a first step toward addressing the deformation problem is to parametrize continuous maps $D\colon |\widetilde{\mathcal{T}}| \to \overline{\mathbb{H}}^5$ mapping the vertices of $|\widetilde{\mathcal{T}}|$ to ideal points in $\partial\mathbb{H}^5$ and each ideal tetrahedron of $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$ to a geodesic ideal tetrahedron in $\mathbb{H}^5$, with adjacent tetrahedra intersecting consistently along their shared faces.

This higher-dimensional setting requires additional care because, unlike in dimension three, prescribing the isometry class of two adjacent ideal tetrahedra and fixing the image of a face they share is not enough to determine them completely in $\mathbb{H}^5$. Indeed, fixing the image of a shared face leaves a 2-parameter family of rotations around a 2-dimensional subspace, allowing adjacent tetrahedra to lie in distinct 3-dimensional hyperbolic subspaces without altering their respective isometry classes. We must therefore be able to construct developing maps where adjacent tetrahedra lie in different 3-dimensional hyperbolic subspaces.

The classical case In the classical, three-dimensional case, by identifying the boundary at infinity $\partial\mathbb{H}^3$ with the Riemann sphere $\widehat{\mathbb{C}}$, the isometry class of every ideal tetrahedron is uniquely determined by a *cross-ratio* of its vertices. Once the isometry class of two adjacent tetrahedra is fixed by these invariants, their gluing along a common face is realized by a unique isometry.

In his work on the hyperbolic Dehn filling theorem, Thurston used this fact to parametrize developing maps $D\colon |\widetilde{\mathcal{T}}| \to \overline{\mathbb{H}}^3$ and holonomy representations. Specifically, he assigned a complex parameter, called the *complex modulus* (which corresponds to a cross-ratio of the vertices), to each edge of each ideal tetrahedron in the triangulation, satisfying some internal conditions in each tetrahedron (a single parameter on one edge of a tetrahedron in fact determines the other five). This assignment induces a well-defined hyperbolic structure on M minus the 1-skeleton of the triangulation.

The structure smoothly extends to the 1-skeleton if and only if the ideal tetrahedra in $\mathbb{H}^3$ glue consistently around every edge. For this to happen, the complex moduli must satisfy a system of compatibility relations known as the *consistency equations*. Specifically, for every edge e of the triangulation, the complex moduli $w_1, \dots, w_k$ around e must satisfy:

$$\prod_{i=1}^{k} w_i = 1, \quad \sum_{i=1}^{k} \arg(w_i) = 2\pi. \tag{1}$$

When the consistency equations (1) hold, the tetrahedra can be glued together step by step to build a developing map into $\mathbb{H}^3$ together with its holonomy representation, thus yielding a (possibly incomplete) hyperbolic structure on the whole manifold M.

The five-dimensional case: quaternionic decorations The main focus of this thesis is the development of a 5-dimensional analogue of Thurston's construction.

We represent hyperbolic space $\mathbb{H}^5$ of dimension 5 via the conformal half-space model $\mathcal{H} \times (0, +\infty)$, where $\mathcal{H}$ is the quaternion algebra, with metric tensor at a point $(q, t) \in \mathcal{H} \times (0, +\infty)$ equal to the Euclidean one rescaled by $1/t^2$. The sphere at infinity $\partial\mathbb{H}^5$ is identified with the compactification $\widehat{\mathcal{H}}$ of the quaternions and there is a natural isomorphism between $\mathrm{Isom}^+(\mathbb{H}^5)$ and the group $\mathrm{PSL}(2, \mathcal{H})$ [Kel03]. The group $\mathrm{PSL}(2, \mathcal{H})$ is the group of 2×2 quaternionic matrices with Dieudonné determinant 1, taken modulo

its center $\{\pm I_2\}$. A matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathcal{H})$$

acts on $\widehat{\mathcal{H}} = \partial\mathbb{H}^5$ as the fractional linear transformation

$$q \mapsto (aq+b)(cq+d)^{-1},$$

and two matrices in $\mathrm{SL}(2,\mathcal{H})$ induce the same transformation if and only if they differ at most by a sign.

The quaternionic formalism has proven useful for studying hyperbolic geometry in dimension 5; for instance, in [Kel03] Kellerhals used it to provide an explicit thick-thin decomposition for hyperbolic 5-manifolds, while the algebraic classification of isometries in $\mathrm{Isom}^+(\mathbb{H}^5) \cong \mathrm{PSL}(2,\mathcal{H})$ has been extensively developed (see [GLM26], [Cao07], [For04]).

In this setting, let T be an oriented ideal tetrahedron in $\mathbb{H}^5$ with ideal vertices $q_1, q_2, q_3, q_4 \in \mathcal{H}$, ordered following the orientation. Given distinct $i, k \in \{1,2,3,4\}$, we assign to the oriented edge from q_i to q_k the quaternion

$$Q_{ik}(T) \coloneqq (q_j - q_k)(q_j - q_i)^{-1}(q_l - q_i)(q_l - q_k)^{-1}, \tag{2}$$

where $j, l \in \{1,2,3,4\}$ are chosen so that the permutation $(i,j,k,l) \mapsto (1,2,3,4)$ is even. The definition easily extends by continuity to the case where one among q_i, q_j, q_l is ∞. The same is not true for the $q_k = \infty$ case: the factors containing q_k in $Q_{ik}(T)$ are not adjacent, hence its limit as q_k goes to ∞ is well-defined only up to conjugation. For the purpose of our construction it is nonetheless convenient to fix a specific element in this conjugacy class, setting

$$Q_{ik}(T) = (q_j - q_i)^{-1}(q_l - q_i), \quad \text{if } q_k = \infty.$$

The quaternion $Q_{ik}(T)$ is called the *quaternionic cross-ratio* of the ordered 4-tuple (q_i, q_j, q_k, q_l).

We point out some major differences from the classical cross-ratio. First, due to the lack of commutativity, the quaternionic cross-ratios assigned to an edge, say $Q_{ik}(T)$, and to the same edge with the opposite orientation, $Q_{ki}(T)$, are not the same. Rather, they are conjugate via a quaternion which we will call the *simple ratio*. The same holds for opposite edges.

The second major difference arises when considering the quaternionic cross-ratios assigned to the common edges of two ideal tetrahedra T_1, T_2 in $\mathbb{H}^5$ sharing a triangular face. In fact, for every common, oriented edge e of that face, the pair $(Q_e(T_1), Q_e(T_2))$ contains information about the relative positions of the two three-dimensional subspaces containing the tetrahedra. Since all common edges must yield consistent information, the six pairs are subject to strict algebraic relations. In particular, the two tetrahedra lie in the same three-dimensional subspace if and only if the cross-ratios $Q_e(T_1)$ and $Q_e(T_2)$ commute for every oriented edge e of the common face.

Lastly, the quaternionic cross-ratio is no longer an isometry invariant, but rather it changes in a controlled way via conjugation [BG11], [GL12]. More precisely, let $f \in \mathrm{PSL}(2,\mathcal{H})$ be a fractional linear transformation. Using local charts, we naturally identify the tangent space $T_p\widehat{\mathcal{H}}$ with $\mathcal{H}$ for every $p \in \widehat{\mathcal{H}}$. With these identifications, the differential of f at any point $p \in \widehat{\mathcal{H}}$ is an orientation-preserving, linear similarity of $\mathcal{H}$. Such a map is obtained by composing an element of the special orthogonal group $\mathrm{SO}(4)$ with a positive dilation. Using the fact that $\mathrm{SO}(4)$ is doubly covered by $S^3 \times S^3$ via the map

$$(u,v) \mapsto (q \mapsto uqv^{-1}),$$

we can encode both the rotation and the dilation by writing the differential $df_p : \mathcal{H} \to \mathcal{H}$ of f at a point p as

$$df_p(h) = u_{f,p} \cdot h \cdot v_{f,p}^{-1},$$

where the coefficients $u_{f,p}, v_{f,p} \in \mathcal{H}^*$ are non-zero quaternions. As we will detail in Chapter 3, they are respectively called the *left* and *right isoclinic* components of the differential, and are defined only up to a common real factor.

Throughout this introduction, we identify $\mathrm{Isom}^+(\mathbb{H}^5)$ with $\mathrm{PSL}(2,\mathcal{H})$. The following result is crucial for our work.

Proposition A. *Let T be an oriented ideal tetrahedron in $\mathbb{H}^5$ and $f \in \mathrm{PSL}(2,\mathcal{H})$ be an orientation-preserving isometry. Then*

$$Q_{ik}(f(T)) = u_{f,q_k} \cdot Q_{ik}(T) \cdot u_{f,q_k}^{-1}.$$

Let us now return to our orientable, complete, non-compact, finite-volume hyperbolic 3-manifold M with its ideal triangulation $\mathcal{T}$. To study how this combinatorial structure can be mapped into $\mathbb{H}^5$, we introduce the notion of a quaternionic developing map.

Definition B. A *quaternionic developing map* for $(M,\mathcal{T})$ is a continuous map

$$D\colon |\widetilde{\mathcal{T}}| \to \overline{\mathbb{H}}^5$$

mapping the vertices of $|\widetilde{\mathcal{T}}|$ to ideal points in $\partial\mathbb{H}^5$ and each ideal tetrahedron of $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$ to a geodesic ideal tetrahedron in $\mathbb{H}^5$, together with a group morphism $\rho : \pi_1(M) \to \mathrm{PSL}(2,\mathcal{H})$, called the *holonomy*, satisfying

$$D(\gamma \cdot \tilde{x}) = \rho(\gamma) D(\tilde{x}) \tag{3}$$

for every $\tilde{x} \in |\widetilde{\mathcal{T}}|$ and for every $\gamma \in \pi_1(M)$.

Just as Thurston used complex parameters and gluing equations to construct developing maps and holonomies in $\mathbb{H}^3$, we introduce the concept of a *quaternionic decoration* on $\mathcal{T}$ to construct developing maps and holonomies in $\mathbb{H}^5$. The choice of the correct algebraic parameters is based on the following observation: given a quaternionic developing map (D,ρ), the images $\tilde{e}_1$, $\tilde{e}_2$ of any two lifts of the same oriented 1-cell in $|\mathcal{T}| \setminus \{\text{vertices}\}$ are related by an isometry $\rho(\gamma)$ for some $\gamma \in \pi_1(M)$. Let $T_1,\dots,T_k$ and $T_1',\dots,T_k'$ be

the ideal tetrahedra of $\mathbb{H}^5$ winding around $\tilde{e}_1$ and $\tilde{e}_2$ respectively. For every $i = 1, \dots, k$, we denote by q_i and q_i' the quaternionic cross-ratios assigned to the edge $\tilde{e}_1$ in T_i and $\tilde{e}_2$ in T_i' respectively. Since $\rho(\gamma)$ maps $\tilde{e}_1$ to $\tilde{e}_2$, Proposition A implies that the two ordered k-tuples of quaternionic cross-ratios $(q_1, \dots, q_k)$ and $(q_1', \dots, q_k')$ are simultaneously conjugate. Consequently, the equivalence class of a developing map naturally assigns a well-defined tuple of quaternions to each oriented edge of $|\mathcal{T}| \setminus \{\text{vertices}\}$, determined uniquely *up to simultaneous conjugation*.

A further subtlety arises in relation to edge orientations. Consider a quaternionic developing map (D, ρ) and take an oriented edge $\tilde{e}$ in $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$. Let $\widetilde{T}_1, \dots, \widetilde{T}_k$ be the ideal tetrahedra incident to $\tilde{e}$, following a positive cyclic order induced by the orientation of the edge and the tetrahedron. If $q_1, \dots, q_k$ denote the quaternionic cross-ratios of $D(\widetilde{T}_1), \dots, D(\widetilde{T}_k)$ assigned to $D(\tilde{e})$ in $\mathbb{H}^5$, then it is easy to see that

$$q_1 \cdot \ldots \cdot q_k = 1,$$

which is the exact analogue of the 3-dimensional case. However, if $-\tilde{e}$ denotes the same edge with the opposite orientation, the positive cyclic order induced by $-\tilde{e}$ traverses the incident tetrahedra in the reverse direction, namely $\widetilde{T}_k, \dots, \widetilde{T}_1$. The corresponding cross-ratios form a new k-tuple $p_k, \dots, p_1$ (where p_i and q_i refer to the same tetrahedron $\widetilde{T}_i$). The product $p_k \cdot \ldots \cdot p_1$ is again equal to 1, and we will see that these two k-tuples satisfy strong algebraic constraints.

Motivated by this, we introduce the concept of a *quaternionic decoration* of $\mathcal{T}$, which assigns:

- complex parameters to the oriented edges of each ideal tetrahedron of $\mathcal{T}$ (to fix its isometry class). These parameters correspond to the complex moduli in the classical case, and are subject to the same internal conditions;
- to each oriented edge e of valence k in $|\mathcal{T}| \setminus \{\text{vertices}\}$, a *quaternionic lift* around e, *i.e.* a *generic* k-tuple $(q_1, \dots, q_k)$ of quaternions, considered up to simultaneous conjugation, with the property that every q_i is conjugate to the corresponding complex parameter. Here, generic means that adjacent entries q_i and q_{i+1} (as well as q_k and q_1) do not commute. Geometrically, as mentioned above, this non-commutativity corresponds to adjacent tetrahedra not lying in the same 3-dimensional hyperbolic subspace.

The consistency of these parameters is ensured by two families of requirements:

- the *Edge relations*, which are a generalization of the consistency equations in dimension 3. They guarantee that the tetrahedra in $\mathbb{H}^5$ incident to a given edge glue together properly around it. Furthermore, they dictate how the quaternionic lifts assigned to an oriented edge and to the same one with the opposite orientation are algebraically related;
- the *Face relations*, which guarantee that the quaternionic lifts assigned to edges of adjacent ideal tetrahedra can actually arise as the quaternionic cross-ratios of adjacent, ideal tetrahedra in $\mathbb{H}^5$.

To state our main result, we restrict our attention to generic quaternionic developing maps, meaning that adjacent tetrahedra always lie in distinct 3-dimensional hyperbolic subspaces. The main theorem proves that the combinatorial data of a decoration perfectly construct such a developing map and recover its holonomy.

Theorem C (Correspondence Theorem)**.** *There is a canonical one-to-one correspondence between:*

1. *quaternionic decorations on $\mathcal{T}$;*
2. *equivalence classes of generic quaternionic developing maps for $(M, \mathcal{T})$.*

Under this correspondence, the quaternionic lifts of the decoration are precisely realized by the geometric quaternionic cross-ratios of the mapped tetrahedra in $\mathbb{H}^5$.

Structure of the thesis This thesis is organized as follows.

Chapter 1 provides the foundational background. We recall some basic notions about Möbius transformations and hyperbolic geometry. We then review ideal triangulations for oriented, finite-volume, non-compact hyperbolic 3-manifolds and Thurston's consistency equations. The exposition of this first part is mainly based on books by Ratcliffe [Rat19], Benedetti and Petronio [BP92], and Martelli [Mar16], alongside Thurston's notes [Thu02]. We conclude by giving the basic algebraic setup for quaternions, following the chapter on Hamilton's quaternions by Koecher and Remmert in the book *Numbers* [KR91].

In Chapter 2 we expand the algebraic framework of the thesis. We start by studying the action of invertible quaternions on tuples of quaternions by simultaneous conjugation. We then define the Dieudonné determinant of 2×2 quaternionic matrices, and introduce the group $\mathrm{PSL}(2, \mathcal{H})$, which acts on $\widehat{\mathcal{H}}$ by quaternionic fractional linear transformations. This is done following the work of Wilker [Wil93]. Lastly, we define the *quaternionic cross-ratio* of four distinct ordered points in $\widehat{\mathcal{H}}$ and study its behavior under the action of $\mathrm{PSL}(2, \mathcal{H})$. This part was strongly inspired by the papers of Bisi and Gentili [BG11] and of Gwinne and Libine [GL12], although we include slight variations to match our geometric purposes. Some of the results presented here have their counterparts in the aforementioned works, while others, such as the explicit computation of the differential and its relation to the cross-ratio, are original contributions of this thesis.

In the third chapter we apply the quaternionic framework developed in the previous one to 5-dimensional hyperbolic geometry. We identify $\partial\mathbb{H}^5$ with $\widehat{\mathcal{H}}$ and $\mathrm{Isom}^+(\mathbb{H}^5)$ with $\mathrm{PSL}(2, \mathcal{H})$. After a partial classification of the orientation-preserving isometries of $\mathbb{H}^5$ (following [GLM26] and [For04]), we adapt the results of the previous chapter to study the action of $\mathrm{Isom}^+(\mathbb{H}^5)$ on oriented, ideal tetrahedra in $\mathbb{H}^5$.

Chapter 4 contains the main results of this work. We formally define generic quaternionic developing maps and quaternionic decorations on an ideal triangulation of an oriented, complete, non-compact hyperbolic 3-manifold with finite volume. We then prove the Correspondence Theorem, establishing a natural bijection between these two notions. We also apply this combinatorial framework to explicitly compute the peripheral

holonomy, providing conditions to detect boundary-parabolic and discrete representations.

Chapter 1

Preliminaries

In this chapter, we introduce the foundational concepts and mathematical objects that will be used throughout this work.

The first section is devoted to Möbius transformations and their properties; for a comprehensive treatment of this topic, we refer the reader to [Ahl81, BP92].

In the second and third sections, we recall several classical notions and fundamental results in hyperbolic geometry. Standard references for this material include [BP92, Mar16, Rat19].

The fourth section focuses on complete, non-compact, 3-dimensional hyperbolic manifolds with finite volume. We introduce ideal triangulations and detail the formulation of Thurston's consistency equations, which translate the problem of explicitly constructing a non-singular hyperbolic structure on a 3-manifold into a system of equations. The foundational framework for this theory can be found in [BP92, Mar16, Thu02].

Finally, we conclude the chapter by establishing the basic algebraic setup for quaternions, which will play a crucial role in the subsequent development of this work. A general reference for this algebraic background is [KR91].

1.1 Möbius transformations

1.1.1 Inversions in spheres

Let $n \geq 2$ be an integer. We describe a family of self-homeomorphisms of $\widehat{\mathbb{R}}^n = \mathbb{R}^n \cup \{\infty\}$ that will play a crucial role in the rest of the work: *inversions in spheres*.

Consider the sphere $S \subset \mathbb{R}^n$ with center $c \in \mathbb{R}^n$ and radius $r > 0$:

$$S = \{x \in \mathbb{R}^n : ||x - c|| = r\}.$$

We show how to construct a map $i_S \colon \mathbb{R}^n \setminus \{c\} \to \mathbb{R}^n \setminus \{c\}$ which fixes S pointwise and swaps the points inside of S with those outside. Take a point $p \in \mathbb{R}^n \setminus \{c\}$. If $p \in S$, we simply set $i_S(p) \coloneqq p$; otherwise, consider the unique line l passing through p and c. There are two possible scenarios:

- if p lies outside of S (*i.e.* $||p-c|| > r$), consider the cone of half-lines starting from p that are tangent to the sphere. The points of tangency form an $(n-2)$-dimensional sphere on S, which spans an affine hyperplane orthogonal to l. This hyperplane intersects l in a unique point q. We set $i_S(p) = q$;
- if p lies inside of S (*i.e.* $||p-c|| < r$), we do the opposite construction: we consider the affine hyperplane orthogonal to l and passing through p. Its intersection with S is an $(n-2)$-dimensional sphere. The cone of lines tangent to S along this $(n-2)$-sphere all intersect l outside of S in a unique point q. We set $i_S(p) = q$.

An explicit formula for i_S is given by

$$i_S(p) = c + \left(\frac{r}{||p-c||}\right)^2 (p-c), \tag{1.1}$$

and we extend it continuously to a map $\widehat{\mathbb{R}}^n \to \widehat{\mathbb{R}}^n$ setting $i_S(c) \coloneqq \infty$ and $i_S(\infty) \coloneqq c$. We call i_S the *inversion in* S.

Lemma 1.1.1. *Inversions in spheres are smooth involutions of* $\widehat{\mathbb{R}}^n$*, and they map the set consisting of all* k*-spheres and* k*-planes in* $\mathbb{R}^n$ *to itself for all* $k = 1, \dots, n-1$.

Proof. Let S be a sphere in $\mathbb{R}^n$. The fact that i_S is an involution follows directly from the definition, while smoothness can be easily checked using the explicit formula on $\mathbb{R}^n \setminus \{c\}$ and passing to local charts at c and ∞. We omit these standard computations.

We now prove the rest of the statement. First, notice that up to dilations and translations, we can assume that S is the unit sphere centered at the origin of $\mathbb{R}^n$, so that $i_S(x) = \frac{x}{||x||^2}$. Now a submanifold $T \subset \mathbb{R}^n$ of codimension 1 is a sphere or a hyperplane if and only if it satisfies

$$A||x||^2 - 2B \cdot x + C = 0 \tag{1.2}$$

for some real numbers $A, C \in \mathbb{R}$ and for some vector $B \in \mathbb{R}^n$ satisfying $||B||^2 - AC > 0$. Here $B \cdot x$ denotes the usual scalar product in $\mathbb{R}^n$. In particular, if $A \neq 0$ then T is a sphere, and if $A = 0$ it is a hyperplane.

Now if a point $x \in \mathbb{R}^n$ satisfies equation (1.2), it is easily checked that $i_S(x)$ satisfies

$$C||i_S(x)||^2 - 2B \cdot i_S(x) + A = 0,$$

proving the statement for $k = n-1$. To deduce the result for a general $k < n-1$, it suffices to observe that any k-sphere or k-plane in $\mathbb{R}^n$ can be expressed as the transverse intersection of $n-k$ hyperspheres or hyperplanes. Since i_S is a bijection of $\widehat{\mathbb{R}}^n$ that preserves the collection of codimension-1 spheres and planes, it must also preserve their intersections. This implies that i_S maps k-spheres and k-planes to objects of the same kind, concluding the proof. □

Along the proof of the previous result, we have discovered the following property.

Corollary 1.1.2. *Let S be any sphere of $\mathbb{R}^n$ centered at a point c, and let $T \subset \mathbb{R}^n$ be a hyperplane or a sphere. Then*

- *if $c \in T$, the inversion in S maps T to a hyperplane;*
- *if $c \notin T$, the inversion in S maps T to a sphere.*

There is another fundamental class of smooth involutions of $\widehat{\mathbb{R}}^n$ exhibiting similar geometric properties, namely *reflections across hyperplanes*. Let $H \subset \mathbb{R}^n$ be a hyperplane. The classical Euclidean reflection $r_H \colon \mathbb{R}^n \to \mathbb{R}^n$ across H continuously extends by simply setting $r_H(\infty) \coloneqq \infty$. Since r_H is an affine isometry of Euclidean space $\mathbb{R}^n$, it trivially preserves the class of k-planes and k-spheres. Furthermore, just like inversions in spheres, extended reflections are smooth involutions of $\widehat{\mathbb{R}}^n$.

Remark 1.1.3. The geometric construction of inversions involving tangent lines needs the assumption $n \geq 2$. However, the concepts of inversions and reflections make sense also in dimension 1. In $\mathbb{R}$, a sphere of center c and radius r is simply the pair of points $S = \{c - r, c + r\}$, and equation (1.1) yields the fractional linear transformation

$$i_S(x) = c + \frac{r^2}{x - c}.$$

Similarly, a hyperplane in $\mathbb{R}$ is a single point c, and the reflection across it is the central symmetry $x \mapsto 2c - x$.

With this understanding, we consider inversions and reflections for $n = 1$ case as well.

Together, inversions and reflections generate a rich and rigid transformation group.

Definition 1.1.4. A *Möbius transformation* of $\widehat{\mathbb{R}}^n$ is a diffeomorphism obtained as a finite composition of inversions in spheres and reflections across hyperplanes.

Möbius transformations form a group under composition, which we denote by $\mathrm{Möb}(\widehat{\mathbb{R}}^n)$ and we call the *Möbius group*. Since both inversions and reflections reverse the orientation, we denote by $\mathrm{Möb}^+(\widehat{\mathbb{R}}^n)$ the index-two subgroup of orientation-preserving transformations, which are exactly those obtained by composing an even number of maps.

Remark 1.1.5. Translations and rotations are orientation-preserving Möbius transformations because they are obtained by composing an even number of reflections. More generally, since reflections across hyperplanes generate the isometry group of $\mathbb{R}^n$, the Möbius group contains all Euclidean isometries. Furthermore, dilations are also Möbius transformations (they can be obtained by composing two inversions in concentric spheres), therefore the group $\mathrm{Möb}(\widehat{\mathbb{R}}^n)$ actually contains all Euclidean similarities.

Proposition 1.1.6. *For every $n \geq 2$, the group $\mathrm{Möb}(\widehat{\mathbb{R}}^n)$ is 3-transitive, i.e. it acts transitively on ordered triples of distinct points in $\widehat{\mathbb{R}}^n$.*

Proof. We show that for each triple of distinct points $p_1, p_2, p_3 \in \widehat{\mathbb{R}}^n$ there exists a Möbius transformation f such that

$$f(p_1) = 0, \quad f(p_2) = e_1, \quad f(p_3) = \infty,$$

where $e_1 \coloneqq (1, 0, \dots, 0) \in \mathbb{R}^n$. We proceed as follows. If $p_3 \neq \infty$, we apply an inversion in a sphere centered at p_3. This maps p_3 to ∞. Let p_1' and p_2' be the images of p_1 and p_2 respectively (if $p_3 = \infty$, we simply set $p_1' = p_1$ and $p_2' = p_2$).

We then apply the translation $x \mapsto x - p_1'$, which fixes ∞ and maps p_1' to 0. Let p_2'' be the image of p_2'.

Finally, if $p_2'' \neq e_1$, we apply a rotation centered at the origin to map p_2'' to $(|p_2''|, 0, \dots, 0)$. Now a dilation by $1/|p_2''|$ maps this point to e_1.

Since both the rotation and the dilation are Möbius transformations that fix 0 and ∞, the proof is complete. □

The result is true also for orientation-preserving Möbius transformations.

Corollary 1.1.7. *For every $n \geq 2$, the group $\mathrm{Möb}^+(\widehat{\mathbb{R}}^n)$ is 3-transitive.*

Proof. Let p_1, p_2, p_3 and q_1, q_2, q_3 be two triples of distinct points in $\widehat{\mathbb{R}}^n$. By Proposition 1.1.6, there exists a Möbius transformation f such that $f(p_i) = q_i$ for every $i = 1, 2, 3$.

If f preserves the orientation we are done. Otherwise, we can post-compose f with an orientation-reversing transformation that fixes q_1, q_2, and q_3.

If one of the q_i, say q_1, is the point ∞, we apply a reflection across any hyperplane passing through q_2, q_3.

If $q_i \neq \infty$ for every $i = 1, 2, 3$, we apply an inversion in any sphere passing through q_1, q_2, q_3. The proof is complete. □

Remark 1.1.8. Proposition 1.1.6 holds true also in dimension $n = 1$. Indeed, by applying an inversion followed by a translation, we can always map three distinct points $p_1, p_2, p_3 \in \widehat{\mathbb{R}}$ to $0, x, \infty$ respectively, for some non-zero $x \in \mathbb{R}$.

Now we have two cases: if $x > 0$, we simply rescale it to get $1 \in \mathbb{R}$; otherwise we need to apply a reflection through the origin first.

This argument also shows why the action of orientation-preserving Möbius transformations of $\widehat{\mathbb{R}}$ is not 3-transitive: for instance, it is not possible to map the triple $0, -1, \infty$ to $0, 1, \infty$ respectively without inverting the orientation.

1.1.2 Möbius groups in dimensions one and two

The groups of Möbius transformations of $\widehat{\mathbb{R}}$ and of $\widehat{\mathbb{R}}^2$ are famously isomorphic to some well-known groups of 2×2 matrices. We now review these classical correspondences, starting with the $n = 2$ case.

Every element of the complex, special linear group

$$\mathrm{SL}(2, \mathbb{C}) = \{A \in \mathrm{Mat}(2, \mathbb{C}) : \det(A) = 1\}$$

acts on the Riemann sphere $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ by fractional linear transformations. More precisely, every matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{C})$$

determines the orientation-preserving diffeomorphism $\varphi_A : \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ defined by

$$\varphi_A(z) \coloneqq \frac{az + b}{cz + d}$$

for $z \neq \infty, -d/c$, which we extend continuously setting $A(\infty) = a/c$, $A(-d/c) = \infty$. Two matrices induce the same map if and only if they differ by a sign; therefore, this action descends to the projective special linear group

$$\mathrm{PSL}(2, \mathbb{C}) = \mathrm{SL}(2, \mathbb{C})/\{\pm I\},$$

yielding a well-defined and injective map $[A] \mapsto \varphi_A$ from $\mathrm{PSL}(2, \mathbb{C})$ into the group of self-diffeomorphisms of $\widehat{\mathbb{C}}$.

Similarly, every such matrix $A \in \mathrm{SL}(2, \mathbb{C})$ induces an orientation-reversing diffeomorphism $\psi_A \colon \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ by pre-composing with complex conjugation:

$$\psi_A(z) \coloneqq \frac{a\bar{z} + b}{c\bar{z} + d}.$$

Again, two matrices induce the same map if and only if they differ by an overall sign; thus, this correspondence descends to $\mathrm{PSL}(2, \mathbb{C})$ as well, providing a well-defined and injective map $[A] \mapsto \psi_A$ from $\mathrm{PSL}(2, \mathbb{C})$ into the group of self-diffeomorphisms of $\widehat{\mathbb{C}}$.

By identifying $\mathbb{R}^2$ with $\mathbb{C}$, we get the following result.

Proposition 1.1.9. *The group of Möbius transformations of $\widehat{\mathbb{R}}^2$ is given by*

$$\{\varphi_A : [A] \in \mathrm{PSL}(2, \mathbb{C})\} \cup \{\psi_A : [A] \in \mathrm{PSL}(2, \mathbb{C})\}. \tag{1.3}$$

In particular, $\mathrm{Möb}^+(\widehat{\mathbb{R}}^2) \cong \mathrm{PSL}(2, \mathbb{C})$

Proof. We start by proving that the whole group $\mathrm{Möb}(\widehat{\mathbb{R}}^2)$ is contained in the set defined by (1.3). Recall that $\mathrm{Möb}(\widehat{\mathbb{R}}^2)$ is, by definition, generated by reflections across lines and inversions in circles. Under the standard identification $\mathbb{R}^2 \cong \mathbb{C}$, the reflection across the real axis corresponds to the complex conjugation $z \mapsto \bar{z}$, which is precisely ψ_{Id}. The inversion in the unit circle centered at the origin is given by $z \mapsto 1/\bar{z}$, which is ψ_J for the matrix

$$J = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \in \mathrm{SL}(2, \mathbb{C}).$$

Both these maps belong to the second set of our statement. Now every line reflection and circle inversion is obtained by conjugating these maps with dilations $z \mapsto \lambda z$ and translations $z \mapsto z + b$. These transformations belong to the first set, being induced by upper-triangular matrices via φ_A.

To conclude this first inclusion, we must verify that the set (1.3) is closed under composition. A direct algebraic computation shows that for every pair of matrices $A, B \in \mathrm{SL}(2,\mathbb{C})$ we have:

$$\varphi_A \circ \varphi_B = \varphi_{AB}, \quad \varphi_A \circ \psi_B = \psi_{AB}, \quad \psi_A \circ \varphi_B = \psi_{A\bar{B}}, \quad \text{and} \quad \psi_A \circ \psi_B = \varphi_{A\bar{B}},$$

where $\bar{B}$ denotes the matrix obtained by complex-conjugating the entries of B. Therefore, depending on whether we compose an even or an odd number of orientation-reversing generators, we respectively preserve or reverse the orientation, thus obtaining a map in the first or in the second set. This proves

$$\mathrm{Möb}(\widehat{\mathbb{R}}^2) \subseteq \{\varphi_A : [A] \in \mathrm{PSL}(2,\mathbb{C})\} \cup \{\psi_A : [A] \in \mathrm{PSL}(2,\mathbb{C})\}.$$

Conversely, consider a matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{C}).$$

We want to show that the associated transformations φ_A and ψ_A are a Möbius transformation. If $c = 0$, then $\varphi_A(z) = \frac{a}{d}z + \frac{b}{d}$, which is a composition of a similarity (i.e., a rotation and a dilation) and a translation, all of which are Möbius transformations. If $c \neq 0$, algebraic manipulation yields

$$\varphi_A(z) = \frac{a}{c} - \frac{1}{c(cz+d)}.$$

This shows that φ_A can be obtained as the composition of the similarity $z \mapsto cz + d$, the standard inversion $z \mapsto -1/z$, the similarity $z \mapsto \frac{1}{c}z$, and a final translation $z \mapsto z + \frac{a}{c}$. Since each of these self-diffeomorphism is a Möbius transformation by Remark 1.1.5, we get $\varphi_A \in \mathrm{Möb}^+(\widehat{\mathbb{R}}^2)$. Furthermore, since $\psi_A(z) = \varphi_A(\bar{z})$, the map ψ_A is also a Möbius transformation (reversing the orientation). This establishes the equality of the two sets. □

In Corollary 1.1.7, we showed that the group $\mathrm{Möb}^+(\widehat{\mathbb{R}}^n)$ acts transitively on triples of distinct points. In the 2-dimensional case, via the isomorphism $\mathrm{Möb}^+(\widehat{\mathbb{R}}^2) \cong \mathrm{PSL}(2,\mathbb{C})$ proved in Proposition 1.1.9, we can easily see that the action on triples is also free.

Corollary 1.1.10. *The action of* $\mathrm{PSL}(2,\mathbb{C})$ *on ordered triples of distinct points in* $\widehat{\mathbb{C}}$ *is free.*

Proof. Since the action is transitive by Corollary 1.1.7, it suffices to show that the stabilizer of the triple $0, 1, \infty$ is trivial. This is immediate: a fractional linear transformation

$$z \mapsto \frac{az+b}{cz+d}$$

fixes $0, 1$ and ∞ if and only if $c = b = 0$ and $a = d$. This corresponds precisely to the identity element in $\mathrm{PSL}(2,\mathbb{C})$. □

Remark 1.1.11. By combining Proposition 1.1.7 with Corollary 1.1.10, we obtain that given three distinct points $p_1, p_2, p_3 \in \widehat{\mathbb{C}}$ there exists a unique transformation $f \in \mathrm{PSL}(2,\mathbb{C})$ mapping p_1, p_2, p_3 to $0, 1, \infty$ respectively.

We now move to the case $n = 1$. The special linear group with real coefficients

$$\mathrm{SL}(2,\mathbb{R}) = \{A \in \mathrm{Mat}(2,\mathbb{R}) : \det(A) = 1\}$$

is naturally a subgroup of $\mathrm{SL}(2,\mathbb{C})$, and as such it acts on the Riemann sphere $\widehat{\mathbb{C}}$ by fractional linear transformations. The action descends to $\mathrm{PSL}(2,\mathbb{R}) = \mathrm{SL}(2,\mathbb{R})/\{\pm I\}$, leaving $\widehat{\mathbb{R}} \subset \widehat{\mathbb{C}}$ invariant.

In this way, we have obtained a family of diffeomorphisms of $\widehat{\mathbb{R}}$. As in the complex case, these maps will correspond to orientation-preserving Möbius transformations of $\widehat{\mathbb{R}}$. To capture the full group, we need to identify the orientation-reversing transformations, which can be done by looking at the entire stabilizer of $\widehat{\mathbb{R}}$ in $\mathrm{PSL}(2,\mathbb{C})$.

Remark 1.1.12. Notice that for every matrix $A \in \mathrm{SL}(2,\mathbb{C})$, the induced maps φ_A and ψ_A coincide when restricted to $\widehat{\mathbb{R}}$. This is why, unlike the 2-dimensional case, pre-composing with complex conjugation does not yield any new transformations on $\widehat{\mathbb{R}}$.

Lemma 1.1.13. *Given a matrix $A \in \mathrm{SL}(2,\mathbb{C})$, the Möbius transformation φ_A preserves $\widehat{\mathbb{R}}$ if and only if A has either real or purely imaginary coefficients.*

Proof. The transformation φ_A preserves $\widehat{\mathbb{R}}$ if and only if A and its complex conjugate $\bar{A}$ induce the same transformation on $\widehat{\mathbb{C}}$. Indeed, since $\bar{x} = x$ for all $x \in \widehat{\mathbb{R}}$, we clearly have $\overline{\varphi_A(x)} = \varphi_{\bar{A}}(x)$. Thus, φ_A preserves $\widehat{\mathbb{R}}$ if and only if φ_A and $\varphi_{\bar{A}}$ coincide on $\widehat{\mathbb{R}}$, and hence on all of $\widehat{\mathbb{C}}$ (since the action of the Möbius group is free on triples of distinct points, see Corollary 1.1.10).

This yields two possibilities: either $\bar{A} = A$ and $A \in \mathrm{SL}(2,\mathbb{R})$, or $\bar{A} = -A$, which implies that the coefficients of A are purely imaginary. □

Suppose that $A \in \mathrm{SL}(2,\mathbb{C})$ has purely imaginary coefficients. Then $A = iB$ where B is a real matrix with $\det(B) = -1$. While the map φ_A preserves the orientation of $\widehat{\mathbb{C}}$, it is easy to see that its restriction to $\widehat{\mathbb{R}}$ is orientation reversing. This allows us to fully classify the one-dimensional case.

Corollary 1.1.14. *The group of Möbius transformations of $\widehat{\mathbb{R}}$ is given by*

$$\{\varphi_A|_{\widehat{\mathbb{R}}} : A \in \mathrm{SL}(2,\mathbb{R})\} \cup \{\varphi_{iB}|_{\widehat{\mathbb{R}}} : B \in \mathrm{Mat}(2,\mathbb{R}), \det(B) = -1\}.$$

In particular, $\mathrm{Möb}^+(\widehat{\mathbb{R}}) \cong \mathrm{PSL}(2,\mathbb{R})$.

Remark 1.1.15. Unlike the complex case where every invertible matrix can be scaled to have determinant 1, the real groups $\mathrm{PGL}(2,\mathbb{R})$ and $\mathrm{PSL}(2,\mathbb{R})$ do not coincide. The projectivization of the subset of matrices with negative determinant forms the non trivial coset:

$$\mathrm{PGL}(2,\mathbb{R}) \setminus \mathrm{PSL}(2,\mathbb{R}) = \{A \in \mathrm{Mat}(2,\mathbb{R}) : \det(A) = -1\}/\{\pm I\}.$$

By Corollary 1.1.14, for every matrix B in this coset, the maps φ_{iB} provide the orientation-reversing Möbius transformations of $\widehat{\mathbb{R}}$. From this one can easily prove the isomorphism $\mathrm{Möb}(\widehat{\mathbb{R}}) \cong \mathrm{PGL}(2,\mathbb{R})$

1.1.3 Conformal maps

Definition 1.1.16. Let (M, g) and (N, h) be two Riemannian manifolds. A diffeomorphism $f\colon M \to N$ is *conformal* if for each point $p \in M$ and for every pair of tangent vectors $u, v \in T_pM$ we have

$$h(df_p(u), df_p(v)) = \lambda_p^2 g(u, v),$$

where $\lambda_p > 0$ is a positive real number which varies smoothly with p.

If both M and N are oriented, we will sometimes refer to orientation-reversing conformal maps as *anticonformal maps*.

If $f\colon M \to N$ is a conformal map between two Riemannian manifolds, for every point $p \in M$ the map $A_p \coloneqq \frac{1}{\lambda_p} df_p$ is a linear isometry between the tangent spaces. Conversely, it is straightforward to check that any diffeomorphism whose differential decomposes in this way is a conformal map.

When $M = N$, the set of conformal diffeomorphisms $M \to M$ forms a group under composition, and we denote it with $\mathrm{Conf}(M)$. When M is oriented, the subgroup of orientation-preserving conformal maps is denoted by $\mathrm{Conf}^+(M)$.

Remark 1.1.17. Conformal maps preserve the angle between two tangent vectors but not necessarily the length of a vector.

Example 1.1.18. The stereographic projection is a conformal map. Indeed, let $S^n \subset \mathbb{R}^{n+1}$ be the unit sphere centered at the origin, and identify $\mathbb{R}^n$ with the horizontal hyperplane $\{x_{n+1} = 0\}$. Recall that the stereographic projection at the north pole $N = (0, \dots, 0, 1) \in S^n$ is the diffeomorphism $f\colon S^n \setminus \{N\} \to \mathbb{R}^n$ given by

$$f(x_1, \dots, x_{n+1}) = \frac{1}{1 - x_{n+1}}(x_1, \dots, x_n).$$

We can explicitly compute the differential of f in coordinates. By adopting the notation $x = (x', x_{n+1})$ where $x' = (x_1, \dots, x_n) \in \mathbb{R}^n$, we have

$$\frac{\partial f}{\partial x_i} = \frac{e_i}{1 - x_{n+1}} \quad \forall i = 1, \dots, n, \quad \text{and} \quad \frac{\partial f}{\partial x_{n+1}} = \frac{x'}{(1 - x_{n+1})^2},$$

where $e_1, \dots, e_n$ sono i vettori della base canonica di $\mathbb{R}^n$. Now for every tangent vector $v = (v', v_{n+1}) \in T_xS^n$ we obtain

$$df_x(v) = \frac{(1 - x_{n+1})v' + v_{n+1}x'}{(1 - x_{n+1})^2}. \tag{1.4}$$

To verify conformality, we take another tangent vector $w = (w', w_{n+1}) \in T_xS^n$ and we use (1.4) to compute the inner product of $df_x(v)$ and $df_x(w)$:

$$\langle df_x(v), df_x(w)\rangle = \frac{(1 - x_{n+1})^2\langle v', w'\rangle + (1 - x_{n+1})(v_{n+1}\langle x', w'\rangle + w_{n+1}\langle v', x'\rangle) + v_{n+1}w_{n+1}\|x'\|^2}{(1 - x_{n+1})^4}.$$

Since $v, w \in T_x S^n$, they are orthogonal to x. This means $\langle x, v\rangle = 0$, which yields $\langle x', v'\rangle = -x_{n+1}v_{n+1}$, and similarly $\langle x', w'\rangle = -x_{n+1}w_{n+1}$. Moreover, since $x \in S^n$, we have $\|x'\|^2 = 1 - x_{n+1}^2$. Substituting these identities into the numerator, we obtain:

$$\begin{aligned}
&(1-x_{n+1})^2\langle v', w'\rangle - 2x_{n+1}(1-x_{n+1})v_{n+1}w_{n+1} + (1-x_{n+1}^2)v_{n+1}w_{n+1}\\
&= (1-x_{n+1})^2\langle v', w'\rangle + (1-x_{n+1})v_{n+1}w_{n+1}(-2x_{n+1}+1+x_{n+1})\\
&= (1-x_{n+1})^2\langle v', w'\rangle + (1-x_{n+1})^2 v_{n+1}w_{n+1}\\
&= (1-x_{n+1})^2(\langle v', w'\rangle + v_{n+1}w_{n+1})\\
&= (1-x_{n+1})^2\langle v, w\rangle_{\mathbb{R}^{n+1}}.
\end{aligned}$$

Dividing by the denominator $(1-x_{n+1})^4$, we finally get:

$$\langle df_x(v), df_x(w)\rangle_{\mathbb{R}^n} = \frac{1}{(1-x_{n+1})^2}\langle v, w\rangle,$$

which proves that f is a conformal map.

Two Riemannian metrics g and g' on a smooth manifold M are said to be *conformally equivalent* if $g' = \lambda^2 g$ for some strictly positive smooth function $\lambda\colon M \to \mathbb{R}$. A *conformal structure* on M is an equivalence class of Riemannian metrics under this relation. If two manifolds M and N are endowed only with a conformal structure, it follows immediately from the definition that the notion of a conformal diffeomorphism $f\colon M \to N$ is still well-defined.

We can endow the one-point compactification $\widehat{\mathbb{R}}^n$ with a conformal structure by identifying it with the unit sphere $S^n \subset \mathbb{R}^{n+1}$. This is done via the inverse of the stereographic projection of Example 1.1.18, extended to a bijection by declaring that ∞ corresponds to the north pole N. Through this identification, $\widehat{\mathbb{R}}^n$ naturally inherits the conformal structure of the sphere, and inversions and reflections are anticonformal maps.

Proposition 1.1.19. *Inversions in spheres and reflections across hyperplanes are anticonformal maps of* $\widehat{\mathbb{R}}^n$.

Proof. The statement is immediate for reflections. For inversions, up to translations and dilations (which are conformal maps by definition), it is enough to prove the statement for the inversion in the unit sphere centered at the origin, which is given by

$$i(x) = \frac{x}{||x||^2}.$$

It is readily checked that the differential at a point $x \in \mathbb{R} \setminus \{0\}$ is given by

$$di_x(v) = \frac{1}{||x||^2}\left(v - 2\frac{x\cdot v}{||x||^2}x\right),$$

where v is a tangent vector at x. Notice that the linear map

$$R_{x^\perp}(v) \coloneqq v - 2\frac{\langle x, v\rangle}{||x||^2}x$$

is precisely the orthogonal reflection across the hyperplane $x^\perp$. Therefore, we can write

$$di_x = \frac{1}{||x||^2} R_{x^\perp}.$$

Since $R_{x^\perp}$ is a linear isometry with determinant -1, the inversion i is anticonformal on $\mathbb{R} \setminus \{0\}$.

The computation of the differentials at 0 and ∞ is easier: the inversion $i(x) = x/||x||^2$ is precisely the transition map from the chart $\mathbb{R}^n$ around 0 and the chart $\widehat{\mathbb{R}}^n \setminus \{0\}$ around ∞. Consequently, in local coordinates, the inversion becomes $i \circ i$ which is the identity because inversions are involutions. □

The previous result implies that

$$\text{Möb}(\widehat{\mathbb{R}}^n) \subset \text{Conf}(S^n), \quad \text{Möb}^+(\widehat{\mathbb{R}}^n) \subset \text{Conf}^+(S^n).$$

We will now prove that these inclusions are equalities for all $n \geq 2$. Let us begin with the case $n = 2$. By identifying S^2 with the Riemann sphere $\widehat{\mathbb{C}}$ via the stereographic projection, the sphere naturally inherits a complex structure In this dimension, the result follows directly from complex analysis.

Proposition 1.1.20. *The group* $\text{Conf}(S^2)$ *consists precisely of biholomorphisms and anti-biholomorphisms of* $\widehat{\mathbb{C}}$. *In particular,* $\text{Conf}^+(S^2) = Aut(\widehat{\mathbb{C}})$.

Proof. A diffeomorphism $f\colon S^2 \to S^2$ is conformal if and only if, at each point $p \in S^2$, its differential df_p is a positive scalar multiple of a linear isometry between the tangent spaces. Under the identification $S^2 = \widehat{\mathbb{C}}$, we can express this differential in the standard complex charts (using z on $\mathbb{C}$ and $w = 1/z$ around ∞), which naturally identify the tangent spaces with $\mathbb{R}^2 \cong \mathbb{C}$. In these coordinates, the condition translates into two possibilities for the differential, depending on the orientation. If f is orientation-preserving, the differential df_z is $\mathbb{C}$-linear everywhere. This is equivalent to f being locally holomorphic; since f is a global diffeomorphism of $\widehat{\mathbb{C}}$, it must be a biholomorphism. Similarly, when f is orientation-reversing, the differential df_z is $\mathbb{C}$-antilinear. This is equivalent to f being locally anti-holomorphic, and we conclude as before that f is an anti-biholomorphism.

Conversely, the differential of any (anti-)biholomorphism is everywhere $\mathbb{C}$-(anti-)linear, which corresponds to a positive scalar multiple of an isometry. Hence, any such map preserves angles and is automatically a conformal diffeomorphism, completing the proof. □

It is a classical and well-known result (see, for instance, [Don11]) that the group of biholomorphisms of $\widehat{\mathbb{C}}$ - also called automorphisms - coincides with the group of fractional linear transformations $\text{PSL}(2, \mathbb{C})$, which we have already identified with $\text{Möb}^+(\widehat{\mathbb{R}}^2)$ in Proposition 1.1.9. Similarly, since any anti-biholomorphism of $\widehat{\mathbb{C}}$ can be obtained by composing a fractional linear transformation with the complex conjugation $z \mapsto \bar{z}$, these maps naturally correspond to the orientation-reversing elements of the Möbius group. We sum up these observations in the following result.

Corollary 1.1.21. *The following equalities hold:*

$$\mathrm{Conf}(S^2) = \mathrm{Möb}(\widehat{\mathbb{R}}^2), \quad \mathrm{Conf}^+(S^2) = \mathrm{Möb}^+(\widehat{\mathbb{R}}^2).$$

For $n \geq 3$, an analogue of Corollary 1.1.21 remains true, but it is due to a general result known as Liouville's theorem. A proof can be found in [BP92].

Theorem 1.1.22 (Liouville's Theorem)**.** *Let $n \geq 3$ be an integer. Every conformal map between two domains of $\mathbb{R}^n$ is a composition of translations, dilations, reflections and inversions.*

Corollary 1.1.23. *For every $n \geq 3$, the following equalities hold:*

$$\mathrm{Conf}(S^n) = \mathrm{Möb}(\widehat{\mathbb{R}}^n), \quad \mathrm{Conf}^+(S^n) = \mathrm{Möb}^+(\widehat{\mathbb{R}}^n).$$

Proof. We already know that Möbius transformations are conformal diffeomorphisms by Proposition 1.1.19. Conversely, let $f\colon S^n \to S^n$ be a conformal map. By conjugating f with the stereographic projection (which is itself conformal, see Example 1.1.18), we can view f locally as a conformal map $\mathbb{R}^n \to \mathbb{R}^n$. Liouville's Theorem 1.1.22 now implies that f is a composition of translations, dilations, reflections and inversions. All these maps can be extended to the compactification $\widehat{\mathbb{R}}^n$, yielding a Möbius transformation. □

Remark 1.1.24. The inclusion $\mathrm{Möb}(\widehat{\mathbb{R}}) \subset \mathrm{Conf}(S^1)$ is strict. Indeed, every self-diffeomorphism of S^1 is a conformal map, meaning that $\mathrm{Conf}(S^1)$ coincides with the infinite-dimensional group $\mathrm{Diffeo}(S^1)$. On the other hand, the group $\mathrm{Möb}(\widehat{\mathbb{R}})$ is a finite-dimensional Lie group by Corollary 1.1.14.

1.2 Hyperbolic space

The n-dimensional hyperbolic space, denoted by $\mathbb{H}^n$, is the unique, simply connected Riemannian n-manifold with constant sectional curvature equal to -1. We briefly describe some models for $\mathbb{H}^n$. Details can be found in [Mar16], [BP92] and [Rat19].

The hyperboloid model Consider the Lorentzian scalar product on $\mathbb{R}^{n+1}$ given by

$$\langle x, y\rangle_L := \sum_{i=1}^{n} x_i y_i - x_{n+1} y_{n+1}.$$

The set of vectors $x \in \mathbb{R}^{n+1}$ such that $\langle x, x\rangle_L = -1$ forms a subset of $\mathbb{R}^{n+1}$ with two connected components, known as the *two-sheeted hyperboloid*. The hyperboloid model I^n of hyperbolic space $\mathbb{H}^n$ is obtained by choosing the sheet with $x_{n+1} > 0$:

$$I^n = \{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} : \ \langle x, x\rangle_L = -1, \ x_{n+1} > 0\}.$$

We immediately see that I^n is a smooth manifold. The function $f\colon \mathbb{R}^{n+1} \to \mathbb{R}$ given by $f(x) = \langle x, x\rangle_L$ is smooth, with differential $df_x(y) = 2\langle x, y\rangle_L$. Since $df_x \neq 0$ for all

$x \in f^{-1}(-1)$, the differential is surjective on the whole level set, and hence I^n is a smooth submanifold of $\mathbb{R}^{n+1}$ having codimension 1.

The restriction of the Lorentzian scalar product to each tangent space naturally induces a Riemannian metric on I^n. Indeed, the tangent space at any point $x \in I^n$ is the n-dimensional vector subspace

$$T_x I^n = \{y \in \mathbb{R}^{n+1} : \langle x, y\rangle_L = 0\} = x^\perp$$

of $\mathbb{R}^{n+1}$, which allows us to split $\mathbb{R}^{n+1}$ as the orthogonal direct sum $\mathbb{R}^{n+1} = \mathrm{Span}(x) \oplus T_x I^n$. Since $\langle x, x\rangle_L = -1 < 0$, the restriction of the scalar product to $\mathrm{Span}(x)$ is negative-definite. Because the Lorentzian scalar product has signature $(n, 1)$, its restriction to the orthogonal complement $x^\perp = T_x I^n$ must be positive-definite.

The Poincaré disc model Let

$$D^n = \{x \in \mathbb{R}^n : |x|^2 < 1\}$$

be the open, unit ball in $\mathbb{R}^n$ centered at the origin. The stereographic projection with respect to the point $(0, \dots, 0, -1) \in \mathbb{R}^{n+1}$ onto the subspace $\mathbb{R}^n \times \{0\}$ is defined on the upper-half space $\{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : x_{n+1} > 0\}$ as

$$f(x_1, \dots, x_{n+1}) = \frac{(x_1, \dots, x_n, 0)}{1 + x_{n+1}}.$$

By identifying $\mathbb{R}^n$ with the horizontal hyperplane $\{x_{n+1} = 0\} \subset \mathbb{R}^{n+1}$, the restriction of the projection f to the hyperboloid I^n yields a diffeomorphism $f\colon I^n \to D^n$. The metric tensor of I^n is pushed by f to a metric tensor g on D^n. Similarly to what we did in Example 1.1.18, via standard calculus and linear algebra arguments we can compute the pushed metric tensor g at a point $x \in D^n$, obtaining

$$g_x = \left(\frac{2}{1 - |x|^2}\right)^2 g^E$$

where g^E is the induced Euclidean metric on D^n.

The half-space model Let H^n denote the upper half-space of $\mathbb{R}^n$, that is

$$H^n = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_n > 0\}.$$

We now show that H^n can be equipped with a hyperbolic metric constructed from the one on the Poincaré disc.

Let i_S denote the inversion in the sphere $S \subset \mathbb{R}^n$ centered at the point $(0, \dots, 0, -1)$ and with radius $\sqrt{2}$. We know by Lemma 1.1.1 and Proposition 1.1.19 that i_S is a smooth involution $\widehat{\mathbb{R}}^n \to \widehat{\mathbb{R}}^n$, which is also an anticonformal map.

Since the center of S is a point on the boundary ∂D^n of D^n, and S itself intersects ∂D^n on the horizontal plane $\{x_n = 0\}$, Corollary 1.1.2 ensures that the boundary ∂D^n is mapped to the hyperplane $\{x_n = 0\} = \partial H^n$, compactified with the point at infinity.

We can actually say more: the map i_S restricts to a diffeomorphism between the open ball D^n and the half-space H^n. Again, we can use i_S to push the metric tensor of D^n to a metric tensor g on H^n. At a point $x \in H^n$ we get

$$g_x = \frac{1}{x_n^2} g^E, \tag{1.5}$$

where g^E denotes the induced Euclidean metric on H^n.

Remark 1.2.1. The metric tensor g defined in (1.5) restricts to a metric that is proportional to the Euclidean one on horizontal hyperplanes $\{x_n = c > 0\}$.

The Poincaré disc model D^n and the half-space model H^n are called *conformal models* because they are diffeomorphic to Euclidean domains of $\mathbb{R}^n$ via conformal maps.

1.2.1 Subspaces and geodesics

We start with a definition that does not depend on the specific model.

Definition 1.2.2. A k-dimensional subspace of $\mathbb{H}^n$ is a complete, connected, totally geodesic submanifold of $\mathbb{H}^n$ having dimension k.

The following propositions describe how subspaces look in the three models of hyperbolic space $\mathbb{H}^n$ that we just introduced. For the hyperboloid model I^n, the result follows directly by looking at the restriction of the Lorentzian scalar product.

Proposition 1.2.3. *A k-dimensional subspace of I^n is the intersection of a $(k+1)$-dimensional vector subspace of $\mathbb{R}^{n+1}$ with I^n (when non-empty).*

For the conformal models D^n and H^n, the result follows by looking at the images of subspaces of I^n by the respective conformal maps (see, for instance, [Mar16]).

Proposition 1.2.4. *The k-subspaces of D^n are the intersection of D^n with Euclidean k-spheres and k-planes orthogonal to its boundary ∂D^n.*

Proposition 1.2.5. *The k-subspaces of H^n are the intersections of H^n with Euclidean k-planes and k-spheres of $\mathbb{R}^n$ orthogonal to ∂H^n.*

Similarly to Euclidean space, the support of each geodesic of $\mathbb{H}^n$ is contained in its subspaces of dimension 1 (see, for instance, [Mar16]), which are called *lines*. In the half-space model H^n, it follows directly from Proposition 1.2.5 that these lines correspond to open vertical half-lines and open Euclidean semicircles orthogonal to the boundary hyperplane $\{x_n = 0\}$.

We can explicitly parametrize the vertical half-lines by arc-length as follows.

Proposition 1.2.6. *Fix a point $x = (x_1, \dots, x_n) \in H^n$ and let $v \in T_x H^n$ be the unit, upward-pointing, vertical tangent vector at x. The unique geodesic $\gamma\colon \mathbb{R} \to H^n$ with $\gamma(0) = x$ and $\gamma'(0) = v$ is the vertical half-line given by*

$$\gamma(t) = (x_1, \dots, x_{n-1}, x_n e^t).$$

1.2.2 Boundary at infinity

The conformal models D^n and H^n for hyperbolic space $\mathbb{H}^n$ are realized as open subsets of $\mathbb{R}^n$, and they naturally come with a topological compactification (obtained by adding the unit sphere ∂D^n to the disc D^n and the extended hyperplane $\partial H^n \cup \{\infty\} = \widehat{\mathbb{R}}^{n-1}$ to the half-space H^n). The boundary points added through this process are called *points at infinity*.

We recall the intrinsic definition, that does not depend on the chosen model.

Definition 1.2.7. We say that two geodesic half-lines $\gamma_1, \gamma_2 \colon [0, +\infty) \to \mathbb{H}^n$ are *equivalent*, and we write $\gamma_1 \sim \gamma_2$, if and only if

$$\sup_{t \geq 0} d(\gamma_1(t), \gamma_2(t)) < +\infty. \tag{1.6}$$

The set of *points at infinity* $\partial\mathbb{H}^n$ of $\mathbb{H}^n$ is the set of geodesic half-lines up to equivalence.

We will sometimes call $\partial\mathbb{H}^n$ the *boundary at infinity* of $\mathbb{H}^n$, and we set $\overline{\mathbb{H}}^n = \partial\mathbb{H}^n \cup \mathbb{H}^n$.

As anticipated, the boundary at infinity can be explicitly identified with the topological boundary of the two compactified models. We describe the identification for the half-space model H^n, and for D^n is analogous.

A geodesic half-line γ in the half-space model H^n is the intersection of H^n with either a Euclidean circle or a Euclidean line orthogonal to the hyperplane $\{x_n = 0\}$ in $\mathbb{R}^n$. Therefore, the limit $\lim_{t \to +\infty} \gamma(t)$ is either ∞ or a point in $\{x_n = 0\}$ and γ naturally determines a unique point in $\partial H^n \cup \{\infty\} = \widehat{\mathbb{R}}^{n-1}$. Using the explicit arc-length parametrization of geodesics, one can prove that two half-lines are equivalent if and only if they converge to the same boundary point (where vertical half-lines converge to the point ∞). Consequently, each point in $\widehat{\mathbb{R}}^{n-1}$ uniquely determines the equivalence class of all geodesic half-lines converging to it, and this provides a canonical bijection between $\partial\mathbb{H}^n$ and $\widehat{\mathbb{R}}^{n-1}$.

This map extends to the compactification, yielding a bijection between $\overline{\mathbb{H}}^n$ and $\bar{H}^n = H^n \cup \widehat{\mathbb{R}}^{n-1}$. We endow $\overline{\mathbb{H}^n}$ and $\partial\mathbb{H}^n$ with the topology of $\bar{H}^n$ and $\widehat{\mathbb{R}}^{n-1}$ respectively. Moreover, the canonical conformal structure of $\widehat{\mathbb{R}}^{n-1}$ naturally endows $\partial\mathbb{H}^n$ with a conformal structure. As we shall see below, the isometries of $\mathbb{H}^n$ will continuously extend to the boundary precisely as conformal diffeomorphisms of $\widehat{\mathbb{R}}^{n-1}$.

We conclude by defining an important family of submanifolds of hyperbolic space.

Definition 1.2.8. Let $p \in \partial\mathbb{H}^n$ be a point at infinity. A *horosphere centered at* p is a connected, complete hypersurface in $\mathbb{H}^n$ that is orthogonal to all geodesics converging to p.

In the half-space model H^n, the geodesics converging to ∞ are vertical half-lines, hence horospheres centered at ∞ are simply horizontal, Euclidean hyperplanes contained in H^n. If $p \neq \infty$ is another point at infinity, every geodesic converging to p is a semi-circle orthogonal to the boundary $\{x_n = 0\}$ at p. The connected and complete hypersurfaces orthogonal to all such semicircles are precisely the Euclidean spheres contained in H^n and tangent to the boundary at x_0.

1.2.3 Isometries of hyperbolic space

Let $\mathrm{O}(n,1)$ denote the orthogonal group with respect to the Lorentzian scalar product, *i.e.* the set of linear isomorphisms f of $\mathbb{R}^{n+1}$ satisfying

$$\langle f(x), f(y)\rangle_L = \langle x, y\rangle_L \quad \forall\, x, y \in \mathbb{R}^{n+1}.$$

By definition, $\mathrm{O}(n,1)$ preserves the entire two-sheeted hyperboloid. The subgroup consisting of those linear maps that preserve the upper sheet I^n has index two, and we denote it by $\mathrm{O}^+(n,1)$.

It is a classical result that the restriction of these linear transformations to I^n gives the group $\mathrm{Isom}(I^n)$ of isometries of I^n (see [Mar16], [Rat19], [BP92]).

Proposition 1.2.9. *The group $\mathrm{O}^+(n,1)$ is isomorphic to $\mathrm{Isom}(I^n)$ via restriction.*

The group $\mathrm{O}^+(n,1)$ contains a special class of maps: reflections across subspaces. Indeed, for every subspace S of I^n we can define an isometry r_S, which is called the *reflection across S*. The construction goes as follows: by definition, we can write S as the intersection $S = I^n \cap V$ where V is a $(k+1)$-dimensional vector subspace of $\mathbb{R}^{n+1}$, and we have $\mathbb{R}^{n+1} = V \oplus V^\perp$. Since V intersects the hyperboloid, the restriction of the Lorentzian form to V is non-degenerate. We set

$$r_S|_V \coloneqq id, \quad r_S|_{V^\perp} \coloneqq -id.$$

Lemma 1.2.10. *The restriction of r_S to I^n is an isometry of I^n. It is orientation-preserving if and only if S has even codimension.*

We now focus on the case where S is a *hyperplane*, *i.e.* it has codimension 1. The following result is standard and a proof can be found in [BP92].

Theorem 1.2.11. *The group $\mathrm{O}^+(n,1)$ is generated by reflections across hyperplanes.*

Since $\mathrm{Isom}(\mathbb{H}^n) \cong \mathrm{O}^+(n,1)$, every isometry of $\mathbb{H}^n$ is the composition of a finite number of reflections in hyperbolic hyperplanes. These maps are easily described in the half-space model H^n. By Proposition 1.2.5, there are two possibilities for a hyperbolic hyperplane $W \subset H^n$:

- if W is the intersection of H^n with a vertical hyperplane $P \subset \mathbb{R}^n$, the hyperbolic reflection across W is the restriction of the Euclidean reflection across P. Since P is orthogonal to $\{x_n = 0\}$, this reflection preserves the boundary.
- if W is the intersection of H^n with a Euclidean sphere S of codimension 1 orthogonal to $\mathbb{R}^{n-1} = \{x_n = 0\}$, the hyperbolic reflection across W is the restriction of the Euclidean inversion in S. Since S is orthogonal to $\{x_n = 0\}$, its center lies in this hyperplane and therefore the inversion preserves $\{x_n = 0\} \cup \{\infty\}$.

In both cases, the reflection in W naturally extends to the compactification $\bar{H}^n$, and we can consider its restriction to the boundary at infinity.

Lemma 1.2.12. *Each $\varphi \in \mathrm{Isom}(\mathbb{H}^n)$ extends to a homeomorphism of $\overline{\mathbb{H}}^n$. The restriction of φ to $\partial\mathbb{H}^n \cong \widehat{\mathbb{R}}^{n-1}$ is a Möbius transformation of $\widehat{\mathbb{R}}^{n-1}$.*

Proof. We adopt the half-space model H^n. By the previous analysis each hyperbolic hyperplane $W \subset H^n$ uniquely determines a subset of $\mathbb{R}^{n-1}$ which is either a hyperplane or a sphere of codimension 1, and the hyperbolic reflection in W restricts on the boundary to either a Euclidean reflection or a Euclidean inversion. Since every $\varphi \in \mathrm{Isom}(\mathbb{H}^n)$ is a finite composition of hyperbolic reflections, the isometry φ admits a continuous extension to the compactification $\bar{H}^n$, which does not depend on the chosen decomposition.

Lastly, the restriction of φ to $\widehat{\mathbb{R}}^{n-1}$ is a finite composition of Euclidean reflections and inversions, which by definition is a Möbius transformation. □

Remark 1.2.13. The extension of an isometry $\varphi \in \mathrm{Isom}(\mathbb{H}^n)$ to $\overline{\mathbb{H}}^n$ can also be defined intrinsically: a boundary point is the equivalence class $[\gamma]$ of a geodesic half-line γ, and we set $\varphi([\gamma]) := [\varphi(\gamma)]$. Since isometries map geodesics to geodesics and preserve distances, the extension is well-defined.

Conversely, every Möbius transformation of the boundary at infinity induces an isometry of $\mathbb{H}^n$. We describe this correspondence for the generators of the Möbius group:

- every Euclidean hyperplane $P \subset \mathbb{R}^{n-1}$ uniquely determines a hyperbolic hyperplane $\widetilde{P} = P \times (0, +\infty)$ in the half-space H^n. The Euclidean reflection across P extends to the hyperbolic reflection across $\widetilde{P}$.
- a sphere $S \subset \mathbb{R}^{n-1}$ with radius $r > 0$ and center $c \in \mathbb{R}^{n-1}$ uniquely bounds the upper hemisphere $\widetilde{S} \subset H^n$ with the same radius and center $(c, 0)$. By definition, $\widetilde{S}$ is a hyperbolic hyperplane in H^n. The Euclidean inversion in S extends to the hyperbolic reflection across $\widetilde{S}$ (that is, the Euclidean inversion in $\widetilde{S}$ restricted to H^n).

Remark 1.2.14. The correspondence we have just described is often referred to as the *Poincaré extension*. As we have seen in Remark 1.1.5, every orientation-preserving similarity f of $\mathbb{R}^{n-1}$ is a Möbius transformation of $\widehat{\mathbb{R}}^{n-1}$, and we can easily give an explicit description of the Poincaré extensions of these maps in coordinates. For instance, assume $f\colon \mathbb{R}^{n-1} \to \mathbb{R}^{n-1}$ is given by

$$f(x) = \lambda A x + b$$

with $\lambda \in \mathbb{R}_{>0}$, $A \in S\,\mathrm{O}(n-1)$ and $b \in \mathbb{R}^{n-1}$. Then its extension to a point $(x, t) \in H^n$ is given by

$$(x, t) \mapsto (f(x), \lambda t).$$

Since this correspondence preserves composition and acts as the inverse to the restriction map of Lemma 1.2.12, we obtain the following result.

Proposition 1.2.15. *For every integer $n \geq 2$, the following group isomorphisms hold:*

$$\mathrm{Isom}(\mathbb{H}^n) \cong \mathrm{Möb}(\widehat{\mathbb{R}}^{n-1}), \quad \mathrm{Isom}^+(\mathbb{H}^n) \cong \mathrm{Möb}^+(\widehat{\mathbb{R}}^{n-1}).$$

In particular, by Corollary 1.1.14 and Proposition 1.1.9, we have the following isomorphisms in dimensions 2 and 3:

$$\mathrm{Isom}^+(\mathbb{H}^2) \cong \mathrm{PSL}(2,\mathbb{R}), \quad \mathrm{Isom}^+(\mathbb{H}^3) \cong \mathrm{PSL}(2,\mathbb{C}).$$

Remark 1.2.16. As observed in Remark 1.2.1, in the half-space model H^n the metric tensor restricts to a metric that is proportional to the Euclidean one on horizontal hyperplanes, making them isometric to $\mathbb{R}^{n-1}$. These hypersurfaces are precisely horospheres centered at ∞. Now let $p \neq \infty$ be another point at infinity: every inversion in a sphere centered at p extends to an isometry of $\mathbb{H}^n$ that maps p to ∞. Because isometries map geodesics to geodesics and preserve orthogonality, they naturally map horospheres to horospheres. Since they also preserve the intrinsic metric, we conclude that all horospheres in $\mathbb{H}^n$ are isometric to Euclidean space $\mathbb{R}^{n-1}$.

1.2.4 Classification of isometries

Given an isometry $f \in \mathrm{Isom}(\mathbb{H}^n)$, its extension to the compactification $\overline{\mathbb{H}}^n$ is continuous and hence admits a fixed point by Brouwer's theorem. Furthermore, the following proposition limits the number of fixed points on the boundary (for a proof, see [BP92]).

Proposition 1.2.17. *Let $f \in \mathrm{Isom}(\mathbb{H}^n)$ be a non-trivial isometry. If f does not fix any point in the interior of $\mathbb{H}^n$, then it has at most two fixed points on the boundary $\partial\mathbb{H}^n$.*

We can thus provide the following definition.

Definition 1.2.18. We say that a non-trivial isometry $f \in \mathrm{Isom}(\mathbb{H}^n)$ is

- *elliptic* if it fixes a point in $\mathbb{H}^n$;
- *parabolic* if it does not fix any point in $\mathbb{H}^n$, and it fixes exactly 1 point on the boundary $\partial\mathbb{H}^n$;
- *hyperbolic* if it does not fix any point in $\mathbb{H}^n$, and it fixes exactly 2 points on the boundary $\partial\mathbb{H}^n$.

In odd dimension, every orientation-preserving elliptic isometry has fixed points on the boundary as well.

Lemma 1.2.19. *Every orientation-preserving elliptic isometry of $\mathbb{H}^n$ fixes a point in $\partial\mathbb{H}^n$ if and only if n is odd.*

Proof. We identify $\mathbb{H}^n$ with the hyperboloid I^n, and its orientation-preserving isometry group with $\mathrm{SO}^+(n,1)$. Let $f \in \mathrm{SO}^+(n,1)$ be any elliptic, orientation-preserving isometry, and let $x \in I^n$ be a fixed point for f. As we have seen, the Lorentzian scalar product restricts to a positive-definite one on $T_xI^n = x^\perp$. Thus, the differential

$$df_x \colon T_xI^n \to T_xI^n$$

acts as an orientation-preserving orthogonal transformation, meaning $df_x \in \mathrm{SO}(n)$.

When n is odd, every element of $\mathrm{SO}(n)$ has 1 as an eigenvalue, hence df_x fixes a non-zero tangent vector $v \in T_x I^n$. If $\gamma\colon \mathbb{R} \to I^n$ is the unique geodesic starting with $\gamma(0) = x$ and $\gamma'(0) = v$, the map f pointwise fixes $\gamma(t)$ for every $t \in \mathbb{R}$. As a consequence, it fixes the two endpoints of γ on the boundary at infinity.

Conversely, when n is even, we can find orientation-preserving elliptic isometries without fixed points on the boundary. For instance, consider the central symmetry $f\colon I^n \to I^n$ with respect to the point $p = (1, 0, \dots, 0) \in I^n$, defined as

$$f((x_1, x_2, \dots, x_{n+1})) = (x_1, -x_2, \dots, -x_{n+1}).$$

This map fixes p, and its differential at p is $df_p = -Id$. Since n is even, $\det(-Id) = (-1)^n = +1$, meaning that this symmetry preserves the orientation. However, it is easy to see that f does not fix any point on the boundary: indeed, for every geodesic $\gamma\colon \mathbb{R} \to I^n$ passing through p, f swaps the two endpoints of γ on the boundary at infinity. □

The argument used in the proof of Lemma 1.2.19 also establishes the following result.

Corollary 1.2.20. *When n is odd, every elliptic, orientation-preserving isometry of $\mathbb{H}^n$ fixes at least two points in $\partial\mathbb{H}^n$.*

In the low-dimensional cases $n = 2$ and $n = 3$, we have made the identifications

$$\mathrm{Isom}^+(\mathbb{H}^2) = \mathrm{PSL}(2, \mathbb{R}), \quad \mathrm{Isom}^+(\mathbb{H}^3) = \mathrm{PSL}(2, \mathbb{C}).$$

In these cases, we can detect whether an orientation-preserving isometry is elliptic, parabolic or hyperbolic by looking at its trace, as stated in the following results. For a proof, see, for instance [Mar16].

Theorem 1.2.21. *Let $f \in \mathrm{Isom}^+(\mathbb{H}^2)$ be a non-trivial, orientation-preserving isometry represented by a matrix*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R}).$$

Then f is:

- *elliptic if and only if $|tr(A)| < 2$;*
- *parabolic if and only if $|tr(A)| = 2$;*
- *hyperbolic if and only if $|tr(A)| > 2$.*

Theorem 1.2.22. *Let $f \in \mathrm{Isom}^+(\mathbb{H}^3)$ be a non-trivial, orientation-preserving isometry represented by a matrix*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{C}).$$

Then f is:

- *elliptic if and only if* $tr(A) \in (-2, 2) \subset \mathbb{R}$*;*
- *parabolic if and only if* $tr(A) = \pm 2$*;*
- *hyperbolic if and only if* $tr(A) \in \mathbb{C} \setminus [-2, 2]$.

Remark 1.2.23. Throughout this work, we will almost exclusively use the upper half-space model. For this reason, we will drop the notation H^n from now on, and simply use $\mathbb{H}^n$ to denote both the abstract hyperbolic space and its half-space model.

1.3 Hyperbolic manifolds and developing maps

In this section we recall the definition of the *developing map* and the *holonomy representation* assigned to a hyperbolic structure on a smooth manifold M.

Definition 1.3.1. A hyperbolic manifold M of dimension n is a Riemannian manifold which is locally isometric to hyperbolic space $\mathbb{H}^n$.

We state the following result for simply connected hyperbolic manifolds. A proof can be found in [Mar16].

Proposition 1.3.2. *Let M be a simply connected, hyperbolic n-manifold. There exists a local isometry*

$$D\colon M \to \mathbb{H}^n$$

which is unique up to post-composition with an isometry of $\mathbb{H}^n$. More precisely, given any two local isometries $D_1, D_2\colon M \to \mathbb{H}^n$, there exists a unique global isometry $g \in \mathrm{Isom}(\mathbb{H}^n)$ such that $D_2 = g \circ D_1$.

Let now M be any hyperbolic n-manifold, and consider its universal cover

$$\pi\colon \widetilde{M} \to M.$$

The manifold $\widetilde{M}$ is simply connected and admits a hyperbolic structure: by Proposition 1.3.2 we obtain a local isometry

$$D\colon \widetilde{M} \to \mathbb{H}^n, \tag{1.7}$$

which is called a *developing map* for M. We use the map D to define a group morphism $\rho\colon \pi_1(M) \to \mathrm{Isom}(\mathbb{H}^n)$ which is uniquely determined — up to conjugation — by the hyperbolic structure on M.

Recall that the fundamental group $\pi_1(M)$ naturally acts on the universal cover $\widetilde{M}$ by deck transformations. For every element $\gamma \in \pi_1(M)$ and $x \in \widetilde{M}$, we denote this action simply by $\gamma \cdot x$. Sometimes, when it is convenient to consider the action as a map, we denote by $f_\gamma\colon \widetilde{M} \to \widetilde{M}$ the deck transformation corresponding to γ.

Proposition 1.3.3. *There exists a group morphism $\rho\colon \pi_1(M) \to \mathrm{Isom}(\mathbb{H}^n)$ making the developing map $D\colon \widetilde{M} \to \mathbb{H}^n$ equivariant with respect to the action of $\pi_1(M)$. More precisely, for every $\gamma \in \pi_1(M)$ and $x \in \widetilde{M}$ we have*

$$\rho(\gamma)(D(x)) = D(\gamma \cdot x). \tag{1.8}$$

Proof. Start with an element $\gamma \in \pi_1(M)$. The corresponding deck transformation $f_\gamma \colon \widetilde{M} \to \widetilde{M}$ is an isometry of $\widetilde{M}$, so the composition

$$D \circ f_\gamma \colon \widetilde{M} \to \mathbb{H}^n$$

is still a local isometry. Since $\widetilde{M}$ is a simply connected hyperbolic manifold, by Proposition 1.3.2 the maps D and $D \circ f_\gamma$ coincide after composing with a unique isometry of $\mathbb{H}^n$, which we call $\rho(\gamma)$. More precisely, we get the required relation (1.8):

$$\rho(\gamma) \circ D = D \circ f_\gamma.$$

Finally, we need to show that ρ is a group morphism: given $\gamma, \delta \in \pi_1(M)$, by repeatedly applying (1.8) we have

$$\begin{aligned} \rho(\gamma\delta) \circ D &= D \circ f_{\gamma\delta} \\ &= D \circ f_\gamma \circ f_\delta \\ &= \rho(\gamma) \circ D \circ f_\delta \\ &= \rho(\gamma) \circ \rho(\delta) \circ D \end{aligned}$$

and we conclude by Proposition 1.3.2 □

The group morphism $\rho \colon \pi_1(M) \to \mathrm{Isom}(\mathbb{H}^n)$ we just found is called the *holonomy representation*.

Corollary 1.3.4. *The holonomy representation is uniquely determined — up to conjugation — by the hyperbolic structure on M.*

Proof. Let $D_1, D_2 \colon \widetilde{M} \to \mathbb{H}^n$ be two developing maps — possibly the same — inducing two holonomy representations $\rho_1, \rho_2 : \pi_1(M) \to \mathrm{Isom}(\mathbb{H}^n)$. By Proposition 1.3.2, there exists an isometry $\varphi \in \mathrm{Isom}(\mathbb{H}^n)$ such that $D_1 = \varphi \circ D_2$. Applying the equivariance equation (1.8) for every $\gamma \in \pi_1(M)$ yields

$$\rho_1(\gamma) \circ \varphi \circ D_2 = \varphi \circ D_2 \circ f_\gamma.$$

Now we post-compose with φ^{-1} and we get

$$\begin{aligned} \rho_2(\gamma) \circ D_2 &= D_2 \circ f_\gamma \\ &= (\varphi^{-1} \circ \rho_1(\gamma) \circ \varphi) \circ D_2. \end{aligned}$$

This shows that the isometries $\rho_2(\gamma)$ and $\varphi^{-1} \circ \rho_1(\gamma) \circ \varphi$ coincide on the image of D_2. Since D_2 is a local isometry, its image is an open subset of $\mathbb{H}^n$. Because two isometries of $\mathbb{H}^n$ that agree on a non-empty open set must coincide everywhere, we conclude that

$$\rho_2(\gamma) = \varphi^{-1} \circ \rho_1(\gamma) \circ \varphi \quad \forall\, \gamma \in \pi_1(M).$$

The proof is complete. □

The pair (D, ρ) is often called a *developing pair* for M. Conversely, such a pair completely determines a hyperbolic structure on M. Indeed, since D is a local diffeomorphism, we can pull back the hyperbolic metric of $\mathbb{H}^n$ to the universal cover $\widetilde{M}$. The equivariance property (1.8) ensures that the deck transformations act as isometries for this pull-back metric, hence the metric descends to a well-defined hyperbolic structure on M.

Remark 1.3.5. When M is complete, the developing map $D\colon \widetilde{M} \to \mathbb{H}^n$ is a covering. On the other hand, $\mathbb{H}^n$ is contractible and so D is a global isometry. In this case, the holonomy representation $\rho\colon \pi_1(M) \to \mathrm{Isom}(\mathbb{H}^n)$ is discrete and faithful, and M is isometric to the quotient $\mathbb{H}^n/\rho(\pi_1(M))$. As a consequence, the image of ρ does not contain any elliptic isometries of $\mathbb{H}^n$.

Remark 1.3.6. Let M, N be two complete, hyperbolic n-manifolds with finite volume. Suppose moreover that their fundamental groups are isomorphic. When $n \geq 3$, Mostow-Prasad rigidity theorem (see, for instance, [BP92] or [Rat19]) implies that M and N are isometric. As a consequence, the hyperbolic structure on a complete, hyperbolic manifold M with finite volume is unique. In particular, there exists a unique discrete and faithful representation $\pi_1(M) \to \mathrm{Isom}(\mathbb{H}^n)$ up to conjugation.

1.4 Ideal triangulations of hyperbolic 3-manifolds

We now focus on the case where M is an oriented, complete hyperbolic 3-manifold. We assume moreover that M is non-compact but has finite volume. By the Margulis Lemma, such a manifold has a finite number of *cusps*, which are geometrically shrinking ends homeomorphic to $T^2 \times [0, \infty)$, where T^2 denotes the 2-torus (see [Mar16]). Consequently, M is homeomorphic to the interior of a compact 3-manifold N whose boundary consists of a disjoint union of tori.

As we have seen in the previous section, the hyperbolic structure on M determines (and is determined by) a developing pair (D, ρ). Sometimes it is possible to provide an explicit description of the hyperbolic structures on M by defining the developing pair piecewise on the tetrahedra of an *ideal triangulation*. This fundamental construction is due to Thurston [Thu02].

Definition 1.4.1. A *triangulation* $\mathcal{T}$ is the data of n identical copies $T_1, \dots, T_n$ of the standard 3-simplex, together with a partition of the $4n$ faces into $2n$ pairs. Moreover, we require that for every pair of faces $\{F, F'\}$ of the partition, we are given two simplicial isomorphisms

$$\sigma_{F,F'}\colon F \to F', \quad \sigma_{F',F}\colon F' \to F$$

such that $\sigma_{F',F} = \sigma_{F,F'}^{-1}$.

We say that $\mathcal{T}$ is an *oriented* triangulation if each of the simplicial isomorphisms $\sigma_{F,F'}$ reverses the orientation, where F and F' are oriented as faces of the tetrahedra they belong to, and the tetrahedra are equipped with the canonical orientation of the standard 3-simplex.

Given a triangulation $\mathcal{T}$, we denote by $|\mathcal{T}|$ the topological space obtained by gluing the tetrahedra T_i along their faces via the prescribed isomorphisms. More precisely

$$|\mathcal{T}| = \bigsqcup_{i=1}^{n} T_i \Big/ \sim$$

where $\sim$ is the equivalence relation generated by requiring that $x \sim y$ if and only if $x = y$ or there exist two paired faces F, F' of the partition such that $x \in F$, $y \in F'$ and $y = \sigma_{F,F'}(x)$. Sometimes, we will refer to $|\mathcal{T}|$ as the *geometric realization* of $\mathcal{T}$.

We denote by

$$\pi_{\mathcal{T}} \colon \bigsqcup_{i=1}^{n} T_i \to |\mathcal{T}|$$

the projection, and by $V(\mathcal{T})$ the set consisting of the vertices of $T_1, \dots, T_n$. Since the faces are glued via simplicial isomorphisms, the equivalence class of every vertex is contained in $V(\mathcal{T})$. We define the set of vertices, or 0-cells, of $|\mathcal{T}|$ as the subset $\pi_{\mathcal{T}}(V(\mathcal{T}))$. Moreover, by construction, $\pi_{\mathcal{T}}$ is injective on the interior of every tetrahedron $T_1, \dots, T_n$ as well as on the interior of every face. However, it is not guaranteed that the interior of an edge of $\mathcal{T}$ projects injectively onto $|\mathcal{T}|$. We now show that it does when the triangulation is oriented.

Lemma 1.4.2. *Let $\mathcal{T}$ be an oriented triangulation and $|\mathcal{T}|$ be its geometric realization. For every edge e of $\mathcal{T}$, the restriction of $\pi_{\mathcal{T}}$ to the interior of e is injective. Moreover, $|\mathcal{T}| \setminus \{\text{vertices}\}$ is an oriented manifold.*

Proof. Fix an edge e of $\mathcal{T}$ and a point x in the interior of e. Let y be another point in the interior of e such that $x \sim y$: we want to prove that $x = y$. By definition, there exists a finite sequence of points

$$x = x_0, x_1, \dots, x_k = y$$

such that for every $j = 1, \dots, k$, there exist two paired faces F_j, F_j' with $x_{j-1} \in F_j$, $x_j \in F_j'$ and

$$\sigma_{F_j,F_j'}(x_{j-1}) = x_j.$$

Since the maps $\sigma_{F_j,F_j'}$ are simplicial isomorphisms, every point x_j lives in the interior of an edge e_j for all $j = 0, \dots, k$. Of course, by definition, $e_0 = e_k = e$ and every $\sigma_{F_j,F_j'}$ maps e_{j-1} to e_j. We denote by $f_j \colon e_{j-1} \to e_j$ the map induced by $\sigma_{F_j,F_j'}$ for $j \geq 1$.

Now the composition

$$f = f_k \circ \dots \circ f_1$$

of all these restrictions is a simplicial isomorphism $f \colon e \to e$ such that $f(x) = y$. We will show that f is the identity. First, we assume, without loss of generality, that the sequence $e_0, \dots, e_k$ of edges is *minimal*, meaning that $e_i \neq e_j$ for every $i \neq j \in \{0, \dots, k-1\}$ (indeed, if two intermediate edges coincide, we replace e by this edge and repeat the same argument).

Minimality implies that for every $j = 1, \dots, k-1$, F'_j and F_{j+1} are two distinct faces of the same tetrahedron of $\mathcal{T}$ and they both contain the edge e_j. Furthermore, we also have that F'_k and F_1 are two distinct faces of one and the same tetrahedron of $\mathcal{T}$ both containing the edge e (unless $k = 2$, where we have $f = f_1^{-1} \circ f_1 = id$).

Now, since $\mathcal{T}$ is an oriented triangulation, every simplicial isomorphism σ_{F_j,F'_j} reverses the orientation induced by the faces on their edges. Moreover, any two faces of a tetrahedron induce opposite orientations on the edge they share. Therefore, tracking the orientation along the entire cycle requires exactly k gluing maps and k face transitions within the abstract tetrahedra. Since each of these $2k$ operations reverses the orientation, the composition f preserves it. We conclude that f is the identity and $x = y$.

We now prove that $|\mathcal{T}| \setminus \{\text{vertices}\}$ is a manifold. Since the projection $\pi_{\mathcal{T}}$ is injective on the interior of every tetrahedron we only need to check that if x is a point in the interior of a face or an edge, then $\pi_{\mathcal{T}}(x)$ has a neighborhood homeomorphic to $\mathbb{R}^3$. Suppose first that x is a point in the interior of a face F of some tetrahedron T. By definition, x is identified via $\sigma_{F,F'}$ with a unique point y lying in the interior of another face F' of some tetrahedron T', where T and T' are possibly the same, while F and F' are distinct. Consider two sufficiently small open half-balls $B_x \subset T$ and $B_y \subset T'$ centered at x and y respectively. Their intersections with the faces F and F' are two open disks. The simplicial isomorphism $\sigma_{F,F'}$ glues the two half-balls along these boundary disks, resulting in a neighborhood of $\pi_{\mathcal{T}}(x)$ homeomorphic to an open ball in $\mathbb{R}^3$.

We move to the case where x is a point in the interior of an edge. Consider the finite set of points lying in the fiber of $\pi_{\mathcal{T}}(x)$. By proceeding exactly as above, we can cyclically order them into a finite sequence

$$x = x_1, \dots, x_k.$$

For every $i = 1, \dots, k$, consider a sufficiently small closed sphere S_i centered at x_i in the corresponding tetrahedron. The intersection of S_i with the tetrahedron is a bigon with two vertices lying on the edge e_i containing x_i, and two boundary arcs (which we call edges) lying on the two adjacent faces. The partition of the faces of $\mathcal{T}$ thus induces a partition of the $2k$ edges of the bigons in k pairs. By gluing the bigons along their edges via restrictions of the corresponding simplicial isomorphisms, we get a closed surface S with the same number k of 1-cells and 2-cells. Lastly, the number of vertices, or 0-cells, is 2, since we just proved that $\pi_{\mathcal{T}}$ is injective on the interior of every edge. Thus the Euler characteristic of S is $\chi(S) = 2 - k + k = 2$, which means that S is homeomorphic to a sphere. As a consequence, the neighborhood of $\pi_{\mathcal{T}}(x)$ in $|\mathcal{T}| \setminus \{\text{vertices}\}$ bounded by S is a 3-ball.

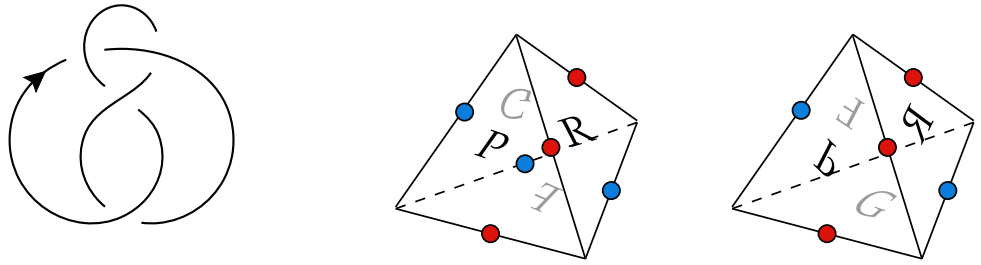
We conclude the proof by noticing that, whether x lies in the interior of a face or of an edge, the 3-ball neighborhood of $\pi_{\mathcal{T}}(x)$ constructed above is obtained by gluing subsets of the tetrahedra T_i along their faces. Since each $\sigma_{F,F'}$ reverses the orientation of the faces, the canonical orientations of the tetrahedra match coherently across the gluings, making $|\mathcal{T}| \setminus \{\text{vertices}\}$ an oriented manifold. □

Recall that we have fixed an oriented, complete, non-compact hyperbolic 3-manifold M with finite volume.

Definition 1.4.3. We say that an oriented triangulation $\mathcal{T}$ is an *ideal triangulation* of M if M is homeomorphic to $|\mathcal{T}| \setminus \{\text{vertices}\}$, and the orientations of the two manifolds match.

Remark 1.4.4. Let $\mathcal{T}$ be an ideal triangulation of M. In what follows, we will often identify M with $|\mathcal{T}| \setminus \{\text{vertices}\}$. Under this identification, by a slight abuse of notation, we will continue to denote by $T_1, \ldots, T_n$ the tetrahedra with their vertices removed, viewing them both in $|\mathcal{T}| \setminus \{\text{vertices}\}$ and in M, and we will refer to them as *ideal tetrahedra*. Keep in mind that, in M, an ideal tetrahedron T_i may fail to be embedded along its boundary due to self-identifications of its faces or edges. We will also talk about the *1-skeleton* of M to mean the subset given by the projection of the interiors of the edges of $\mathcal{T}$ onto $|\mathcal{T}| \setminus \{\text{vertices}\}$.

Example 1.4.5. Let M_8 denote the figure-eight knot complement in S^3, see Figure 1.1a. It is a well-known fact (see [Thu02]) that M_8 admits a complete, hyperbolic structure with finite volume and with a single cusp. The manifold M_8 admits an oriented, ideal triangulation $\mathcal{T}_8$ with two ideal tetrahedra, as illustrated in Figure 1.1b. Every face is labeled with a letter, either F, G, P, or R, and faces with the same letter are glued so that the letters match. After the gluing, $|\mathcal{T}_8| \setminus \{\text{vertices}\} \cong M_8$ has 4 faces, 2 edges (the red one and the blue one), and a single ideal vertex, which corresponds to the cusp of the manifold.



(a) A diagram for the figure-eight knot. (b) The ideal triangulation $\mathcal{T}_8$ of M_8.

Figure 1.1: On the left, an oriented diagram of the figure-eight knot in S^3. On the right, an ideal triangulation $\mathcal{T}_8$ for the figure-eight knot complement M_8. Faces with the same letters are identified so that the letters match. Every edge is marked with either a red dot or a blue dot. Edges with the same colors are identified after the gluing.

Induced 2-dimensional triangulation Let $\mathcal{T}$ be an ideal triangulation of M. Recall that M has a finite number of *cusps*, which are shrinking ends homeomorphic to $T^2 \times [0, \infty)$, where T^2 denotes the 2-torus.

Consider a vertex $v \in |\mathcal{T}|$, and let $v_1, \dots, v_{k_v}$ be the vertices in the fiber $\pi_{\mathcal{T}}^{-1}(v)$. For each $i = 1, \dots, k_v$, we intersect the tetrahedron containing v_i with a sufficiently small sphere centered at v_i: this yields a collection of 2-simplices

$$\Delta_1^{(v)}, \dots, \Delta_{k_v}^{(v)},$$

called the *links* of the vertices $v_1, \dots, v_{k_v}$. Since every edge of $\Delta_1^{(v)}, \dots, \Delta_{k_v}^{(v)}$ is contained in exactly one face of $T_1, \dots, T_n$, we deduce that:

- the partition of the faces of $T_1, \dots, T_n$ into pairs induces a partition of the $3k_v$ edges of $\Delta_1^{(v)}, \dots, \Delta_{k_v}^{(v)}$ into pairs. In particular, k_v is even and the partition consists of $3k_v/2$ pairs;
- the simplicial isomorphisms $\sigma_{F,F'}$ between the paired faces of the tetrahedra induce simplicial isomorphisms between the corresponding paired edges of $\Delta_1^{(v)}, \dots, \Delta_{k_v}^{(v)}$.

We call $\mathcal{T}_v$ the collection of the k_v 2-simplices above, together with the edge pairings and the simplicial isomorphisms between them. The topological space obtained by gluing the simplices $\Delta_1^{(v)}, \dots, \Delta_{k_v}^{(v)}$ along their edges via the prescribed simplicial isomorphisms is a closed, connected surface which we denote by L_v. By definition of the quotient topology, the cone

$$C(L_v) = (L_v \times [0,1])/(L_v \times \{1\})$$

over L_v is homeomorphic to a closed neighborhood of v in $|\mathcal{T}|$. By removing the vertex v, we get a cylinder $L_v \times [0,1)$ which is contained in $|\mathcal{T}| \setminus \{\text{vertices}\}$, and which corresponds to a cusp of M. Since every cusp is homeomorphic to a closed, infinite cylinder over the torus, we conclude that L_v is homeomorphic to a torus. In the rest of the work, we will sometimes refer to $\mathcal{T}_v$ as the *induced 2-dimensional triangulation* of the truncated cusp of M corresponding to the vertex $v \in |\mathcal{T}|$.

The lifted triangulation Let $p\colon \widetilde{M} \to M$ denote the universal cover of M. Recall that $\pi_1(M)$ acts on $\widetilde{M}$ by deck transformations, and that for every $\gamma \in \pi_1(M)$ we have denoted by $f_\gamma\colon \widetilde{M} \to \widetilde{M}$ the deck transformation corresponding to γ.

We show that $\widetilde{M}$ is obtained by gluing infinitely many tetrahedra along their faces, in a way that is induced by the ideal triangulation $\mathcal{T}$ of M. We start with the following observation. For every tetrahedron T_i of $\mathcal{T}$, let

$$q_i\colon T_i \setminus \{\text{vertices}\} \to |\mathcal{T}| \setminus \{\text{vertices}\}$$

be the restriction of $\pi_{\mathcal{T}}$ to T_i with its vertices removed. Since $T_i \setminus \{\text{vertices}\}$ is simply connected, q_i lifts to a continuous map $\tilde{q}_i$ from $T_i \setminus \{\text{vertices}\}$ to the universal cover of $|\mathcal{T}| \setminus \{\text{vertices}\}$, which is homeomorphic to $\widetilde{M}$. Moreover, every lift is obtained as the composition $f_\gamma \circ \tilde{q}_i$ as γ varies in $\pi_1(M)$.

Now consider two tetrahedra T_i and T_j of $\mathcal{T}$, possibly the same, such that a face F of T_i is paired with a face F' of T_j (recall that F and F' cannot coincide). By definition,

$$q_i(\text{int}(F)) = q_j(\text{int}(F')).$$

Since the interiors of the faces are connected, we obtain the following result, whose proof we omit.

Lemma 1.4.6. *In the above setting, for every $\gamma \in \pi_1(M)$ there exists a unique $\gamma' \in \pi_1(M)$ such that*

$$f_\gamma \circ \tilde{q}_i(\mathrm{int}(F)) = f_{\gamma'} \circ \tilde{q}_j(\mathrm{int}(F')).$$

Definition 1.4.7. The *lifted triangulation* $\widetilde{\mathcal{T}}$ of $\mathcal{T}$ is the data of:

- a collection
$$\{T_{i,\gamma} : i = 1, \dots, n,\ \gamma \in \pi_1(M)\}$$
of 3-simplices, where $T_{i,\gamma}$ is a copy of T_i for every $\gamma \in \pi_1(M)$ and for every $i = 1, \dots, n$;
- a partition of the faces of the tetrahedra into pairs, equipped with simplicial isomorphisms defined as follows: whenever a face F of T_i is paired to a face F' of $T_{i'}$ via $\sigma_{F,F'}$ in $\mathcal{T}$, the corresponding faces F_γ of $T_{i,\gamma}$ and $F'_{\gamma'}$ of $T_{i',\gamma'}$ are paired via $\sigma_{F,F'}$, where γ' is the unique element provided by Lemma 1.4.6.

Given a lifted triangulation $\widetilde{\mathcal{T}}$ of $\mathcal{T}$, we denote by $|\widetilde{\mathcal{T}}|$ the topological space obtained by gluing the tetrahedra $T_{i,\gamma}$ along their faces via the prescribed isomorphisms. As the gluing maps of $\widetilde{\mathcal{T}}$ are exactly the simplicial isomorphisms of $\mathcal{T}$, the lifted triangulation is naturally oriented. Consequently, the same conclusions of Lemma 1.4.2 apply to $\widetilde{\mathcal{T}}$: the projection is injective on the interior of every abstract edge, and the geometric realization $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$ is an oriented manifold.

Notice that the union
$$\bigsqcup_{\substack{i \in \{1,\dots,n\} \\ \gamma \in \pi_1(M)}} T_{i,\gamma}$$
of the tetrahedra of $\widetilde{\mathcal{T}}$ is naturally endowed with an action of $\pi_1(M)$, obtained by left-multiplication on the second index. More precisely, given an element $\eta \in \pi_1(M)$ and a point x in some $T_{i,\gamma}$, its image $\eta \cdot x$ is the copy of x in $T_{i,\eta\gamma}$. This action is compatible with the simplicial isomorphisms between the faces. Indeed, suppose that a face F_γ of $T_{i,\gamma}$ is paired with the face $F'_{\gamma'}$ of $T_{j,\gamma'}$. By definition, this happens precisely when

- the faces F_γ and $F'_{\gamma'}$ correspond to the paired faces F of T_i and F' of T_j respectively;
- the following equality holds
$$f_\gamma \circ \tilde{q}_i(\mathrm{int}(F)) = f_{\gamma'} \circ \tilde{q}_j(\mathrm{int}(F')). \tag{1.9}$$

Applying the deck transformation f_η to both sides of (1.9) yields $f_{\eta\gamma} \circ \tilde{q}_i = f_{\eta\gamma'} \circ \tilde{q}_j$. Consequently, the face $F_{\eta\gamma}$ is paired to $F'_{\eta\gamma'}$.

As a result, the action of $\pi_1(M)$ descends to a well-defined action by homeomorphisms on the quotient space $|\widetilde{\mathcal{T}}|$. We remark that this action preserves the cellular structure

of $|\widetilde{\mathcal{T}}|$, and hence restricts to an action on $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$. It is easy to see that this restricted action is free and properly discontinuous, yielding a regular covering

$$|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\} \to {}^{|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}}\Big/_{\pi_1(M)}.$$

Moreover, the quotient space is naturally identified with $|\mathcal{T}| \setminus \{\text{vertices}\} \cong M$. The previous arguments imply the following result.

Proposition 1.4.8. *The universal cover $\widetilde{M}$ is homeomorphic to $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$.*

We say that $\widetilde{\mathcal{T}}$ is an *ideal triangulation* of $\widetilde{M}$, and that $\mathcal{T}$ lifts to $\widetilde{\mathcal{T}}$.

Remark 1.4.9. The quotient

$$|\widetilde{\mathcal{T}}|\Big/_{\pi_1(M)}$$

is naturally homeomorphic to $|\mathcal{T}|$. The action of $\pi_1(M)$ on the entire space $|\widetilde{\mathcal{T}}|$ is neither free nor properly discontinuous. In fact, as we will mention later, the non-trivial stabilizer of a vertex $\tilde{v}$ of $|\widetilde{\mathcal{T}}|$ is obtained as follows: the projection v of $\tilde{v}$ in $|\mathcal{T}|$ corresponds to a cusp of M, and to a torus L_v. The stabilizer $\mathrm{Stab}(\tilde{v})$ is naturally isomorphic to $\pi_1(L_v) \cong \mathbb{Z} \times \mathbb{Z}$.

1.4.1 Complex moduli of tetrahedra

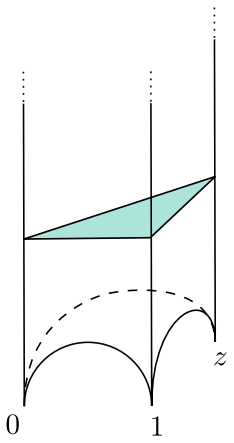
Let T be an ideal tetrahedron in hyperbolic space $\mathbb{H}^3$. By adopting the upper half-space model, the ideal vertices of T can be identified with four distinct points x_1, x_2, x_3, x_4 on the boundary at infinity $\widehat{\mathbb{C}}$. We endow T with the standard orientation of $\mathbb{H}^3$. In what follows, we will assume that the ordering (x_1, x_2, x_3, x_4) induces the *negative* orientation on T.

Remark 1.4.10. The group of orientation-preserving isometries of the ideal tetrahedron T acts transitively on its ideal vertices. Moreover, for every edge e of T there exists $\varphi \in \mathrm{Isom}^+(T)$ such that $\varphi(e)$ is the edge opposite to e. Indeed, the rotation of angle π around the unique geodesic orthogonal to two opposite edges x_ix_j and x_kx_l has the following effects:

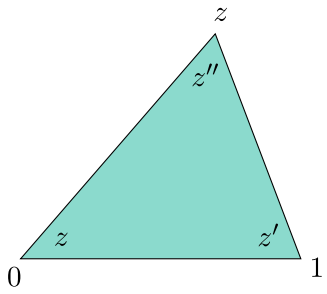
- it preserves the edges x_ix_j and x_kx_l as sets, swapping their endpoints ($x_i \leftrightarrow x_j$ and $x_k \leftrightarrow x_l$);
- it swaps the edge x_ix_k with its opposite x_jx_l;
- it swaps the edge x_ix_l with its opposite x_jx_k.

The group of orientation-preserving isometries of $\mathbb{H}^3$, which is isomorphic to $\mathrm{PSL}(2,\mathbb{C})$ by Proposition 1.2.15, acts freely and transitively on ordered triples of distinct points (see Corollary 1.1.7 and Corollary 1.1.10). As a consequence, there exists a unique element $[A] \in \mathrm{PSL}(2,\mathbb{C})$ such that the induced orientation-preserving isometry φ_A satisfies

$$\varphi_A(x_1) = 0, \quad \varphi_A(x_2) = 1, \quad \varphi_A(x_3) = \infty.$$



(a) The ideal tetrahedron



(b) Triangular section

Figure 1.2: On the left, an ideal tetrahedron in the half-space model of $\mathbb{H}^3$, with vertices at 0, 1, ∞, and $z \in \mathbb{C}$. On the right, the triangular section highlighted within the tetrahedron on the left, along with the complex angles associated with each vertex, see Remark 1.4.13 for the notation.

The remaining vertex x_4 is mapped by φ_A to a certain complex number $z \in \mathbb{C}$. Since we have assumed that the ordering (x_1, x_2, x_3, x_4) of the ideal vertices induces the negative orientation on T, z has strictly positive imaginary part. We call T' the transformed tetrahedron with vertices $0, 1, \infty$ and z. Every small enough horosphere centered at ∞ intersects T' in a Euclidean triangle (see Figure 1.2a), which we can represent in the complex plane $\mathbb{C}$ as in Figure 1.2b.

Definition 1.4.11. Let $\Delta \subset \mathbb{C}$ be a Euclidean triangle with vertices w_1, w_2, w_3 ordered counterclockwise. The *complex angles* z_i of Δ at the vertices w_i for $i = 1, 2, 3$ are defined as:

$$z_1 = \frac{w_3 - w_1}{w_2 - w_1}, \quad z_2 = \frac{w_1 - w_2}{w_3 - w_2}, \quad z_3 = \frac{w_2 - w_3}{w_1 - w_3}.$$

Notice that $z_1 z_2 z_3 = -1$.

In the specific case depicted in Figure 1.2b, where the vertices of the cross-section can be placed at $0, 1$, and z, the complex angles are precisely:

$$z_1 = z, \quad z_2 = \frac{1}{1 - z}, \quad z_3 = 1 - \frac{1}{z}. \tag{1.10}$$

Remark 1.4.12. Let Δ be a triangle in $\mathbb{C}$ with vertices w_1, w_2, w_3 as in Definition 1.4.11. Then the map

$$x \mapsto \frac{x - w_1}{w_2 - w_1}$$

is an orientation-preserving similarity of $\mathbb{C}$ mapping w_1 and w_2 to 0 and 1 respectively. The last vertex w_3 is mapped to the complex angle of Δ at w_1. As a consequence, if two Euclidean triangles in $\mathbb{C}$ have the same complex angles then they are (orientation-preservingly) similar. Conversely, every orientation-preserving similarity of $\mathbb{C}$ is of the form $x \mapsto ax+b$ with $a \in \mathbb{C}^*$, $b \in \mathbb{C}$, and thus preserves the complex angles by definition.

Remark 1.4.13 (Notation)**.** From now on, given a non-real complex number z, we will set

$$z' \coloneqq \frac{1}{1-z}, \quad z'' \coloneqq 1 - \frac{1}{z}.$$

Now we want to assign a complex number to each edge of T. We start with the edges incident to x_3. Recall that the orientation-preserving isometry φ_A maps x_3 to ∞, while the images of the other vertices form a Euclidean triangle $\Delta \subset \mathbb{C}$ with vertices $0, 1, z$, as in Figure 1.2b. Every edge of T incident to x_3 naturally corresponds to a corner of Δ, and we assign to it the corresponding complex angle, which is either z, z' or z''. We extend this assignment to the remaining three edges of T by declaring that opposite edges receive the same complex parameter.

We must verify that this assignment is well-defined, meaning that it does not depend on the choice of the vertices mapped to $0, 1$, and ∞ (provided the ordering preserves the orientation). We proceed as follows:

- first, recall from Remark 1.4.10 that $\mathrm{Isom}^+(T)$ contains isometries that swap every given edge with its opposite. So, for instance, there exists a symmetry $\psi \in \mathrm{Isom}^+(T)$ such that $\psi(x_3) = x_1, \psi(x_1) = x_3, \psi(x_2) = x_4$, and $\psi(x_4) = x_2$. The map $\varphi_A \circ \psi$ is an orientation-preserving isometry of $\mathbb{H}^3$ mapping x_1 to ∞, x_3 to 0 and x_4 to 1. Because our rule assigns the same complex number to opposite edges, this alternative map yields the exact same assignment of complex parameters. We can repeat this argument using the other double transpositions to freely choose any of the four vertices as the one mapped to ∞;

- lastly, we must consider what happens if we change the cyclic ordering of the remaining three vertices. For instance, suppose we choose the ordering (x_2, x_4, x_3, x_1) so that x_3 still maps to ∞, but $x_2 \mapsto 0$ and $x_4 \mapsto 1$. This configuration is achieved by post-composing φ_A with the orientation-preserving similarity

$$f(w) = \frac{1-w}{1-z}.$$

  Indeed, it is readily checked that $f \circ \varphi_A$ maps x_2, x_4, and x_3 to $0, 1$, and ∞ respectively, while x_1 is mapped to $f(0) = \frac{1}{1-z} = z'$. Now when truncating with a small enough horosphere, we get a Euclidean triangle Δ' with vertices $0, 1, z'$. As a result, the edge x_2x_3 receives the complex angle of Δ' at 0, which is exactly z'. This perfectly coincides with the parameter assigned to x_2x_3 in the original configuration, where it corresponded to the corner of Δ at 1. A similar check holds for the other edges, confirming that the assignment is independent of the chosen cyclic ordering.

With this procedure, to each edge of T is assigned a complex number, which is either z, z' or z'', called the *complex modulus* of T along the edge (see Figure 1.3).

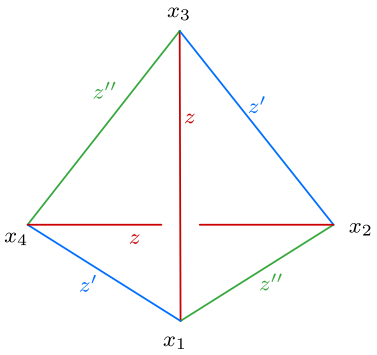


Figure 1.3: An ideal tetrahedron T with its complex moduli z, z' and z''. Pairs of opposite edges are drawn with the same colour, as their complex moduli coincide.

Proposition 1.4.14. *Consider two ideal tetrahedra T_1 and T_2 in $\mathbb{H}^3$ with ideal vertices x_1, x_2, x_3, x_4 and y_1, y_2, y_3, y_4 respectively, ordered such that they induce the negative orientation. The following facts are equivalent:*

1. *there exists an orientation-preserving isometry $f \in \mathrm{Isom}^+(\mathbb{H}^3)$ such that $f(x_i) = y_i$ for each $i = 1, \dots, 4$;*
2. *there exist two indices $i \neq j$ such that the complex modulus of the edge $x_i x_j$ in T_1 coincides with the complex modulus of $y_i y_j$ in T_2;*
3. *for every pair of indices $i \neq j$, the complex modulus of the edge $x_i x_j$ in T_1 coincides with the complex modulus of $y_i y_j$ in T_2;*

Moreover, whenever such an isometry f exists, it is unique.

Proof. Let $z_1, z_2 \in \mathbb{C}$ be the two complex numbers with $\mathrm{Im}(z_i) > 0$ such that:

- there exists an orientation-preserving isometry A_1 mapping x_1, x_2, x_3 and x_4 to $0, 1, \infty$ and z_1 respectively;
- there exists an orientation-preserving isometry A_2 mapping y_1, y_2, y_3 and y_4 to $0, 1, \infty$ and z_2 respectively.

Note that both these isometries are unique. By construction, the complex modulus associated to the edge of T_1 connecting x_1 and x_3 is z_1, and similarly z_2 is the complex modulus of the edge connecting y_1 and y_3 in T_2. The remaining edges of T_1 and T_2 are labeled with either z_i, z_i' or z_i'' as in Figure 1.4. From this explicit labeling, it follows immediately that (2) and (3) are equivalent.

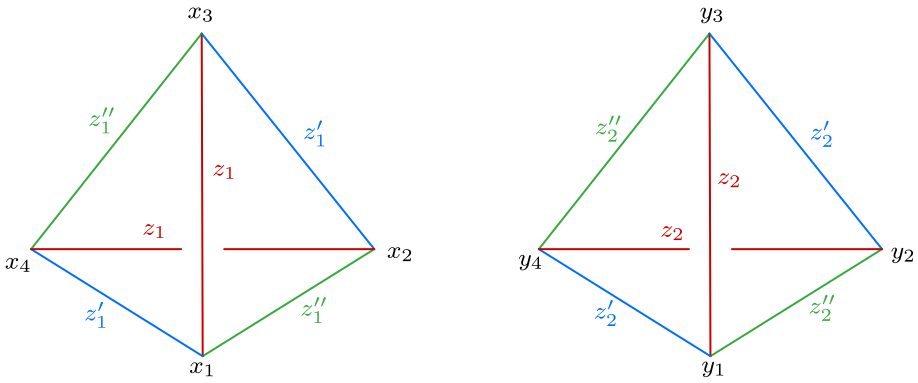


Figure 1.4: The two oriented, ideal tetrahedra T_1 (on the left) and T_2 (on the right) in $\mathbb{H}^3$. Each edge is labeled with its complex modulus.

If there exists $f \in \mathrm{Isom}^+(\mathbb{H}^3)$ mapping x_i to y_i for every $i = 1, 2, 3, 4$, then the map $A_2 \circ f \circ A_1^{-1}$ maps $0, 1, \infty, z_1$ to $0, 1, \infty, z_2$. Since the unique orientation-preserving isometry fixing $0, 1$ and ∞ is the identity, we must have $z_1 = z_2$ and so (1) implies (2).

Finally, we show that (3) implies (1). If (3) holds, then $z_1 = z_2$, and the orientation-preserving isometry $f = A_2^{-1} \circ A_1$ satisfies the requirements of (1).

The uniqueness of f follows immediately from the fact that the action of $\mathrm{Isom}^+(\mathbb{H}^3)$ on triples of distinct points of $\partial\mathbb{H}^3$ is free. □

Remark 1.4.15. The complex moduli defined above can be explicitly computed using the complex *cross-ratio* of the ideal vertices of T. Recall that T is endowed with the standard orientation of $\mathbb{H}^3$, and its ideal vertices x_1, x_2, x_3, x_4 are ordered such that they induce the negative orientation. For simplicity, we assume moreover that none of the x_i is the point ∞. The unique orientation-preserving isometry mapping x_1, x_2, x_3 to $0, 1, \infty$ respectively is the Möbius transformation

$$f(z) = \frac{(x_2 - x_3)(z - x_1)}{(x_2 - x_1)(z - x_3)}. \tag{1.11}$$

By the previous construction, the complex modulus assigned to the edge connecting the ideal vertices x_1 and x_3 is precisely $f(x_4)$, which is the cross-ratio

$$\frac{(x_2 - x_3)(x_4 - x_1)}{(x_2 - x_1)(x_4 - x_3)}.$$

This procedure can be applied to any edge: let e be the directed edge from x_i to x_k, and let x_j and x_l be the other two vertices such that the permutation (i, j, k, l) of $(1, 2, 3, 4)$ is even. Then the complex modulus assigned to e is exactly the cross-ratio

$$z_{ik}(T) := \frac{(x_j - x_k)(x_l - x_i)}{(x_j - x_i)(x_l - x_k)}. \tag{1.12}$$

By the symmetries of the cross-ratio, the expression on the right-hand side of (1.12) remains unchanged if we reverse the direction of the edge (which requires swapping j and l to maintain an even permutation) or if we compute it for the opposite edge (x_j, x_l).

Moreover, notice that the quantity in (1.12) is well-defined by taking limits if one of the vertices is ∞. For instance, if $x_l = \infty$, the terms involving x_l cancel out in the limit, yielding

$$z_{ik}(T) = \frac{x_j - x_k}{x_j - x_i}.$$

Motivated by the previous remark, we will sometimes use the notation $z_{ik}(T)$ to denote the complex modulus assigned to the edge $x_i x_k$ of T. Note that $z_{ik}(T) = z_{ki}(T)$ and $z_{ik}(T) = z_{jl}(T)$ where $x_j x_l$ is the opposite edge.

The knowledge of the complex moduli of an ideal tetrahedron T in $\mathbb{H}^3$, together with three of its vertices, uniquely determines the fourth.

Lemma 1.4.16. *Let $z \in \mathbb{C}$ be a complex number with $\mathrm{Im}(z) > 0$, and let*

$$z' = \frac{1}{1-z}, \quad z'' = 1 - \frac{1}{z}.$$

Let F be an ideal triangle in $\mathbb{H}^3$ with vertices x_1, x_2, x_3. Then there exists a unique point $x_4 \in \widehat{\mathbb{C}}$ such that the ideal tetrahedron T with ordered vertices (x_1, x_2, x_3, x_4) has the following complex moduli:

$$z_{13}(T) = z_{31}(T) = z_{24}(T) = z_{42}(T) = z, \tag{1.13}$$
$$z_{23}(T) = z_{32}(T) = z_{14}(T) = z_{41}(T) = z', \tag{1.14}$$
$$z_{43}(T) = z_{34}(T) = z_{12}(T) = z_{21}(T) = z''. \tag{1.15}$$

Proof. Assume for simplicity that $x_1, x_2, x_3 \in \mathbb{C}$. As observed in Remark 1.4.15, the map

$$f(x) = \frac{(x_2 - x_3)(x - x_1)}{(x_2 - x_1)(x - x_3)}$$

is the unique orientation-preserving Möbius transformation mapping x_1, x_2, x_3 to $0, 1, \infty$ respectively. In particular, f is a biholomorphism $\widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$. Therefore, for any given $z \in \mathbb{C}$, there exists a unique point $x_4 \in \widehat{\mathbb{C}}$ such that $f(x_4) = z$. The ideal tetrahedron T with vertices x_1, x_2, x_3, x_4 satisfies the required properties: indeed, by construction, z coincides with the complex modulus assigned to the edge connecting x_1 and x_3, and the complex moduli assigned to all other edges automatically match either z, z' or z'' from the well-known symmetries of the cross-ratio. □

1.4.2 Consistency equations

Let M be an oriented, complete, hyperbolic 3-manifold with finite volume, and let $\mathcal{T}$ be an ideal triangulation of M. Suppose that $\mathcal{T}$ consists of n tetrahedra $T_1, \dots, T_n$. We realize $T_1, \dots, T_n$ as hyperbolic ideal tetrahedra in $\mathbb{H}^3$, and we pair their faces by

orientation-reversing isometries. This way, we get a well-defined hyperbolic structure on M minus the 1-skeleton of the triangulation, as well as a decoration of the edges of the tetrahedra of $\mathcal{T}$ with complex numbers. The latter is obtained by assigning to the edges of T_i the complex moduli of the chosen realization of T_i in $\mathbb{H}^3$. For every T_i we randomly pick an edge, and we call z_i the complex modulus assigned to said edge. The other edges are labeled with either $z_i, 1/(1-z_i)$ or $1-1/z_i$. We recall that these complex numbers all have strictly positive imaginary parts, and only depend on the (orientation-preserving) isometry classes of the geometric realizations of $T_1, \dots, T_n$ in $\mathbb{H}^3$.

Extension to the 1-skeleton We now want to understand under which conditions this hyperbolic structure extends smoothly to the whole manifold M.

Consider an edge e of $|\mathcal{T}| \setminus \{\text{vertices}\} \cong M$, and let $T(e)_1, \dots, T(e)_{r_e}$ be the ideal tetrahedra incident to e after the gluing. Notice that the same tetrahedron can appear multiple times. For every $i = 1, \dots, r_e$, we denote by $w(e,i)$ the complex modulus of the edge corresponding to e in $T(e)_i$. The hyperbolic structure extends to e if and only if $T(e)_1, \dots, T(e)_{r_e}$ glue properly around e. By this, we mean that the tetrahedra close up around the edge without leaving a gap, overlapping, or inducing a translational shift along the edge itself.

This happens if and only if the following conditions hold:

$$\sum_{i=1}^{r_e} \arg(w(e,i)) = 2\pi, \qquad \prod_{i=1}^{r_e} w(e,i) = 1, \tag{1.16}$$

where we choose the argument of each $w(e,i)$ to lie in $(0,\pi)$.

Remark 1.4.17. Up to orientation-preserving isometries of $\mathbb{H}^3$, we can assume that the ideal tetrahedra $T(e)_1, \dots, T(e)_{r_e}$ are realized as geodesic, vertical ideal tetrahedra in $\mathbb{H}^3$ sharing the line from 0 to ∞ as an edge — corresponding to e (see Figure 1.5). Any small enough horosphere centered at ∞ intersects the ideal tetrahedra in r_e Euclidean triangles $\Delta_1, \dots, \Delta_{r_e}$, and we know that every complex modulus $w(e,i)$ coincides with the complex angle of Δ_i at the common vertex (see Figure 1.6).

A priori, the second equation in (1.16) does not imply the first, but it only guarantees that $\sum_{i=1}^{r_e} \arg(w(e,i))$ is an integer multiple of 2π. We now show that, in this specific case, the algebraic product condition alone is actually sufficient.

Since M is homeomorphic to the interior of a compact manifold with toric boundary, its Euler characteristic vanishes. As a consequence, the number of edges of $|\mathcal{T}| \setminus \{\text{vertices}\}$ equals the number n of ideal tetrahedra. Indeed, denote by c_k the number of k-cells of $|\mathcal{T}| \setminus \{\text{vertices}\}$ for $k = 1, 2, 3$. Since the faces of the tetrahedra of $\mathcal{T}$ are glued in pairs, we have $c_2 = 2c_3$, therefore

$$\chi(M) = c_1 - c_2 + c_3 = c_1 - c_3,$$

and the vanishing of the Euler characteristic implies $c_1 = c_3 = n$. Given that each parameter z_i has $\mathrm{Im}(z_i) > 0$, for every edge e we find a *positive* integer $n(e) \in \mathbb{N}$ such

that

$$\sum_{i=1}^{r_e} \arg(w(e,i)) = 2\pi \cdot n(e), \tag{1.17}$$

where $w(e,1), \dots, w(e,r_e)$ are the complex moduli around e.

Now we compute the sum of the internal angles of all the Euclidean triangles in the horospherical sections in two ways.

(a) By tetrahedra: each of the n ideal tetrahedra contributes with four horospherical triangles. In total, we need to sum the internal angles of $4n$ Euclidean triangles, and we get

$$4n \cdot \pi. \tag{1.18}$$

(b) By edges: each of the n edges connects two ideal vertices, so every angle in the sum (1.17) appears exactly twice in the triangulated horospherical sections, once at each endpoint. Summing over all edges yields

$$2 \cdot 2\pi \cdot \sum_{e} n(e). \tag{1.19}$$

By equating (1.18) and (1.19) we conclude that $n(e) = 1$ for every edge e.

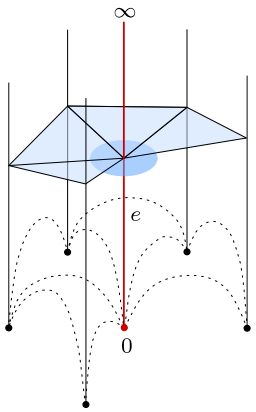


Figure 1.5: The star around the realization of the edge e in $\mathbb{H}^3$.

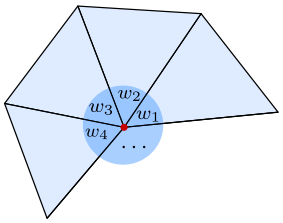


Figure 1.6: The horospherical section. Here we have set $w_i := w(e,i)$ to ease notation.

Summing up, the hyperbolic structure extends to the whole manifold M if and only if for every edge e we have

$$\prod_{i=1}^{r_e} w(e,i) = 1. \tag{1.20}$$

Equation (1.20) is called the *consistency equation* around the edge e.

Remark 1.4.18. Since the number of edges equals the number of ideal tetrahedra in $|\mathcal{T}| \setminus \{\text{vertices}\}$, the consistency equations yield a system of n equations in n complex variables.

Definition 1.4.19. If the consistency equations are satisfied around every edge, we say that $\mathcal{T}$ is a *geometric ideal triangulation* for M.

Example 1.4.20. Let M_8 be the figure-eight knot complement, and consider the ideal triangulation $\mathcal{T}_8$ of Example 1.4.5 depicted in Figure 1.1b. We call T_1 and T_2 the tetrahedra on the left and on the right of Figure 1.1b respectively. We assign the parameters

$$z_1 = z,\ z_2 = \frac{1}{1-z},\ z_3 = 1 - \frac{1}{z} \quad \text{and} \quad w_1 = w,\ w_2 = \frac{1}{1-w},\ w_3 = 1 - \frac{1}{w} \tag{1.21}$$

with strictly positive imaginary parts to the edges of T_1 and T_2 respectively. The assignment is carried out so that the horospherical sections around the two edges, after the gluing, are decorated as in Figure 1.7. The consistency equations (1.20) are therefore

$$z_2^2 z_3 w_2^2 w_3 = 1, \qquad z_1^2 z_3 w_1^2 w_3 = 1,$$

and since $z_1 \cdot z_2 \cdot z_3 = w_1 \cdot w_2 \cdot w_3 = -1$, using the formulas (1.21) for z_i and w_i, they are equivalent to

$$zw(1-z)(1-w) = 1. \tag{1.22}$$

Every solution (z, w) to this equation with strictly positive imaginary parts yields a valid, though possibly incomplete, hyperbolic structure on the manifold M_8.

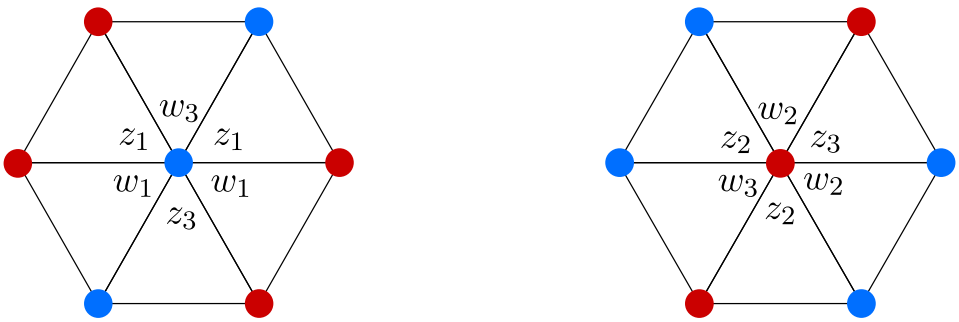


Figure 1.7: The local picture of the 2-dimensional triangulation around each of the two edges of the figure-eight knot complement. Both the blue edge (on the right) and the red edge (on the left) have valence six, and each internal angle is decorated by the corresponding complex parameter.

Remark 1.4.21. While it is a well-known fact that every complete non-compact hyperbolic 3-manifold with finite volume admits an ideal triangulation, it is in general not possible to realize this triangulation as a geometric one. Furthermore, the very existence

of a geometric ideal triangulation is an open problem. By the canonical Epstein-Penner decomposition, every complete non-compact hyperbolic 3-manifold M with finite volume is partitioned into convex ideal polyhedra identified isometrically along their faces (see [EP88]). Each of these polyhedra can be further partitioned into ideal tetrahedra, without adding vertices, but in order to match the induced triangulations on the shared faces we might need to add *flat* tetrahedra (that have real complex moduli). Triangulations of this kind are called *partially flat*, and every hyperbolic 3-manifold M as above admits a partially flat ideal triangulation. The consistency equations make sense also in this case, and in fact Petronio and Weeks [PW00] showed that such partially flat ideal triangulations still determine the hyperbolic structures of M.

Explicit construction of the developing map Every solution $(z_1, \dots, z_n)$ to the consistency equations yields a — possibly incomplete — hyperbolic structure on M. We now show how to explicitly reconstruct the corresponding developing pair (D, ρ).

Recall that the ideal triangulation $\mathcal{T}$ of M lifts to an ideal triangulation $\widetilde{\mathcal{T}}$ of its universal cover $\widetilde{M}$, and that by definition $\widetilde{M}$ is homeomorphic to $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$. We define

$$D\colon |\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\} \to \mathbb{H}^3$$

by prescribing the image of every ideal tetrahedron. Since a geodesic, ideal tetrahedron in $\mathbb{H}^3$ is uniquely determined by its four ideal vertices in $\partial\mathbb{H}^3 = \hat{\mathbb{C}}$, we actually define D as a map

$$|\widetilde{\mathcal{T}}| \to \overline{\mathbb{H}}^3,$$

where the vertices of $|\widetilde{\mathcal{T}}|$ are mapped to ideal points in $\partial\mathbb{H}^3$. Every time we fix the image in $\partial\mathbb{H}^3$ of the vertices of a lifted tetrahedron $\widetilde{T}$, we continuously extend D to $\widetilde{T} \setminus \{\text{vertices}\}$ mapping every cell to the corresponding unique totally geodesic ideal simplex. We moreover require this extension to be coherent across shared faces.

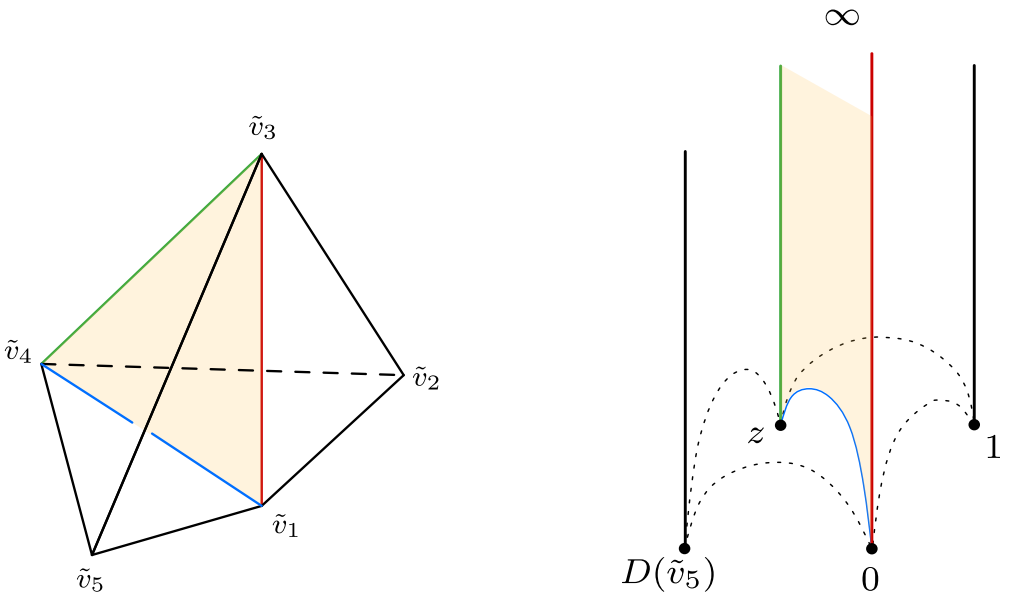
Let T_0 be an ideal tetrahedron and fix a lift $\widetilde{T}_0$ with ideal vertices $\tilde{v}_1, \tilde{v}_2, \tilde{v}_3, \tilde{v}_4$. Let z be the complex label assigned to the edge joining the projections of $\tilde{v}_1$ and $\tilde{v}_3$ in T_0. We start by defining D on the ideal vertices of $\widetilde{T}_0$ as follows:

$$D(\tilde{v}_1) = 0, \quad D(\tilde{v}_2) = 1, \quad D(\tilde{v}_3) = \infty, \quad D(\tilde{v}_4) = z.$$

By construction, the complex moduli of $D(\widetilde{T}_0)$ coincide with the labels assigned to the edges of T_0.

Next, we proceed to an adjacent lifted tetrahedron $\widetilde{T}_1$. Suppose $\widetilde{T}_1$ shares the face spanned by $\tilde{v}_1, \tilde{v}_3, \tilde{v}_4$ with $\widetilde{T}_0$, and has a fourth vertex $\tilde{v}_5$, as in Figure 1.8a. The images $D(\tilde{v}_1), D(\tilde{v}_3)$, and $D(\tilde{v}_4)$ are already determined. By Lemma 1.4.16, there exists a unique point $D(\tilde{v}_5) \in \widehat{\mathbb{C}}$ such that the complex moduli of $D(\widetilde{T}_1)$ consistently match the complex parameters assigned to the edges of T_1 (see Figure 1.8b).

We can propagate this construction one tetrahedron at a time across the whole universal cover. Since $\widetilde{M}$ is simply connected and the complex moduli satisfy the consistency equations (1.20), this piecewise construction is independent of the chosen path, and it closes up around the edges of the 1-skeleton. We get a developing map $D\colon \widetilde{M} \to \mathbb{H}^3$.



(a) The adjacent tetrahedra $\widetilde{T}_0$ and $\widetilde{T}_1$ in $|\widetilde{\mathcal{T}}|$.

(b) The image of $\widetilde{T}_0$ and $\widetilde{T}_1$ i $\mathbb{H}^3$.

Figure 1.8: A step in the definition of the developing map D given a solution to the consistency equations.

We now show how to construct the holonomy representation $\rho\colon \pi_1(M) \to \mathrm{PSL}(2,\mathbb{C})$. Recall that we want the pair (D,ρ) to satisfy

$$D(\gamma \cdot \tilde{x}) = \rho(\gamma) D(\tilde{x}) \tag{1.23}$$

for every $\tilde{x} \in |\widetilde{\mathcal{T}}|$ and for every $\gamma \in \pi_1(M)$.

Given an element $\gamma \in \pi_1(M)$, the tetrahedron $\gamma \cdot \widetilde{T}_0$ is another lift of T_0. By construction, the geodesic ideal tetrahedra $D(\widetilde{T}_0)$ and $D(\gamma \cdot \widetilde{T}_0)$ have the same complex moduli, and therefore there exists a unique orientation-preserving isometry $\rho(\gamma)$ mapping $D(\tilde{v}_i)$ to $D(\gamma \cdot \tilde{v}_i)$ for each $i = 1, \dots, 4$. In particular, the vertices of $\widetilde{T}_0$ satisfy equation (1.23). We now show that the equation is satisfied for every vertex of the lifted triangulation. By repeating the same argument for the tetrahedron $\widetilde{T}_1$, we find another isometry $\rho'(\gamma)$ mapping the vertices of $D(\widetilde{T}_1)$ to those of $D(\gamma \cdot \widetilde{T}_1)$. Since the two isometries coincide on the three vertices of the shared face and every isometry of $\mathbb{H}^3$ is uniquely determined by its action on three ideal points, we must have $\rho(\gamma) = \rho'(\gamma)$.

By the connectedness of $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$, we can propagate this argument across the entire triangulation, concluding that the isometry $\rho(\gamma)$ satisfies equation (1.23) for every vertex of $|\widetilde{\mathcal{T}}|$. We furthermore notice that the extension of D to every tetrahedron can be chosen so that the equation (1.23) holds for every $\tilde{x} \in |\widetilde{\mathcal{T}}|$.

Finally, making a different initial choice for the images of the vertices of $\widetilde{T}_0$ in $\widehat{\mathbb{C}}$ modifies the developing map by a global isometry of $\mathbb{H}^3$, which precisely corresponds to conjugating the representation ρ.

1.4.3 Completeness equations

By the analysis of the previous section, every solution $(z_1, \dots, z_n)$ to the consistency equations yields a (possibly incomplete) hyperbolic structure on the 3-manifold M. We want to understand whether it is complete.

Suppose that M has $k \geq 1$ cusps, and let $L_1, \dots, L_k$ denote the boundaries of the truncated cusps. Recall that each of the L_i is a torus. We now show that the hyperbolic structure on M induced by $(z_1, \dots, z_n)$ endows every L_i with a similarity structure. We know that every L_i corresponds to a vertex v_i of $|\mathcal{T}|$, and that it admits a triangulation $\mathcal{T}_{v_i}$ induced by $\mathcal{T}$. Every triangle

$$\Delta_1^{(v_i)}, \dots, \Delta_{k_{v_i}}^{(v_i)}$$

of $\mathcal{T}_{v_i}$ is obtained as the link of a vertex in $\mathcal{T}$, hence every corner of $\Delta_i^{(v_i)}$ corresponds to an edge of $\mathcal{T}$. We decorate every corner of the triangles in $\mathcal{T}_{v_i}$ with the complex modulus of the corresponding edge of $\mathcal{T}$.

Since the gluing is performed along the edges, the labeling descends to the angles of $L_i \cong |\mathcal{T}_{v_i}|$. By construction, we can view every triangle of $|\mathcal{T}_{v_i}|$ as a Euclidean triangle arising as the horospherical section of an ideal tetrahedron in $\mathbb{H}^3$ (for example, as we did in Remark 1.4.17). Therefore, every triangle of L_i is naturally equipped with a similarity structure. The fact that $(z_1, \dots, z_n)$ satisfies the consistency equations implies that these similarity structures glue together to a global one on L_i.

For a proof of the following result, we refer to [Mar16] or [BP92].

Proposition 1.4.22. *The hyperbolic structure on M is complete if and only if the induced similarity structure on each L_i is Euclidean.*

We want to translate this condition into algebraic equations.

Given a torus L_i, we represent every element $\gamma \in \pi_1(L_i)$ as a closed curve in the 1-skeleton of L_i, and define

$$\mu_i(\gamma) = (-1)^{|\gamma|} \prod_{w \in Z_\gamma} w \in \mathbb{C}^* \tag{1.24}$$

where $|\gamma|$ is the number of edges of the curve, and Z_γ is the collection of complex angles that γ encounters on its left. We conclude by stating the following result, whose proof can be found in [Mar16]. Remark 1.4.24 contains an idea of the proof.

Proposition 1.4.23. *The element $\mu_i(\gamma)$ is well-defined for every $\gamma \in \pi_1(L_i)$, and $\mu_i \colon \pi_1(L_i) \to \mathbb{C}^*$ is a group homomorphism. Moreover, μ_i is trivial if and only if the similarity structure on L_i is a Euclidean structure.*

Remark 1.4.24. Let $\rho \colon \pi_1(M) \to \mathrm{PSL}(2, \mathbb{C})$ be the holonomy of the hyperbolic structure induced by the parameters $(z_1, \dots, z_n)$. Recall that every torus L_i is the boundary of a truncated cusp of M, and corresponds to a vertex v_i of $|\mathcal{T}|$. Since M is a finite-volume hyperbolic 3-manifold, the inclusion of L_i into M is π_1-injective (see [Mar16]),

meaning we can naturally identify $\pi_1(L_i)$ with a subgroup of $\pi_1(M)$ (once we choose a common basepoint in L_i). In the universal cover, the subgroup $\pi_1(L_i)$ acts by deck transformations that stabilize a lifted vertex $\tilde{v}_i$ of v_i in $|\widetilde{\mathcal{T}}|$ (where $\tilde{v}_i$ is uniquely determined by a choice of a lift of the basepoint).

By the equivariance of the developing pair (D, ρ), the subgroup $\rho(\pi_1(L_i))$ of $\rho(\pi_1(M))$ must fix the point $D(\tilde{v}_i) \in \partial\mathbb{H}^3 = \widehat{\mathbb{C}}$. Up to conjugation, we can assume that $D(\tilde{v}_i) = \infty$, so that for every $\gamma \in \pi_1(L_i)$ the orientation-preserving isometry $\rho(\gamma)$ is represented by an upper-triangular matrix

$$\begin{pmatrix} a_\gamma & b_\gamma \\ 0 & a_\gamma^{-1} \end{pmatrix} \in \mathrm{SL}(2, \mathbb{C}).$$

In particular, the restriction of $\rho(\gamma)$ to $\mathbb{C}$ acts as the similarity

$$z \mapsto a_\gamma^2 z + a_\gamma b_\gamma.$$

A crucial step in the proof of Proposition 1.4.23 is that $\mu_i(\gamma)$ coincides precisely with the derivative a_γ^2 of the restriction of $\rho(\gamma)$ to $\mathbb{C}$. To obtain a Euclidean structure on the smooth torus L_i, the action of $\rho(\pi_1(L_i))$ on $\mathbb{C}$ must consist of pure translations whose translation vectors generate a 2-dimensional lattice. The algebraic condition $\mu_i(\gamma) = 1$ detects exactly that $\rho(\gamma)$ is a translation. The fact that these translations actually generate a lattice is a consequence of our assumptions: since every parameter z_j has $\mathrm{Im}(z_j) > 0$, L_i is obtained by gluing together triangles with positive areas, thus the translation vectors cannot be linearly dependent.

By combining Proposition 1.4.22 and Proposition 1.4.23, we deduce that the hyperbolic structure on M is complete if and only if μ_i is trivial for every $i = 1, \dots, k$. Since L_i is a torus, its fundamental group $\pi_1(L_i)$ is isomorphic to $\mathbb{Z}^2$. We fix a basis for $\pi_1(L_i)$ consisting of two simple closed curves that intersect transversely in a single point, namely a meridian m_i and a longitude l_i.

Because μ_i is a group homomorphism, it is trivial on $\pi_1(L_i)$ if and only if it evaluates to 1 on its generators. Therefore, the completeness condition reduces to the following $2k$ algebraic constraints, known as the *completeness equations*:

$$\mu_i(m_i) = 1, \quad \mu_i(l_i) = 1 \quad \text{for all } i = 1, \dots, k. \tag{1.25}$$

We can summarize the above considerations with the following result.

Proposition 1.4.25. *If a set of complex parameters $(z_1, \dots, z_n)$ with strictly positive imaginary parts satisfies both the consistency and the completeness equations, then it induces a complete hyperbolic structure of finite volume on M.*

If M admits such a complete hyperbolic metric of finite volume, Mostow-Prasad rigidity guarantees that it is uniquely determined up to isometry. We now illustrate this algebraic machinery on a concrete example.

Example 1.4.26. In Example 1.4.20, we solved the consistency equations for the ideal triangulation $\mathcal{T}_8$ of the figure-8 knot complement M_8 depicted in Figure 1.1b. Since M_8

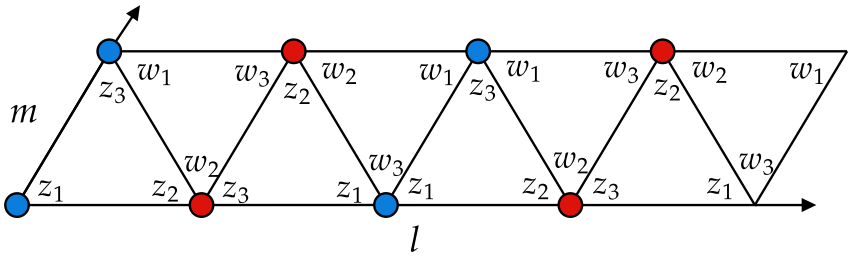


Figure 1.9: The induced triangulation on the boundary of the truncated cusp. Identifying the opposite sides of the parallelogram yields a torus equipped with a meridian m and a longitude l.

has a single cusp, we need to solve the completeness equation for a single boundary torus L. By fixing generators m and l of $\pi_1(L) \cong \mathbb{Z} \times \mathbb{Z}$ as in Figure 1.9, we get

$$\mu(m) = -z_1 z_3 w_1, \qquad \mu(l) = w_1^4 z_2^2 z_3^2 w_2^2 w_3^2,$$

and imposing $\mu = 1$ yields the equation

$$(1-z)w = 1, \quad w^2 = z^2. \tag{1.26}$$

The only pair (z, w) with strictly positive imaginary parts satisfying both the consistency equations (1.22) and the completeness equations (1.26) is given by

$$z = w = e^{\frac{i\pi}{3}},$$

which induces a complete hyperbolic structure on M_8.

1.5 Quaternions

In what follows, we will denote by $\mathcal{H}$ the quaternion algebra. We recall that an element $q \in \mathcal{H}$ is of the form

$$q = q_0 + q_1 i + q_2 j + q_3 k \tag{1.27}$$

where $i^2 = j^2 = k^2 = -1$, the coefficients q_l are real numbers, and

$$ij = k = -ji, \quad jk = i = -kj, \quad ki = j = -ik.$$

By linearly extending the previous relations, one can obtain the usual formulas for sum and product in $\mathcal{H}$. Explicitly, given two quaternions $p = p_0 + p_1 i + p_2 j + p_3 k$ and $q = q_0 + q_1 i + q_2 j + q_3 k$, their sum is

$$p + q = (p_0 + q_0) + (p_1 + q_1)i + (p_2 + q_2)j + (p_3 + q_3)k,$$

and their product is given by

$$\begin{aligned} pq = &(p_0q_0 - p_1q_1 - p_2q_2 - p_3q_3) \\ &+ (p_0q_1 + p_1q_0 + p_2q_3 - p_3q_2)i \\ &+ (p_0q_2 - p_1q_3 + p_2q_0 + p_3q_1)j \\ &+ (p_0q_3 + p_1q_2 - p_2q_1 + p_3q_0)k. \end{aligned} \tag{1.28}$$

Given a quaternion $q \in \mathcal{H}$ as in (1.27), we fix the following notation:

$$\begin{array}{rcll} \mathrm{Re}(q) & = & q_0 & \text{real part} \\ \mathrm{Im}(q) & = & q_1 i + q_2 j + q_3 k & \text{imaginary part} \\ \overline{q} & = & q_0 - q_1 i - q_2 j - q_3 k & \text{(quaternionic) conjugate} \\ |q| & = & (q_0^2 + q_1^2 + q_2^2 + q_3^2)^{1/2} & \text{(quaternionic) norm} \end{array}$$

There are natural inclusions $\mathbb{R} \subset \mathbb{C} \subset \mathcal{H}$. We use the symbol $\mathcal{H}$ instead of $\mathbb{H}$ because the latter is employed for hyperbolic spaces. Of course $q \in \mathbb{R}$ if and only if its imaginary part $\mathrm{Im}(q)$ vanishes. We say that a quaternion $q \in \mathcal{H}$ is *purely imaginary* if $\mathrm{Re}(q) = 0$.

We will often treat $\mathrm{Im}(q)$ as an element of $\mathbb{R}^3$, which we will still denote by $\mathrm{Im}(q)$. Under this identification, the product formula (1.28) yields the following useful relations for the real and imaginary parts of the product of two quaternions $p, q \in \mathcal{H}$:

$$\mathrm{Re}(pq) = \mathrm{Re}(p)\,\mathrm{Re}(q) - \langle \mathrm{Im}(p), \mathrm{Im}(q) \rangle, \tag{1.29}$$

$$\mathrm{Im}(pq) = \mathrm{Re}(p)\,\mathrm{Im}(q) + \mathrm{Re}(q)\,\mathrm{Im}(p) + \mathrm{Im}(p) \times \mathrm{Im}(q), \tag{1.30}$$

where $\langle \cdot, \cdot \rangle$ and $\times$ denote the standard Euclidean inner product and cross product in $\mathbb{R}^3$, respectively. Notice that (1.29) immediately yields $\mathrm{Re}(pq) = \mathrm{Re}(qp)$.

In this section, we review the action of the group of invertible quaternions $\mathcal{H}^*$ on $\mathcal{H}$ by conjugation. Most of the results can be found in [KR91].

1.5.1 Conjugate quaternions

We say that two quaternions $p, q \in \mathcal{H}$ are *conjugate* (or *similar*) if there exists an invertible quaternion $d \in \mathcal{H}^*$ such that $p = dqd^{-1}$. Since $\mathcal{H}$ is non-commutative, the conjugacy class of an element may be non-trivial, and its centralizer may be a proper subset of $\mathcal{H}$. In fact, the following results hold.

Theorem 1.5.1. *Two quaternions $p, q \in \mathcal{H}$ are conjugate if and only if $\mathrm{Re}(p) = \mathrm{Re}(q)$ and $|p| = |q|$.*

Moreover, if both p and q are not real, then the elements $d \in \mathcal{H}$ such that $pd = dq$ form a vector space of dimension two.

Proof. Let $p, q \in \mathcal{H}$. By definition, they are conjugate if and only if there exists a non-zero quaternion d such that

$$pd - dq = 0. \tag{1.31}$$

Viewing d, p and q as vectors in $\mathbb{R}^4$, equation (1.31) corresponds to a linear system with matrix

$$M_{p,q} = \begin{pmatrix} p_0 - q_0 & -(p_1 - q_1) & -(p_2 - q_2) & -(p_3 - q_3) \\ p_1 - q_1 & p_0 - q_0 & -(p_3 + q_3) & p_2 + q_2 \\ p_2 - q_2 & p_3 + q_3 & p_0 - q_0 & -(p_1 + q_1) \\ p_3 - q_3 & -(p_2 + q_2) & p_1 + q_1 & p_0 - q_0 \end{pmatrix}. \tag{1.32}$$

A non-trivial solution exists if and only if $\det(M_{p,q}) = 0$. A direct computation shows:

$$\det(M_{p,q}) = (p_0 - q_0)^4 + 2(p_0 - q_0)^2(|\operatorname{Im}(p)|^2 + |\operatorname{Im}(q)|^2) + (|\operatorname{Im}(p)|^2 - |\operatorname{Im}(q)|^2).$$

Since all terms are non-negative, the determinant vanishes if and only if $p_0 = q_0$ and $|\operatorname{Im}(p)| = |\operatorname{Im}(q)|$, which is equivalent to $\operatorname{Re}(p) = \operatorname{Re}(q)$ and $|p| = |q|$. Lastly, if $p, q \notin \mathbb{R}$ the space of all d such that $pd = dq$ has dimension 2 since it is the kernel of the singular non-trivial 4×4 skew-symmetric matrix $M_{p,q}$ (1.32). □

Corollary 1.5.2. *Every quaternion $q \in \mathcal{H}$ is similar to its quaternionic conjugate $\bar{q}$.*

Corollary 1.5.3. *Let $q \in \mathcal{H}$ be a quaternion. If $q \notin \mathbb{R}$, the conjugacy class of q, given by*

$$\{p \in \mathcal{H} : \operatorname{Re}(p) = \operatorname{Re}(q),\ |p| = |q|\},$$

is diffeomorphic to the 2-sphere S^2.

Proposition 1.5.4. *Two non-real quaternions p, q commute if and only if $\operatorname{Im}(p)$ and $\operatorname{Im}(q)$ are collinear, i.e., $\operatorname{Im}(p) = s \cdot \operatorname{Im}(q)$ for some non-zero $s \in \mathbb{R}$.*

Proof. Since the equality $\operatorname{Re}(pq) = \operatorname{Re}(qp)$ is always satisfied (see (1.29)), the two non-real quaternions p and q commute if and only if $\operatorname{Im}(pq) = \operatorname{Im}(qp)$. Using the formula (1.30) for the imaginary part of the product, we find that

$$\operatorname{Im}(pq) - \operatorname{Im}(qp) = \operatorname{Im}(p) \times \operatorname{Im}(q) - \operatorname{Im}(q) \times \operatorname{Im}(p) = 2(\operatorname{Im}(p) \times \operatorname{Im}(q)),$$

where the last equality follows from the fact that the cross-product is anti-commutative. The two imaginary parts therefore coincide if and only if the cross-product $\operatorname{Im}(p) \times \operatorname{Im}(q)$ vanishes, which—since both quaternions are non-real—happens if and only if $\operatorname{Im}(p)$ and $\operatorname{Im}(q)$ are collinear. □

Corollary 1.5.5. *A quaternion p lies in the center of $\mathcal{H}$ if and only if $p \in \mathbb{R}$.*

As we have seen in Theorem 1.5.1, for a fixed $d \in \mathcal{H}^*$, the conjugation map $q \mapsto dqd^{-1}$ preserves both the real part and the norm of q. Its action on the imaginary part $\operatorname{Im}(q)$ corresponds to a rotation in $\mathbb{R}^3 \cong \operatorname{Im}(\mathcal{H})$. This is formalized by the following result, whose proof can be found in [KR91].

Theorem 1.5.6. *Let $d \in \mathcal{H}^*$ be a non-real unit quaternion, written as $d = \cos\frac{\omega}{2} + \sin\frac{\omega}{2} I$ for some unitary imaginary quaternion I and some angle $\omega \in [0, 2\pi]$. Then the map $q \mapsto dqd^{-1}$ restricted to purely imaginary quaternions is a rotation of angle ω around the axis I.*

1.5.2 Orientation-preserving isometries of $\mathcal{H}$

As we have just seen in Theorem 1.5.1 and Theorem 1.5.6, the conjugation by a fixed non-real quaternion $d \in \mathcal{H}^*$ given by $q \mapsto dqd^{-1}$ preserves the real part and the norm of every quaternion q, while acting as a 3-dimensional rotation on $\mathrm{Im}(q)$. We can therefore view it as an orientation-preserving isometry of $\mathcal{H} \cong \mathbb{R}^4$, pointwise fixing the plane spanned by 1 and $\mathrm{Im}(d)$.

If we decouple the left and right factors, we famously get a broader class of transformations. Indeed, by identifying the unit quaternions with the 3-sphere S^3, we can consider the Lie group homomorphism

$$\begin{array}{ccc} S^3 \times S^3 & \longrightarrow & \mathrm{SO}(4) \\ (u,v) & \longmapsto & (q \mapsto uqv^{-1}) \end{array}. \tag{1.33}$$

It is a well-known fact that the homomorphism defined in (1.33) is a double covering (see for example [Mar16]); in particular

$$\mathrm{SO}(4) \cong S^3 \times S^3/_{\{\pm(1,1)\}}.$$

Thanks to this isomorphism, every orientation-preserving isometry f of $\mathcal{H}$ can be written in the form $f(q) = uqv^{-1}$, where u and v are unit quaternions that are uniquely determined up to a common sign. We now give a geometric description of these maps depending on the properties of u and v.

Left isoclinic rotations We start by considering the case where either u or v is real. For instance, consider the isometry $f\colon \mathcal{H} \to \mathcal{H}$ given by $f(q) = uq$, and write

$$u = e^{\theta I} = \cos\theta + \sin\theta\, I, \tag{1.34}$$

for a purely imaginary, unit quaternion I and an angle $\theta \in (0, \pi)$, which we can suppose upon replacing I by $-I$ if necessary. This exponential notation, completely analogous to Euler's formula for complex numbers, will prove convenient for algebraic manipulations later on.

Choosing a purely imaginary, unit quaternion I_2 that is orthogonal to I, and setting $I_3 = I \cdot I_2$, we get an orthonormal basis $\{1, I, I_2, I_3\}$ of $\mathcal{H}$. Notice that I, I_2, I_3 satisfy the same relations as the standard imaginary units i, j, k:

$$I \cdot I_2 = I_3, \quad I_2 \cdot I_3 = I, \quad I_3 \cdot I = I_2.$$

Denote by P_1, P_2 the planes spanned by $1, I$, and I_2, I_3 respectively, and notice that $\mathcal{H}$ decomposes as the orthogonal direct sum $P_1 \oplus P_2$. A direct computation shows that f leaves the two planes invariant and acts on either of them, with respect to the basis $(1, I)$ or (I_2, I_3), as the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

hence f acts as a rotation of angle θ on both planes. The isometry f is called a *left isoclinic* rotation. Notice moreover that f does not have any non-zero fixed points. For every $q \in \mathcal{H}$, the quaternions q and Iq are orthogonal, they have the same norm and they span a plane in $\mathcal{H}$. By distributing the product, we see that

$$f(q) = (\cos\theta + \sin\theta I)q = \cos\theta\, q + \sin\theta\, Iq,$$

which means that f rotates every non-zero point $q \in \mathcal{H}$ by an angle θ within the plane spanned by $\{q, Iq\}$.

Right isoclinic rotations We now consider an isometry of the form $f(q) = qv^{-1}$, and write $v = \cos\phi + \sin\phi J$ so that

$$v^{-1} = e^{-\phi J} = \cos\phi - \sin\phi J, \tag{1.35}$$

with J a purely imaginary, unit quaternion and $\phi \in (0, \pi)$. As in the previous case, we consider an orthonormal basis $\{1, J, J_2, J_3\}$ where J_2, J_3 are purely imaginary unit quaternions that, together with J, satisfy the same relations as the standard imaginary units: $J \cdot J_2 = J_3$, $J_2 \cdot J_3 = J$, and $J_3 \cdot J = J_2$. We again decompose the space into two orthogonal planes $Q_1 = \mathrm{Span}(1, J)$ and $Q_2 = \mathrm{Span}(J_2, J_3)$. A direct computation shows that f leaves the two planes invariant, and acts on them, with respect to the bases $(1, J)$ and (J_2, J_3), as rotations of angles $-\phi$ and ϕ respectively.

The map f is called a *right isoclinic* rotation. As for left isoclinic rotations, f does not fix any point besides the origin: it rotates any non-zero quaternion q of angle $-\phi$ in the plane spanned by q and qJ.

Double and simple rotations We now consider the case where $f(q) = uqv^{-1}$ and u, v^{-1} are both non-real quaternions written as in (1.34) and (1.35) respectively. We decompose $\mathcal{H}$ as the direct sum of the two orthogonal planes

$$P_A = \{q \in \mathcal{H} : Iq = qJ\}, \quad P_B = \{q \in \mathcal{H} : Iq = -qJ\}.$$

The action of f on both planes is a 2-dimensional rotation. In fact, if $q \in P_A$ we can move q past the exponential as follows:

$$qe^{-\phi J} = q(\cos\phi - \sin\phi J) = \cos\phi\, q - \sin\phi\, Iq = (\cos\phi - \sin\phi I)q = e^{-\phi I}q.$$

Therefore, we can compute the action on P_A as

$$f(q) = e^{\theta I}qe^{-\phi J} = e^{(\theta-\phi)I}q,$$

which is a rotation of angle $\theta - \phi$. Similarly, if $q \in P_B$ we have $Iq = -qJ$, which yields $qe^{-\phi J} = e^{\phi I}q$, so that

$$f(q) = e^{\theta I}qe^{-\phi J} = e^{(\theta+\phi)I}q,$$

which is a rotation of angle $\theta + \phi$.

The geometric nature of the isometry depends on the values of θ and ϕ. In the general case ($\theta \neq \pm\phi$), the map f has no fixed point besides the origin and we say that it is a *double rotation*.

If $\theta = \phi$, then the action on P_A is trivial and we obtain a rotation around a plane, generalizing the isometries given by conjugation discussed in Theorem 1.5.6 (which occur when $I = J$). Similarly, if $\theta = -\phi$ we get a rotation that fixes the entire plane P_B pointwise (again, this case reduces to conjugation if $I = -J$). Since $\theta, \phi \in (0, \pi)$, these last two cases arise if and only if $\cos\theta = \cos\phi$, *i.e.* if and only if $\mathrm{Re}(u) = \mathrm{Re}(v)$. Because u and v have unit norm, the condition $\mathrm{Re}(u) = \mathrm{Re}(v)$ is equivalent to saying that u and v are conjugate quaternions. Such maps are called *simple rotations*.

1.5.3 Linear similarities

If we extend the action defined in (1.33) to all non-zero quaternions by taking $(u, v) \in \mathcal{H}^* \times \mathcal{H}^*$, the linear transformation $q \mapsto uqv^{-1}$ no longer acts as an isometry. Instead, since $|uqv^{-1}| = |u||v|^{-1}|q|$, the norm of q is scaled by a constant positive factor. Consequently, this extended map represents an orientation-preserving linear similarity of $\mathbb{R}^4$. This yields a surjective homomorphism from $\mathcal{H}^* \times \mathcal{H}^*$ onto the group of orientation-preserving linear similarities of $\mathcal{H}$. In fact, this construction exhausts all possible orientation-preserving linear similarities of the 4-dimensional space, as stated in the following proposition.

Proposition 1.5.7. *Every orientation-preserving linear similarity of $\mathcal{H} \cong \mathbb{R}^4$ can be written in the form*

$$q \mapsto uqv^{-1}$$

for some $u, v \in \mathcal{H}^*$. The pair (u, v) is unique up to a common real factor.

The quaternions u and v of Proposition 1.5.7 are respectively called the *left isoclinic* and *right isoclinic* components of the similarity.

Chapter 2

Quaternionic cross-ratio

The primary goal of this chapter is to introduce the *quaternionic cross-ratios* of four distinct points in the compactification $\widehat{\mathcal{H}}$ of the quaternions, and to investigate their behavior under the action of the Möbius group $\mathrm{Möb}(\widehat{\mathcal{H}})$.

The first section is devoted to the study of the action by simultaneous conjugation of the group of invertible quaternions $\mathcal{H}^*$ on ordered tuples of non-real quaternions.

In the second section we introduce the group $\mathrm{PSL}(2,\mathcal{H})$ of 2×2 quaternionic matrices with Dieudonné determinant equal to 1, taken modulo its center. Similarly to the real and complex case, we show that $\mathrm{PSL}(2,\mathcal{H})$ is isomorphic to the Möbius group $\mathrm{Möb}(\widehat{\mathbb{R}}^4)$ in dimension four. Standard references are [Wil93] and [Ahl81]. We conclude the section by explicitly computing the differentials of a Möbius transformation.

Lastly, in the third section, we define the quaternionic cross-ratios and investigate their properties. As we will see, non-commutativity prevents the quaternionic cross-ratios from being invariant under Möbius transformations; instead, they transform by conjugation (see [BG11] and [GL12]). A key observation of this work is that the conjugating factor is precisely determined by the differential of the corresponding Möbius transformation. We will use this property to classify configurations of points in $\widehat{\mathcal{H}}$ up to Möbius transformations.

2.1 Configurations of quaternions

In this section, we study the action of the group of invertible quaternions $\mathcal{H}^*$ on tuples of quaternions via simultaneous conjugation.

2.1.1 Simultaneous conjugation

Definition 2.1.1. Two ordered k-tuples $(q_1,\dots,q_k)$ and $(p_1,\dots,p_k)$ of non-real quaternions are *simultaneously conjugate* (or *simultaneously similar*) if there exists a single $d\in\mathcal{H}^*$ such that $p_i = dq_id^{-1}$ for every $i=1,\dots,k$.

As we have seen in Theorem 1.5.1, two quaternions are conjugate if and only if they have the same real part and the same norm. When studying the action on tuples,

however, we need to take into account how conjugation changes the imaginary parts: by Theorem 1.5.6, this naturally translates into the classification of non-zero vectors in $\mathbb{R}^3$ under the diagonal action of the group of linear rotations SO(3).

Remark 2.1.2. Commutativity is preserved by conjugation. Thus, if the entries of a tuple $(q_1, \dots, q_k)$ are pairwise commuting, any simultaneously conjugate tuple $(p_1, \dots, p_k)$ will also have pairwise commuting entries.

Definition 2.1.3. We say that a k-tuple $(q_1, \dots, q_k)$ of non-real quaternions is

- *commutative* if its entries q_i pairwise commute;
- *generic* if cyclically adjacent entries do not commute ($q_i q_{i+1} \neq q_{i+1} q_i$) and $q_k q_1 \neq q_1 q_k$.

The geometric viewpoint of Theorem 1.5.6 allows us to classify configurations using Gram matrices.

Definition 2.1.4. Let $(q_1, \dots, q_k)$ be a k-tuple of quaternions. The *Gram matrix* of $(q_1, \dots, q_k)$ is the Gram matrix of the vectors $\mathrm{Im}(q_1), \dots, \mathrm{Im}(q_k) \in \mathbb{R}^3$.

By definition, the ij-th entry of the Gram matrix of a k-tuple $(q_1, \dots, q_k)$ is the scalar product $\langle \mathrm{Im}(q_i), \mathrm{Im}(q_j) \rangle$. As a direct consequence of the formula (1.29), this real number is

$$\mathrm{Re}(q_i)\,\mathrm{Re}(q_j) - \mathrm{Re}(q_i q_j). \tag{2.1}$$

Theorem 2.1.5. *Let $k \geq 2$ be an integer and let $(q_1, \dots, q_k)$ and $(p_1, \dots, p_k)$ be two k-tuples of non-real quaternions. If $(q_1, \dots, q_k)$ is simultaneously conjugate to $(p_1, \dots, p_k)$, the following hold:*

1. *for every $i = 1, \dots, k$, we have $\mathrm{Re}(q_i) = \mathrm{Re}(p_i)$;*
2. *the two k-tuples have the same Gram matrices;*
3. *for $k \geq 3$, they satisfy*

$$\langle \mathrm{Im}(q_1) \times \mathrm{Im}(q_2), \mathrm{Im}(q_i) \rangle = \langle \mathrm{Im}(p_1) \times \mathrm{Im}(p_2), \mathrm{Im}(p_i) \rangle, \tag{2.2}$$

for all $i \geq 3$.

Conversely, if $(q_1, \dots, q_k)$ and $(p_1, \dots, p_k)$ are either both commutative or both generic, then these three conditions are also sufficient to guarantee that the two k-tuples are simultaneously conjugate. Furthermore, if the equality in condition 3 holds for an index $i \geq 3$ such that the triple scalar product is non-zero, it automatically holds for all indices.

Notice that the third condition is always satisfied for k-tuples of commuting quaternions, because their imaginary parts are collinear and thus both terms of (2.2) vanish.

Proof. The "only if" direction is immediate, as conjugation preserves real parts and acts as a rotation on the imaginary parts (see Theorem 1.5.1 and Theorem 1.5.6).

Conversely, assume that $(q_1, \dots, q_k)$ and $(p_1, \dots, p_k)$ are either both commutative or both generic k-tuples, and that the three conditions hold. For every index $i = 1, \dots, k$, we set $v_i \coloneqq \operatorname{Im}(q_i) \in \mathbb{R}^3$ and $w_i \coloneqq \operatorname{Im}(p_i) \in \mathbb{R}^3$. The equality of the Gram matrices implies that there exists an isometry $R \in \mathrm{O}(3)$ mapping v_i to w_i for all i. We will now show that R can be chosen in $\mathrm{SO}(3)$.

If the vectors $v_1, \dots, v_k$ (and thus equivalently $w_1, \dots, w_k$) span a proper subspace of $\mathbb{R}^3$ (e.g. a line for the commutative case, see Proposition 1.5.4, or a plane), we can post-compose R with a reflection across any plane containing the subspace to yield a rotation in $\mathrm{SO}(3)$. Notice that in this case, all triple scalar products vanish and the third condition is trivially satisfied.

If the vectors span the whole $\mathbb{R}^3$ (which implies $k \geq 3$ and the tuples are generic), the isometry R is uniquely determined and we need to show $\det(R) = 1$. This is precisely detected by the triple scalar product condition (2.2), which guarantees that the two frames have the same orientation. More precisely, since $v_1, \dots, v_k$, and thus $w_1, \dots, w_k$, span the whole $\mathbb{R}^3$, there exists an index $i \geq 3$ such that $\langle v_1 \times v_2, v_i \rangle \neq 0$. By condition three, we know that $\langle w_1 \times w_2, w_i \rangle = \langle v_1 \times v_2, v_i \rangle$, therefore the equality

$$\langle w_1 \times w_2, w_i \rangle = \det(R) \langle v_1 \times v_2, v_i \rangle$$

implies that $\det(R) = 1$.

Now the rotation R is induced by some unit quaternion $d \in \mathcal{H}^*$ as in Theorem 1.5.6, meaning

$$\operatorname{Im}(dq_id^{-1}) = R(v_i) = w_i = \operatorname{Im}(p_i)$$

for all i. Since conjugation preserves the real part, the first condition guarantees that $dq_id^{-1} = p_i$ for all i. □

Remark 2.1.6. As for the Gram matrices' equality (2.1), equation (2.2) can also be expressed in terms quaternionic multiplication and real parts. Indeed, by the formula for the imaginary part of the product (1.30), the cross product $\operatorname{Im}(q_1) \times \operatorname{Im}(q_2)$ coincides precisely with half the commutator $q_1q_2 - q_2q_1$. Since this quaternion is purely imaginary, by (1.29) we get

$$-\frac{1}{2} \operatorname{Re}((q_1q_2 - q_2q_1)q_i) = \langle \operatorname{Im}(q_1) \times \operatorname{Im}(q_2), \operatorname{Im}(q_i) \rangle$$

Notice that if $d \in \mathcal{H}^*$ is such that $dq_id^{-1} = p_i$, then

$$(p_1p_2 - p_2p_1)p_i = d(q_1q_2 - q_2q_1)q_id^{-1},$$

and so by Theorem 1.5.1 we know that

$$\operatorname{Re}((q_1q_2 - q_2q_1)q_i) = \operatorname{Re}((p_1p_2 - p_2p_1)p_i). \tag{2.3}$$

We can thus replace equation (2.2) with (2.3) in Theorem 2.1.5.

We are now left with the question: if two k-tuples are simultaneously conjugate, how many quaternions realize the conjugation?

Corollary 2.1.7. *Let $(q_1, \dots, q_k)$ and $(p_1, \dots, p_k)$ be two commutative k-tuples of non-real quaternions. Suppose that the two k-tuples are simultaneously conjugate. Then an element $d \in \mathcal{H}^*$ satisfies $dq_1d^{-1} = p_1$ if and only if $dq_id^{-1} = p_i$ for every $i = 1, \dots, k$.*

In particular, the elements $d \in \mathcal{H}$ such that $dq_i = p_id$ for each $i = 1, \dots, k$ form a vector space of dimension 2.

Proof. By hypothesis, there exists $d_0 \in \mathcal{H}^*$ such that $d_0q_id_0^{-1} = p_i$ for each $i = 1, \dots, k$. Assume now that d is an invertible quaternion such that $dq_1d^{-1} = p_1$. We want to show that $dq_id^{-1} = p_i$ for every $i = 1, \dots, k$. By our assumptions, we have

$$dq_1d^{-1} = d_0q_1d_0^{-1},$$

which implies that $d_0^{-1}d$ and q_1 commute. Since the tuple is commutative, $d_0^{-1}d$ also commutes with q_i for every $i = 1, \dots, k$, and we easily get

$$dq_id^{-1} = d_0q_id_0^{-1} = p_i$$

for every $i = 1, \dots, k$, as required. The other implication is trivial, and the last part of the statement regarding the dimension now follows from Theorem 1.5.1. □

Corollary 2.1.8. *Let $(q_1, \dots, q_k)$ and $(p_1, \dots, p_k)$ be two generic k-tuples of non-real quaternions, with $k \geq 2$. Suppose that the two k-tuples are simultaneously conjugate. Then the elements $d \in \mathcal{H}$ such that $dq_i = p_id$ for each $i = 1, \dots, k$ form a vector space of dimension 1.*

Proof. Let d, d' be two invertible quaternions that satisfy

$$dq_id^{-1} = p_i, \quad d'q_id'^{-1} = p_i$$

for every $i = 1, \dots, k$. Then clearly $d^{-1}d'$ and q_i commute for every $i = 1, \dots, k$. If $d^{-1}d'$ were not real, by Proposition 1.5.4, the imaginary part of $d^{-1}d'$ would be proportional to the imaginary part of every q_i, and this is not possible because $(q_1, \dots, q_k)$ is generic. We conclude that $d = \lambda d'$ for some $\lambda \in \mathbb{R}$. The proof is complete. □

Corollary 2.1.9. *Let $(q_1, \dots, q_k)$ be a generic k-tuple of non-real quaternions, with $k \geq 2$, and let $d \in \mathcal{H}^*$ be a quaternion. Then the values of dq_1d^{-1} and dq_2d^{-1} uniquely determine the whole k-tuple $(dq_1d^{-1}, \dots, dq_kd^{-1})$.*

Proof. The result follows directly from Corollary 2.1.8: since q_1 and q_2 do not commute, d is unique up to multiplication by a real number and the resulting conjugate k-tuple is uniquely determined. □

2.1.2 Normal form

As established in Corollary 2.1.9, given the simultaneous conjugacy class of a generic k-tuple, the exact values of any two adjacent entries uniquely determine a specific representative within the class. Here, we want to describe a way to anchor a specific pair to a standard position, which will uniquely fix the remaining entries of the k-tuple. To this end, we introduce the following definition.

Definition 2.1.10. Let $\mathbb{C}^+ = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$. We say that an ordered pair of non-commuting quaternions (p, q) is in *normal form* if

$$p \in \mathbb{C}^+, \quad \langle q, j\rangle > 0, \quad \langle q, k\rangle = 0.$$

Lemma 2.1.11. *Let (p, q) be a pair of non-commuting quaternions. Then there exists $d \in \mathcal{H}^*$ such that the pair (dpd^{-1}, dqd^{-1}) is in normal form. More precisely, we have*

$$\mathrm{Im}(dpd^{-1}) = |\,\mathrm{Im}(p)|i, \quad \mathrm{Im}(dqd^{-1}) = |\,\mathrm{Im}(q)| \cdot (\cos(\theta)\, i + \sin(\theta)\, j)$$

where θ is the angle between $\mathrm{Im}(p)$ and $\mathrm{Im}(q)$. Moreover, d is unique up to multiplication by a real number.

Proof. Suppose that (p, q) is not in normal form (otherwise there is nothing to prove). If $p \notin \mathbb{C}$, identify $\mathrm{Im}(p)$ with $(x, y, z) \in \mathbb{R}^3$ and fix $v^{st} := (|\,\mathrm{Im}(p)|, 0, 0)$. We can then rotate $\mathrm{Im}(p)$ to v^{st} via the unitary quaternion

$$d_0 = \cos\left(\frac{\theta_0}{2}\right) + \sin\left(\frac{\theta_0}{2}\right) I_0$$

where

$$\cos(\theta_0) = \frac{\langle \mathrm{Im}(p), v^{st}\rangle}{|\,\mathrm{Im}(p)|^2} = \frac{x}{|\,\mathrm{Im}(p)|}, \quad I_0 = \frac{\mathrm{Im}(p) \times v^{st}}{|\,\mathrm{Im}(p) \times v^{st}|} = \frac{(0, z, -y)}{\sqrt{y^2 + z^2}}.$$

Now let $q' := d_0 q d_0^{-1}$ and fix $w := |\,\mathrm{Im}(q)|(\cos(\theta), \sin(\theta), 0)$. Notice that both $\mathrm{Im}(q')$ and w lie on the circle

$$\{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 = |\,\mathrm{Im}(q)|\cos(\theta),\ x_2^2 + x_3^2 = \sin^2(\theta) \cdot |\,\mathrm{Im}(q)|^2\}.$$

If $\mathrm{Im}(q') = w$ there is nothing to prove. Otherwise, we can map $\mathrm{Im}(q')$ to w by applying a rotation R with axis $(1, 0, 0)$ and angle θ_1 such that

$$\cos(\theta_1) = \frac{\langle \mathrm{Im}(q'), (0, 1, 0)\rangle}{\sin(\theta) \cdot |\,\mathrm{Im}(q)|}.$$

Since R fixes v^{st}, setting $d_1 := \cos(\frac{\theta_1}{2}) + \sin(\frac{\theta_1}{2})\, i$ and $d := d_1 \cdot d_0$, we get

$$\mathrm{Im}(dpd^{-1}) = |\,\mathrm{Im}(p)|i, \quad \mathrm{Im}(dqd^{-1}) = |\,\mathrm{Im}(q)| \cdot (\cos(\theta)\, i + \sin(\theta)\, j),$$

as desired.

The case where $p \in \mathbb{C}$ is easier: up to reflections, we can assume $\mathrm{Im}(p) = v^{st}$ and we can thus repeat the same argument as above with $d_0 = 1$.

Finally, d is unique up to multiplication by a real number thanks to Corollary 2.1.8, since (p, q) is a generic pair. □

We already know by Theorem 2.1.5 that the action by simultaneous conjugation on tuples of quaternions is extremely rigid. By employing the normal form and Lemma 2.1.11, we have a way to prescribe canonical positions for any generic tuple.

Corollary 2.1.12. *Let $(q_1, \dots, q_k)$ be a generic k-tuple of non-real quaternions. For any index $m \in \{1, \dots, k\}$, with the convention that $q_{k+1} = q_1$, there exists an element $d_m \in \mathcal{H}^*$, unique up to multiplication by a real number, such that the simultaneous conjugation*

$$(d_m q_1 d_m^{-1}, \dots, d_m q_k d_m^{-1})$$

brings the cyclically adjacent pair (q_m, q_{m+1}) into normal form.

Proof. Since the k-tuple is generic, the adjacent quaternions q_m and q_{m+1} do not commute. The existence and uniqueness of the conjugating element $d_m \in \mathcal{H}^*$ follow immediately by applying Lemma 2.1.11 to the pair (q_m, q_{m+1}). □

2.1.3 Adjoint tuples of quaternions

Definition 2.1.13. Two generic tuples of quaternions $(q_1, \dots, q_k)$ and $(p_1, \dots, p_k)$ are *adjoint* if there exist $\alpha_1, \dots, \alpha_k \in \mathcal{H}^*$ such that

$$q_i = \alpha_i \alpha_{i+1}^{-1}, \quad p_i = \alpha_{i+1}^{-1} \alpha_i \quad \forall\, i = 1, \dots, k-1, \quad \text{and} \quad q_k = \alpha_k \alpha_1^{-1}, \quad p_k = \alpha_1^{-1} \alpha_k.$$

Lemma 2.1.14. *Let $(q_1, \dots, q_k)$ and $(p_1, \dots, p_k)$ be two generic, adjoint k-tuples of quaternions. Then:*

1. *we have $q_1 \cdot \ldots \cdot q_k = 1$ and $p_k \cdot \ldots \cdot p_1 = 1$;*
2. *for every $a, b \in \mathcal{H}^*$, the k-tuples $(aq_1a^{-1}, \dots, aq_ka^{-1})$ and $(bp_1b^{-1}, \dots, bp_kb^{-1})$ are adjoint.*

Proof. By definition, there exist $\alpha_1, \dots, \alpha_k \in \mathcal{H}^*$ such that

$$q_i = \alpha_i \alpha_{i+1}^{-1}, \quad p_i = \alpha_{i+1}^{-1} \alpha_i,$$

where the indices are taken modulo k. The first property follows immediately from the telescopic nature of the products: for instance, we have

$$q_1 \cdot q_2 \cdot \ldots \cdot q_k = (\alpha_1 \alpha_2^{-1}) \cdot (\alpha_2 \alpha_3^{-1}) \cdot \ldots \cdot (\alpha_k \alpha_1^{-1}) = 1,$$

and similarly for $p_k \cdot \ldots \cdot p_1$.

For the second property, for each $i = 1, \dots, k$ we set $\beta_i := a\alpha_i b^{-1}$. Then

$$\begin{aligned} aq_ia^{-1} &= a\alpha_i\alpha_{i+1}^{-1}a^{-1} = a\alpha_i b^{-1} b\alpha_{i+1}^{-1}a^{-1} = \beta_i\beta_{i+1}^{-1}, \\ bp_ib^{-1} &= b\alpha_{i+1}^{-1}\alpha_i b^{-1} = b\alpha_{i+1}^{-1}a^{-1}a\alpha_i b^{-1} = \beta_{i+1}^{-1}\beta_i, \end{aligned}$$

so that $a(q_1, \dots, q_k)a^{-1}$ and $b(p_1, \dots, p_k)b^{-1}$ are adjoint. The proof is complete. □

Remark 2.1.15. The definition of adjoint tuples is a "local version" of simultaneous conjugation. On the one hand, it is a strong condition: a direct check shows that $p_i = \alpha_{i+1}^{-1} q_i \alpha_{i+1}$ and $p_{i+1} = \alpha_{i+1}^{-1} q_{i+1} \alpha_{i+1}$. This implies that any pair of consecutive entries (q_i, q_{i+1}) is simultaneously conjugate to (p_i, p_{i+1}) via the quaternion α_{i+1}^{-1}. On the other hand, this is weaker than requiring the whole tuples to be simultaneously conjugate, as the check is only "local": the conjugating element α_{i+1} varies with the index i.

Remark 2.1.16. The reversed order in the product $p_k \cdot \ldots \cdot p_1 = 1$ is not accidental. As we will see in the following sections, adjoint tuples naturally arise as the cross-ratios assigned to an edge of an ideal triangulation when viewed with opposite orientations. The reversed product precisely reflects the fact that looking along an edge from the opposite direction reverses the cyclic order of the incident tetrahedra.

2.2 The group $\mathrm{PSL}(2, \mathcal{H})$

In this section, we will introduce a Lie group that plays a crucial role in our work: the group $\mathrm{PSL}(2, \mathcal{H})$ of 2×2 quaternionic matrices with Dieudonné determinant equal to 1, taken modulo its center. For further details and complete proofs, see [Wil93] and [Die43].

2.2.1 The Dieudonné determinant

We start by reviewing the definition of the *Dieudonné determinant* of a 2×2 quaternionic matrix. Let $q = q_0 + q_1 i + q_2 j + q_3 k \in \mathcal{H}$ be a quaternion. We express q as

$$q = u + vj, \quad \text{where} \quad u = q_0 + q_1 i \in \mathbb{C}, \;\; v = q_2 + q_3 i \in \mathbb{C},$$

and we define the map $\varphi \colon \mathcal{H} \to \mathrm{Mat}(2, \mathbb{C})$ given by

$$\varphi(q) \coloneqq \begin{pmatrix} u & v \\ -\overline{v} & \overline{u} \end{pmatrix}.$$

We state the following result, whose proof is a direct computation (see [Wil93]).

Proposition 2.2.1. *The map $\varphi \colon \mathcal{H} \to Mat(2, \mathbb{C})$ is an embedding of real algebras.*

Remark 2.2.2. For every quaternion $q \in \mathcal{H}$, we have

$$\det(\varphi(q)) = |u|^2 + |v|^2 = |q|^2$$

and hence:

- if $q \in \mathcal{H}^*$ then $\varphi(q) \in \mathrm{GL}(2, \mathbb{C})$;
- if $|q| = 1$ then $\varphi(q) \in \mathrm{SL}(2, \mathbb{C})$.

Notice also that $\operatorname{tr}(\varphi(q)) = u + \overline{u} = 2\operatorname{Re}(q) \in \mathbb{R}$. The map φ actually restricts to a Lie group isomorphism between the unit quaternions S^3 and $\mathrm{SU}(2)$.

We can extend the embedding φ to an embedding of $\mathbb{R}$-algebras $\phi\colon \mathrm{Mat}(2,\mathcal{H}) \to \mathrm{Mat}(4,\mathbb{C})$ simply by setting

$$\phi\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) \coloneqq \begin{pmatrix} \varphi(a) & \varphi(b) \\ \varphi(c) & \varphi(d) \end{pmatrix}.$$

We now compute the determinant of a matrix that lies in the image of ϕ. Set

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{Mat}(2,\mathcal{H}), \quad \widetilde{A} = \phi(A).$$

If $c = 0$, then $\varphi(c) = 0$ and $\widetilde{A}$ is an upper-triangular block matrix. We get

$$\det(\widetilde{A}) = |a|^2|d|^2 = |ad|^2.$$

If $a \neq 0$, we can write

$$\begin{pmatrix} 1 & 0 \\ -ca^{-1} & 1 \end{pmatrix} A = \begin{pmatrix} a & b \\ 0 & -ca^{-1}b + d \end{pmatrix} =: A'$$

and thus we get

$$\det(\widetilde{A}) = \det(\phi(A')) = |ad - aca^{-1}b|^2,$$

where the first equality follows from the fact that the lower-triangular matrix multiplying A maps via ϕ to a matrix having determinant 1. Finally, if $a = 0$, then again

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} A = \begin{pmatrix} c & d \\ 0 & b \end{pmatrix} =: A''$$

and, since the permutation matrix multiplying A is mapped by ϕ to a matrix with determinant 1, we obtain $\det(\widetilde{A}) = \det(\phi(A'')) = |-cb|^2$. We can summarize the above computations with the formula

$$\det(\widetilde{A}) = \begin{cases} |ad - aca^{-1}b|^2 & \text{if } a \neq 0, \\ |-cb|^2 & \text{if } a = 0. \end{cases}$$

Definition 2.2.3. Consider a matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{Mat}(2,\mathcal{H}).$$

The Dieudonné determinant $\Delta(A)$ of A, as defined in [Die43] (see also [Asl96]), is the non-negative real number

$$\Delta(A) \coloneqq \begin{cases} |ad - aca^{-1}b| & \text{if } a \neq 0, \\ |-cb| & \text{if } a = 0. \end{cases}.$$

As we would expect from a determinant, $\Delta(A)$ characterizes invertible matrices and satisfies the Binet formula (see [Wil93]). For the following explicit formulation of the inverse matrix, we adopt the notation used by Kellerhals in [Kel03].

Theorem 2.2.4. *The matrix A is invertible if and only if $\Delta(A) \neq 0$. In that case*

$$A^{-1} = \begin{pmatrix} l_{11}^{-1}d & -l_{12}^{-1}b \\ -l_{21}^{-1}c & l_{22}^{-1}a \end{pmatrix} = \begin{pmatrix} dr_{11}^{-1} & -br_{12}^{-1} \\ -cr_{21}^{-1} & ar_{22}^{-1} \end{pmatrix},$$

where, assuming the coefficients of A are all non-zero, we have

$$\begin{aligned} l_{11} &= da - dbd^{-1}c, & l_{12} &= bdb^{-1}a - bc, \\ l_{21} &= cac^{-1}d - cb, & l_{22} &= ad - aca^{-1}b, \\ r_{11} &= ad - bd^{-1}cd, & r_{12} &= db^{-1}ab - cd, \\ r_{21} &= ac^{-1}dc - bc, & r_{22} &= da - ca^{-1}ba. \end{aligned}$$

Remark 2.2.5. When the coefficients of A are all non-zero, the numbers l_{ij} and r_{ij} defined in Theorem 2.2.4 all have the same norm. In particular

$$\Delta(A) = |l_{ij}| = |r_{ij}| \quad \text{for } 1 \leq i, j \leq 2.$$

If one of the coefficients of A is zero, some of the expressions for l_{ij} and r_{ij} in Theorem 2.2.4 are still well-defined, while others involve division by zero. For these latter cases, we can suitably extend the definitions of l_{ij} and r_{ij}, analogously to the Dieudonné determinant, so that the formula for A^{-1} remains valid. For example, if $a = 0$, we define $l_{22} := -cb$ and $l_{12} := -bc$; the corresponding entries in the inverse matrix then evaluate to $l_{22}^{-1}a = 0$ and $-l_{12}^{-1}b = c^{-1}$, yielding the correct inverse.

Theorem 2.2.6. *For every $A, B \in \mathrm{Mat}(2, \mathcal{H})$, we have $\Delta(AB) = \Delta(A)\Delta(B)$.*

We define the following groups of matrices

$$\begin{aligned} \mathrm{GL}(2, \mathcal{H}) &= \{A \in \mathrm{Mat}(2, \mathcal{H}) : \Delta(A) \neq 0\}, \\ \mathrm{SL}(2, \mathcal{H}) &= \{A \in \mathrm{GL}(2, \mathcal{H}) : \Delta(A) = 1\}. \end{aligned}$$

For every $A \in \mathrm{Mat}(2, \mathcal{H})$ and $\lambda \in \mathbb{R}$, we have $\Delta(\lambda A) = \lambda^2 \Delta(A)$. Therefore every matrix $A \in \mathrm{GL}(2, \mathcal{H})$ lies in $\mathrm{SL}(2, \mathcal{H})$ after rescaling it by $\lambda = 1/\sqrt{\Delta(A)}$.

2.2.2 Quaternionic Möbius transformations

As in the real and complex cases, the group $\mathrm{GL}(2, \mathcal{H})$ naturally acts on the compactification $\widehat{\mathcal{H}} = \mathcal{H} \cup \{\infty\}$. Every invertible matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}(2, \mathcal{H})$$

determines the following homeomorphism $T_A\colon \widehat{\mathcal{H}} \to \widehat{\mathcal{H}}$

$$T_A(q) = (aq+b)(cq+d)^{-1}$$

that is defined on $\mathcal{H} \setminus \{-c^{-1}d\}$ and extended to $\widehat{\mathcal{H}}$ by continuity:

$$T_A(\infty) = \begin{cases} \infty & \text{if } c = 0, \\ ac^{-1} & \text{if } c \neq 0. \end{cases}$$
$$T_A(-c^{-1}d) = \infty \quad \text{if } c \neq 0.$$

A *quaternionic Möbius transformation* of $\widehat{\mathcal{H}}$ is a map T_A constructed in this way.

Remark 2.2.7. The choice of the name "quaternionic Möbius transformations" for the maps induced by $\mathrm{GL}(2,\mathcal{H})$ stems from a direct analogy with the complex case, where fractional linear transformations are classically referred to as Möbius transformations. Strictly speaking, however, these maps do not cover the full Möbius group of $\widehat{\mathbb{R}}^4$ introduced in Section 1.1, but only its orientation-preserving component. We will formally prove this isomorphism in Theorem 2.2.14.

Since quaternions do not commute, we should not take things for granted, and it is instructive to check that this construction yields a well-defined group action.

Proposition 2.2.8. *The map $A \mapsto T_A$ defines an action of $\mathrm{GL}(2,\mathcal{H})$ on $\widehat{\mathcal{H}}$.*

Proof. Given two invertible matrices

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

that induce two quaternionic Möbius transformations T_A and T_B, we get

$$\begin{aligned} T_B(T_A(q)) &= (e(aq+b)(cq+d)^{-1} + f)(g(aq+b)(cq+d)^{-1} + h)^{-1} \\ &= ((e(aq+b)+f(cq+d))(cq+d)^{-1})((g(aq+b)+h(cq+d))(cq+d)^{-1})^{-1} \\ &= ((ea+fc)q + (eb+fd))((ga+hc)q + (gb+hd))^{-1} \end{aligned}$$

that is indeed the quaternionic Möbius transformation defined by the matrix BA. □

Warning 2.2.9. Simple variations on the definition of quaternionic Möbius transformations do not work! The alternative expressions $T(q) = (qa+b)(qc+d)^{-1}$ and $U(q) = (aq+b)^{-1}(cq+d)$ would not define continuous maps on the whole $\widehat{\mathcal{H}}$ in general: for instance, the map $q \mapsto (q+1)(qc)^{-1} = qc^{-1}q^{-1} + c^{-1}q^{-1}$ does not extend continuously at $q = \infty$, since it may converge along different paths to any quaternion that is conjugate to c^{-1}.

Proposition 2.2.10. *Quaternionic Möbius transformations act transitively on ordered triples of distinct points in $\widehat{\mathcal{H}}$.*

Proof. Given three distinct points $q_1, q_2, q_3 \in \mathcal{H}$, we can find a quaternionic Möbius transformation T such that $T(q_1) = 0, T(q_2) = 1, T(q_3) = \infty$. It is enough to define:

$$T(q) = (q_2 - q_3)(q_2 - q_1)^{-1}(q - q_1)(q - q_3)^{-1}.$$

This is indeed the quaternionic Möbius transformation T_A induced the matrix

$$A = \begin{pmatrix} (q_2 - q_3)(q_2 - q_1)^{-1} & -(q_2 - q_3)(q_2 - q_1)^{-1}q_1 \\ 1 & -q_3 \end{pmatrix},$$

where A is invertible because the computation of the Dieudonné determinant yields

$$\Delta(A) = |(q_2 - q_3)(q_2 - q_1)^{-1}(q_3 - q_1)| \neq 0.$$

The case where one of the q_i is ∞ is left as an exercise. □

If we multiply a matrix A by some real number $\lambda \neq 0$ we get another matrix λA that defines the same quaternionic Möbius transformation as A. If we take $\lambda = \sqrt{\Delta(A)}^{-1}$ the new matrix lies in $\mathrm{SL}(2, \mathcal{H})$. The group $\mathrm{SL}(2, \mathcal{H})$ represents all quaternionic Möbius transformations, in a unique way up to sign:

Proposition 2.2.11. *Two matrices $A, B \in \mathrm{SL}(2, \mathcal{H})$ induce the same action on $\widehat{\mathcal{H}}$ if and only if $B = \pm A$.*

Proof. The "if" direction is immediate: for every matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathcal{H}),$$

we have

$$T_{-A}(q) = (-aq - b)(-cq - d)^{-1} = (aq + b)(cq + d)^{-1} = T_A(q),$$

hence $T_A = T_{-A}$. Conversely, assume that $A, B \in \mathrm{SL}(2, \mathcal{H})$ induce the same transformation, meaning $T_A = T_B$. Let $q_1 \coloneqq T_A(0)$, $q_2 \coloneqq T_A(1)$, and $q_3 \coloneqq T_A(\infty)$, which are three distinct points in $\widehat{\mathcal{H}}$. By Proposition 2.2.10, there exists a matrix $C \in \mathrm{GL}(2, \mathcal{H})$ such that the induced transformation T_C maps q_1, q_2, and q_3 to $0, 1$, and ∞ respectively. Up to rescaling, we can assume that C lies in $\mathrm{SL}(2, \mathcal{H})$.

By construction, the quaternionic Möbius transformations $T_{CA} = T_C \circ T_A$ and $T_{CB} = T_C \circ T_B$ coincide, and they both fix 0, 1 and ∞. From this we easily deduce that

$$CA = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}, \quad CB = \begin{pmatrix} b & 0 \\ 0 & b \end{pmatrix}$$

for some $a, b \in \mathcal{H}^*$, and they induce the quaternionic Möbius transformations $q \mapsto aqa^{-1}$ and $q \mapsto bqb^{-1}$ respectively. By hypothesis they coincide for every q, that is, $b^{-1}a$ commutes with every quaternion q, and this is possible only if $b^{-1}a \in \mathbb{R}$. Since $\Delta(CA) = \Delta(CB) = 1$ we have $|a| = |b|$ and therefore $b = \pm a$, so $CB = \pm CA$. Left-multiplying by C^{-1}, we conclude $B = \pm A$. □

The quotient group

$$\mathrm{PSL}(2,\mathcal{H}) = \mathrm{SL}(2,\mathcal{H})/\{\pm \mathrm{Id}\}$$

is the group of all quaternionic Möbius transformations of $\widehat{\mathcal{H}}$, just as in the more familiar real and complex cases. Along the proofs of the previous propositions we have also discovered the following interesting fact.

Proposition 2.2.12. *A quaternionic Möbius transformation T*

- *fixes $\infty \iff$ it is of type $T(q) = aqb + c$ for some $a, b \in \mathcal{H}^*, c \in \mathcal{H}$;*
- *fixes 0 and $\infty \iff$ it is of type $T(q) = aqb$ for some $a, b \in \mathcal{H}^*$;*
- *fixes $0, 1$, and $\infty \iff$ it is of type $T(q) = aqa^{-1}$ for some $a \in \mathcal{H}^*$.*

Remark 2.2.13. Let T_A be the quaternionic Möbius transformation induced by a matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathcal{H}),$$

and suppose that $c \neq 0$. Then we can write

$$\begin{aligned} T_A(q) = (aq+b)(cq+d)^{-1} &= (ac^{-1}cq + ac^{-1}d - ac^{-1}d + b)(cq+d)^{-1} \\ &= ac^{-1} + (b - ac^{-1}d)(cq+d)^{-1}. \end{aligned}$$

As anticipated in Remark 2.2.7, the geometric properties of the quaternionic Möbius transformations are similar to their real and complex analogues. Recall that in Section 1.1 we defined the M"obius group $\mathrm{Möb}(\widehat{\mathbb{R}}^4)$ as the group generated by inversions in spheres and reflections across hyperplanes. Its orientation-preserving subgroup, denoted by $\mathrm{Möb}^+(\widehat{\mathbb{R}}^4)$, consists precisely of the maps obtained by composing an even number of such generators.

Theorem 2.2.14. *The quaternionic Möbius transformations of $\widehat{\mathcal{H}}$ are precisely the compositions of an even number of inversions in spheres and reflections across hyperplanes in $\mathcal{H}$. In particular, the following isomorphism holds:*

$$\mathrm{PSL}(2,\mathcal{H}) \cong \mathrm{Möb}^+(\widehat{\mathbb{R}}^4).$$

Proof. We first prove that a composition of an even number of reflections and inversions is a quaternionic Möbius transformation. Consider the hyperplane

$$H = \{q \in \mathcal{H} : \langle q, n\rangle = b\} \subset \mathcal{H},$$

where $n \in \mathcal{H}^*$ is a unit quaternion and $b \in \mathbb{R}$ is a real number. We have

$$\langle q, n\rangle = \mathrm{Re}(q\overline{n}) = \frac{q\overline{n} + n\overline{q}}{2}.$$

The reflection r_H across H can be written as

$$r_H(q) = q - 2(\langle q, n\rangle - b)n = -n\bar{q}n + 2bn. \tag{2.4}$$

Now given a point $c \in \mathcal{H}$ and a positive real number r, consider the sphere

$$S = \{q \in \mathcal{H} : |q - c| = r\} \subset \mathcal{H}.$$

We can write the inversion i_S in S as

$$i_S(q) = c + \frac{r^2}{|q-c|^2}(q-c) = (c\bar{q} - |c|^2 + r^2)(\bar{q} - \bar{c})^{-1}. \tag{2.5}$$

Notice that the center c is mapped to $\infty \in \widehat{\mathcal{H}}$. The explicit formulations given in (2.4) and (2.5) are almost quaternionic Möbius transformations, except they act on the conjugate variable $\bar{q}$. Therefore, composing any two such map returns an expression in q. This shows that composing an even number of reflections across hyperplanes and inversions in spheres yields a quaternionic Möbius transformation.

Conversely, consider the quaternionic Möbius transformation T_A of $\widehat{\mathcal{H}}$ defined by the matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathcal{H}).$$

If $c = 0$, then $T_A(q) = aqd^{-1} + bd^{-1}$. By Proposition 1.5.7, T_A is the composition of a linear similarity of $\mathcal{H}$ and a translation, hence the thesis follows by Remark 1.1.5. If $c \neq 0$, by Remark 2.2.13, we can write

$$T_A(q) = ac^{-1} + (b - ac^{-1}d)(cq + d)^{-1}.$$

Again, the thesis follows by Remark 1.1.5, provided we prove that $q \mapsto q^{-1}$ is the composition of an even number of inversions and reflections. Since $q^{-1} = \bar{q}/|q|^2$, the map $q \mapsto q^{-1}$ is the composition of the inversion $q \mapsto q/|q|^2$ in the unit sphere centered at the origin with the quaternionic conjugation $q \mapsto \bar{q}$. Notice that this last transformation of $\mathcal{H}$ is the composition of the three reflections across the (pairwise orthogonal) hyperplanes spanned by $\{1, j, k\}$, $\{1, i, k\}$ and $\{1, i, j\}$. Thus, $q \mapsto q^{-1}$ is the composition of an inversions and three reflections, and the proof is complete. □

In what follows, whenever the context is clear, we will simply say *Möbius transformation* to mean a quaternionic one. The following results follows by combining Theorem 2.2.14 with Lemma 1.1.1 and Proposition 1.1.19.

Corollary 2.2.15. *Every quaternionic Möbius transformation is a conformal diffeomorphism and it maps the set consisting of all k-spheres and k-planes in $\mathcal{H}$ to itself for all $k = 1, 2, 3$.*

2.2.3 Differentials of Möbius transformations

In Theorem 2.2.14 we have shown that each element of the group $\mathrm{PSL}(2,\mathcal{H})$ acts as a Möbius transformation on the sphere $S^4 \cong \widehat{\mathcal{H}}$. As we have seen in Proposition 1.1.19, these transformations are conformal diffeomorphisms. To explicitly compute their differentials, we read the maps through the canonical charts of $\widehat{\mathcal{H}}$, namely the identity $q \mapsto q$ on $\mathcal{H}$ and the quaternionic inversion $q \mapsto q^{-1}$ around ∞. This choice naturally identifies the tangent spaces of $\widehat{\mathcal{H}}$ with the vector space $\mathcal{H}$. With a slight abuse of notation, we will still denote by $df_{q_0} \colon \mathcal{H} \to \mathcal{H}$ the differential of the local representative of a quaternionic Möbius transformation f at a point $q_0 \in \widehat{\mathcal{H}}$. Under this identification, df_{q_0} is an orientation-preserving linear similarity of $\mathcal{H}$.

Recall from Proposition 1.5.7 that any orientation-preserving linear similarity of $\mathcal{H}$ can be written in the form

$$q \mapsto uqv^{-1}$$

for some quaternions $u, v \in \mathcal{H}^*$. The parameters u and v are called the *left isoclinic* and *right isoclinic* components respectively, and they are determined only up to a common real factor. Similarly to the complex case, we can write the differential of a quaternionic Möbius transformation in terms of the entries of a matrix representing it.

We briefly recall what happens in the complex case: let φ_A be the Möbius transformation of $\widehat{\mathbb{R}}^2 = \widehat{\mathbb{C}}$ represented by the matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{C}).$$

By definition, φ_A acts on $\widehat{\mathbb{C}}$ as $\varphi_A(z) = \frac{az+b}{cz+d}$. If $c \neq 0$, the differential at a generic point z_0 is

$$\varphi_A'(z_0) = \frac{1}{(cz_0+d)^2}.$$

At the special points ∞ and $-d/c$ (the pole of φ_A) we need to change charts. More precisely, the differential of φ_A at ∞ is the derivative of the map $g_A(z) = \varphi_A(1/z)$ at 0, while the differential of φ_A at the pole $-d/c$ is the derivative of the map $h_A(z) = 1/\varphi_A(z)$ at $-d/c$. By standard calculus, one easily gets $\varphi_A'(\infty) = -1/c^2$ and $\varphi_A'(-d/c) = -c^2$. When $c = 0$, φ_A fixes ∞ and acts as a similarity on $\mathbb{C}$; we get

$$\varphi_A'(\infty) = \frac{d}{a} = d^2, \quad \varphi_A'(z_0) = \frac{a}{d} = d^{-2} \ \text{ if } z_0 \in \mathbb{C}.$$

Suppose now that f is a quaternionic Möbius transformation. Unlike the complex case, where the differential at z_0 is completely determined by multiplication by a single complex number $\varphi_A'(z_0)$, the non-commutativity of $\mathcal{H}$ requires us to explicitly write the action of the differential on a tangent vector. As previously discussed, using local charts we view the differential of f at a point $q_0 \in \widehat{\mathcal{H}}$ as an orientation-preserving linear similarity $df_{q_0} \colon \mathcal{H} \to \mathcal{H}$.

Proposition 2.2.16. *Suppose that $f = T_A$ is represented by the matrix*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathcal{H}).$$

If f fixes ∞, then $c = 0$ and for every $h \in \mathcal{H}$ we have

$$df_{q_0}(h) = \begin{cases} dha^{-1} & \text{if } q_0 = \infty, \\ ahd^{-1} & \text{if } q_0 \neq \infty. \end{cases}$$

If f does not fix ∞, then $c \neq 0$, f has a finite pole $p_f = -c^{-1}d$ and for every $h \in \mathcal{H}$ we have the following equalities:

$$df_{q_0}(h) = \begin{cases} -(b - ac^{-1}d)(q_0 - p_f)^{-1}h(q_0 - p_f)^{-1}c^{-1} & \text{if } q_0 \neq \infty,\ q_0 \neq p_f, \\ (b - ac^{-1}d)hc^{-1} & \text{if } q_0 = \infty, \\ ch(b - ac^{-1}d)^{-1} & \text{if } q_0 = p_f. \end{cases}.$$

Moreover, the differential does not depend on the chosen matrix representing f.

Proof. As in the complex case, we start with the case $c \neq 0$. If q_0 is neither ∞ nor the pole of f, we write

$$\begin{aligned} f(q_0 + h) - f(q_0) &= (b - ac^{-1}d)((cq_0 + ch + d)^{-1} - (cq_0 + d)^{-1}) \\ &= (b - ac^{-1}d)(cq_0 + ch + d)^{-1}(cq_0 + d - cq_0 - ch - d)(cq_0 + d)^{-1} \\ &= (b - ac^{-1}d)(cq_0 + ch + d)^{-1}(-ch)(cq_0 + d)^{-1}, \end{aligned}$$

where the first equality follows from Remark 2.2.13. This implies the statement. The differential at ∞ is again obtained by computing the differential of $g(q) := f(q^{-1})$ at $q_0 = 0$, while the differential at p_f is obtained by computing the differential of the map $h(q) := f(q)^{-1}$ at p_f.

The case $c = 0$ is easier: here f fixes the point at infinity and acts on $\mathcal{H}$ as the similarity

$$f(q) = aqd^{-1} + bd^{-1},$$

so the computation of df_{q_0} is immediate. Lastly, if we choose $-A$ instead of A as a matrix representation for f, then the sign of each parameter changes and it is easy to see that the global sign is unaltered. □

In the rest of the work, it will be convenient to decompose the differential of f into its left isoclinic component u_{f,q_0} and right isoclinic component v_{f,q_0}. Strictly speaking, these quantities are defined only up to a simultaneous sign change, corresponding to the choice of the matrix representative A or $-A$. However, this ambiguity does not affect our results: in all subsequent formulas, $u_{f,\cdot}$ and $v_{f,\cdot}$ appear solely in products where the signs cancel out (e.g. terms of the form $u_{f,p}(\dots)v_{f,q}^{-1}$ and $u_{f,p}(\dots)u_{f,q}^{-1}$). Therefore, we may fix an arbitrary matrix representative $A \in \mathrm{SL}(2, \mathcal{H})$ to compute them without loss of generality. With this understanding, we write $df_{q_0}(h) = u_{f,q_0} \cdot h \cdot v_{f,q_0}^{-1}$.

Remark 2.2.17. The explicit expressions for the isoclinic components u_{f,q_0} and v_{f,q_0} can be directly read off from the formulas in Proposition 2.2.16. For instance, in the generic case where $c \neq 0$ and $q_0 \notin \{\infty, p_f\}$, the equality

$$df_{q_0}(h) = -(b - ac^{-1}d)(q_0 - p_f)^{-1}h(q_0 - p_f)^{-1}c^{-1}$$

immediately yields $u_{f,q_0} = -(b - ac^{-1}d)(q_0 - p_f)^{-1}$ and $v_{f,q_0}^{-1} = (q_0 - p_f)^{-1}c^{-1}$. The components for the special points and the case $c = 0$ are extracted analogously.

Lemma 2.2.18. *Let f be a quaternionic Möbius transformation, and let $q_1, q_2 \in \mathcal{H}$ be two quaternions such that $f(q_1) \neq \infty, f(q_2) \neq \infty$. Then*

$$f(q_2) - f(q_1) = u_{f,q_2}(q_2 - q_1)v_{f,q_1}^{-1}.$$

Proof. Suppose that $f = T_A$ is represented by the matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathcal{H})$. If $c = 0$, we get

$$f(q_2) - f(q_1) = a(q_2 - q_1)d^{-1},$$

and the thesis follows immediately from the first case of Proposition 2.2.16. If $c \neq 0$, by Remark 2.2.13 we can rewrite

$$f(q) = ac^{-1} + (b - ac^{-1}d)(cq + d)^{-1},$$

which yields

$$\begin{aligned} f(q_2) - f(q_1) &= (b - ac^{-1}d)((cq_2 + d)^{-1} - (cq_1 + d)^{-1}) \\ &= (b - ac^{-1}d)(cq_2 + d)^{-1}(cq_1 - cq_2)(cq_1 + d)^{-1} \\ &= -(b - ac^{-1}d)(cq_2 + d)^{-1}c(q_2 - q_1)(cq_1 + d)^{-1}. \end{aligned}$$

The thesis follows from the second case of Proposition 2.2.16. □

2.3 Quaternionic cross-ratio

We want to parametrize configurations of points in $\widehat{\mathcal{H}}$ up to Möbius transformations. In the complex case, the cross-ratio elegantly classifies the possible configurations of four distinct points (see Remark 1.4.15). In the quaternionic case we can still define and use the cross-ratio, but some care is needed due to the lack of commutativity.

Definition 2.3.1. Let $q_1, q_2, q_3, q_4 \in \mathcal{H}$ be four distinct quaternions. The *quaternionic cross-ratio* of the ordered tuple (q_1, q_2, q_3, q_4) is the quaternion

$$Q(q_1, q_2, q_3, q_4) \coloneqq (q_2 - q_3)(q_2 - q_1)^{-1}(q_4 - q_1)(q_4 - q_3)^{-1}. \tag{2.6}$$

Remark 2.3.2. The quaternionic cross-ratio has already appeared in the literature, for instance in [BG11] and [GL12], although defined with slight variations. These definitions are essentially equivalent to ours, differing only by a permutation of the factors. Some of the results presented in this section can also be found in the aforementioned works.

By taking limits in (2.6), Definition 2.3.1 naturally extends to the case where at most one among q_1, q_2 and q_4 is ∞. We get

$$\begin{aligned} Q(\infty, q_2, q_3, q_4) &= (q_2 - q_3)(q_4 - q_3)^{-1}, \\ Q(q_1, \infty, q_3, q_4) &= (q_4 - q_1)(q_4 - q_3)^{-1}, \\ Q(q_1, q_2, q_3, \infty) &= (q_2 - q_3)(q_2 - q_1)^{-1}. \end{aligned}$$

In fact the following holds.

Lemma 2.3.3. *Let $q_1, q_2, q_3, q_4 \in \mathcal{H}$ be distinct. The three maps $\mathcal{H} \to \widehat{\mathcal{H}}$ given by*

$$q \mapsto Q(q, q_2, q_3, q_4), \quad q \mapsto Q(q_1, q, q_3, q_4), \quad q \mapsto Q(q_1, q_2, q_3, q)$$

are restrictions of Möbius transformations that send q_1, q_2, q_3, q_4 to $0, 1, \infty, Q(q_1, q_2, q_3, q_4)$.

Proof. This is easy for the second and third maps, which can be written simply as

$$\begin{aligned} q &\mapsto (q - q_3)(q - q_1)^{-1}(q_4 - q_1)(q_4 - q_3)^{-1}, \\ q &\mapsto (q_2 - q_3)(q_2 - q_1)^{-1}(q - q_1)(q - q_3)^{-1}, \end{aligned}$$

but not so straightforward for the first map. By manipulating its expression for $q \neq q_2$, we get

$$\begin{aligned} q &\mapsto (q_2 - q_3)(q_2 - q)^{-1}(q_4 - q)(q_4 - q_3)^{-1} \\ &= (q_2 - q_3)(q_2 - q)^{-1}(q_4 - q_2 + q_2 - q)(q_4 - q_3)^{-1} \\ &= (q_2 - q_3)(q_4 - q_3)^{-1} + (q_2 - q_3)(q_2 - q)^{-1}(q_4 - q_2)(q_4 - q_3)^{-1}, \end{aligned}$$

which is a Möbius transformation (let us call it T) because it is the composition of a translation with the Möbius transformation represented by the matrix

$$\begin{pmatrix} 0 & q_2 - q_3 \\ -(q_4 - q_3)(q_4 - q_2)^{-1} & (q_4 - q_3)(q_4 - q_2)^{-1} q_2 \end{pmatrix}.$$

Notice that both $q \mapsto Q(q, q_2, q_3, q_4)$ and the resulting Möbius transformation T map q_2 to ∞, therefore they coincide for every $q \in \mathcal{H}$. □

Extending the definition of $Q(q_1, q_2, q_3, q_4)$ to the case $q_3 = \infty$ is subtler: since the factors containing q_3 in (2.6) are not adjacent, the limit of $Q(q_1, q_2, q_3, q_4)$ as q_3 tends to infinity is actually not well-defined in general (see Warning 2.2.9) and only its conjugacy class is. For our purposes it is nevertheless convenient to choose one specific element in this conjugacy class, setting

$$Q(q_1, q_2, \infty, q_4) \coloneqq (q_2 - q_1)^{-1}(q_4 - q_1). \tag{2.7}$$

In particular we have

$$Q(0, 1, \infty, q) = q.$$

We should however keep in mind that Q is not continuous at $q_3 = \infty$ (it is continuous along paths lying in the stabilizer of $(q_2 - q_1)^{-1}(q_4 - q_1)$). We will soon see that q_3 plays a special role also in other circumstances.

Remark 2.3.4. Let $q_1, q_2, q_3 \in \widehat{\mathcal{H}}$ be distinct. The Möbius transformation

$$T(q) = \begin{cases} (q_2 - q_3)(q - q_3)^{-1} & \text{if } q_1 = \infty, \\ (q - q_1)(q - q_3)^{-1} & \text{if } q_2 = \infty, \\ (q_2 - q_1)^{-1}(q - q_1) & \text{if } q_3 = \infty, \\ (q_2 - q_3)(q_2 - q_1)^{-1}(q - q_1)(q - q_3)^{-1} & \text{otherwise} \end{cases}$$

maps q_1, q_2, q_3 to $0, 1, \infty$ respectively, and every other distinct point $q_k \in \widehat{\mathcal{H}} \setminus \{q_1, q_2, q_3\}$ is mapped to the quaternionic cross-ratio $Q(q_1, q_2, q_3, q_k)$. We observe that T is a Möbius transformation because it is represented by an invertible quaternionic matrix. For instance, as we stated during the proof of Proposition 2.2.10, when q_1, q_2, q_3, q_k are all finite, $T = T_A$ is represented by the matrix

$$A = \begin{pmatrix} (q_2 - q_3)(q_2 - q_1)^{-1} & -(q_2 - q_3)(q_2 - q_1)^{-1}q_1 \\ 1 & -q_3 \end{pmatrix}.$$

A direct computation of the Dieudonné determinant yields

$$\Delta(A) = |(q_2 - q_3)(q_2 - q_1)^{-1}(q_3 - q_1)| \neq 0,$$

hence $A \in \mathrm{GL}(2, \mathcal{H})$.

We now examine the effect of a quaternionic Möbius transformation on the cross-ratio $Q(q_1, q_2, q_3, q_4)$ of an ordered tuple (q_1, q_2, q_3, q_4) of pairwise distinct quaternions. As opposed to the complex case, quaternionic cross-ratios in general are *not* invariant under Möbius transformations, but they change in a controlled way.

Lemma 2.3.5. *Let $a, b, c \in \mathcal{H}$ be three distinct quaternions. Then*

$$(a - b)^{-1}(c - b)(c - a)^{-1} = (c - a)^{-1}(c - b)(a - b)^{-1}.$$

Proof. We can easily re-write both sides as follows

$$(a - b)^{-1}(c - b)(c - a)^{-1} = (a - b)^{-1}(c - a + a - b)(c - a)^{-1} = (c - a)^{-1} + (a - b)^{-1},$$
$$(c - a)^{-1}(c - b)(a - b)^{-1} = (c - a)^{-1}(c - a + a - b)(a - b)^{-1} = (a - b)^{-1} + (c - a)^{-1}.$$

The proof is complete. □

Proposition 2.3.6. *Let q_1, q_2, q_3, q_4 be four pairwise distinct points in $\widehat{\mathcal{H}}$, and let f be a quaternionic Möbius transformation. Then*

$$Q(f(q_1), f(q_2), f(q_3), f(q_4)) = u_{f,q_3} Q(q_1, q_2, q_3, q_4) u_{f,q_3}^{-1}.$$

Proof. Suppose that $f = T_A$ is represented by the matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathcal{H}).$$

If $c = 0$, then it is easy to see that

$$Q(f(q_1), f(q_2), f(q_3), f(q_4)) = \begin{cases} aQ(q_1, q_2, q_3, q_4)a^{-1} & \text{if } q_3 \neq \infty, \\ dQ(q_1, q_2, q_3, q_4)d^{-1} & \text{if } q_3 = \infty. \end{cases},$$

and the thesis follows from Proposition 2.2.16. We will henceforth suppose that $c \neq 0$, and we have to consider several cases. Assume that the points q_1, q_2, q_3, q_4 are all finite. If none of them is the pole of f, then the thesis follows by applying Lemma 2.2.18 to each factor of $Q(f(q_1), f(q_2), f(q_3), f(q_4))$. If one of them, say q_1, is the pole of f, then

$$\begin{aligned} Q(f(q_1), f(q_2), f(q_3), f(q_4)) = Q(\infty, f(q_2), f(q_3), f(q_4)) = \\ = (f(q_2) - f(q_3))(f(q_4) - f(q_3))^{-1} \end{aligned}$$

and by applying Lemma 2.2.18 to the last expression we get

$$Q(f(q_1), f(q_2), f(q_3), f(q_4)) = u_{f,q_3}(q_2 - q_3)v_{f,q_2}^{-1}v_{f,q_4}(q_4 - q_3)^{-1}u_{f,q_3}^{-1}.$$

Since q_1 is the pole of f, we can re-write the right isoclinic differentials of f as

$$v_{f,q_2}^{-1} = (q_2 - q_1)^{-1}c^{-1}, \quad v_{f,q_4} = c(q_4 - q_1)$$

and we easily recover the desired statement. The cases where one of the other points is the pole of f can be carried out with a similar strategy. Lastly, it remains to check what happens if one among the q_i is the point ∞. Whether the pole of f is among the q_i's or not, the proof follows by substituting the appropriate definitions of the cross-ratio and of the differentials, with a combination of Lemma 2.2.18 and Proposition 2.2.16. The verification is purely algebraic and confirms that the conjugation formula holds in all cases. □

As a direct consequence of Proposition 2.3.6, we notice that the quaternionic cross-ratios of four points change by conjugation under a Möbius transformation [BG11, Corollary 4.6]. For exactly four points, this condition completely classifies the tuples up to Möbius transformations [GL12, Theorem 8]. The following theorem is crucial for our purposes: it says that we can use cross-ratios up to uniform conjugation to classify n-tuples of points up to Möbius transformations.

Theorem 2.3.7. *Fix $n \geq 4$ and let $q_1, \dots, q_n \in \widehat{\mathcal{H}}$ and $q_1', \dots, q_n' \in \widehat{\mathcal{H}}$ be two n-tuples of distinct points. There exists a Möbius transformation f such that $f(q_i) = q_i'$ for each $i = 1, \dots, n$ if and only if there exists $p \in \mathcal{H}^*$ such that*

$$Q(q_1', q_2', q_3', q_k') = pQ(q_1, q_2, q_3, q_k)p^{-1} \tag{2.8}$$

for all $k = 4, \dots, n$.

Proof. The "only if" part is straightforward: if such an f exists, by Proposition 2.3.6 the cross-ratios change by conjugation via a certain $p \in \mathcal{H}^*$ that only depends on f and

q_3. Conversely, suppose that the cross-ratios are related as in (2.8) for some $p \in \mathcal{H}^*$. We apply the Möbius transformation f of Remark 2.3.4 and send q_1, q_2, q_3 to $0, 1, \infty$, and q_k to $Q(q_1, q_2, q_3, q_k)$ respectively, for every $k \geq 4$. We apply an analogous Möbius transformation f' and send q'_1, q'_2, q'_3 to $0, 1, \infty$ and q'_k to $Q(q'_1, q'_2, q'_3, q'_k)$ respectively. Now the two n-tuples of points differ only by a conjugation by p, so we can send the first to the second via the Möbius transformation $q \mapsto pqp^{-1}$. The proof is complete. □

Corollary 2.3.8. *The cross-ratio of an ordered 4-tuple of distinct points in $\widehat{\mathcal{H}}$ is real if and only if the four points lie on the same line or circle.*

Proof. Let $q_1, q_2, q_3, q_4 \in \widehat{\mathcal{H}}$ be four distinct points, $Q := Q(q_1, q_2, q_3, q_4)$ the quaternionic cross-ratio of the tuple (q_1, q_2, q_3, q_4), and T a Möbius transformation that maps q_1, q_2, q_3, q_4 to $0, 1, \infty, Q$ respectively as in Remark 2.3.4. Since Möbius transformations send lines and circles into lines and circles, we have that q_1, q_2, q_3, q_4 lie in the same line or circle if and only if $0, 1, \infty, Q$ do, and this holds if and only if $Q(q_1, q_2, q_3, q_4)$ is a real number. □

2.3.1 The simple ratio

For later use, we introduce another quaternionic quantity called the *simple ratio*, assigned to ordered triples of distinct quaternions.

Let q_1, q_2, q_3 be three distinct points in $\widehat{\mathcal{H}}$. Since Möbius transformations act transitively on ordered triples of distinct points (as shown in Proposition 2.2.10), we know that there exists a Möbius transformation f mapping $0, 1, \infty$ to q_1, q_2, q_3 respectively. If g is another Möbius transformation with the same property, then we can relate their differentials to each other.

Lemma 2.3.9. *Let f, g be two quaternionic Möbius transformations as above. Then there exists $a \in \mathcal{H}^*$ such that*

$$u_{g,p} = u_{f,apa^{-1}} \cdot a$$

for all $p \in \widehat{\mathcal{H}}$.

Proof. The composition $h = f^{-1} \circ g$ is a Möbius transformation that fixes the points $0, 1, \infty$, so it is of the form $h(q) = aqa^{-1}$ for some invertible quaternion $a \in \mathcal{H}^*$. By Proposition 2.2.16, we know that the left isoclinic component of the differential is given by $u_{h,p} = a$ for every point $p \in \widehat{\mathcal{H}}$. Now we simply apply the composition rule for the differential, and we get

$$u_{g,p} = u_{f \circ h,p} = u_{f,h(p)} \cdot u_{h,p} = u_{f,apa^{-1}} \cdot a.$$

The proof is complete. □

Definition 2.3.10. Let $q_1, q_2, q_3 \in \widehat{\mathcal{H}}$ be three pairwise distinct points. Let $f \in \mathrm{PSL}(2, \mathcal{H})$ be any Möbius transformation mapping $0, 1, \infty$ to q_1, q_2, q_3 respectively. The *simple ratio* of the points q_1, q_2, q_3 is the quaternion

$$S(q_1, q_2, q_3) := u_{f,\infty} \cdot u_{f,0}^{-1}. \tag{2.9}$$

Remark 2.3.11. The simple ratio $S(q_1, q_2, q_3)$ is well-defined. Indeed, if f and g are two Möbius transformations mapping $0, 1, \infty$ to q_1, q_2, q_3 respectively, then by Lemma 2.3.9 we know that

$$u_{g,\infty} = u_{f,\infty} \cdot a, \quad u_{g,0} = u_{f,0} \cdot a$$

for some $a \in \mathcal{H}^*$, so that clearly

$$u_{g,\infty} \cdot u_{g,0}^{-1} = u_{f,\infty} \cdot u_{f,0}^{-1}.$$

If we fix three distinct points q_1, q_2, q_3 in $\widehat{\mathcal{H}}$, there are six possible simple ratios assigned to q_1, q_2, q_3, one for each permutation of $1, 2, 3$. The following lemma explores the relations between these quantities.

Lemma 2.3.12. *Fix three distinct points $q_1, q_2, q_3 \in \widehat{\mathcal{H}}$. The simple ratios assigned to q_1, q_2, q_3 satisfy the following properties:*

1. $S(q_3, q_2, q_1) = S(q_1, q_2, q_3)^{-1}$;
2. *if f is a Möbius transformation mapping $0, 1, \infty$ to q_1, q_2, q_3 respectively, then*

$$S(q_2, q_3, q_1) = u_{f,0} \cdot u_{f,1}^{-1}, \quad S(q_3, q_1, q_2) = -u_{f,1} \cdot u_{f,\infty}^{-1}.$$

   In particular, $S(q_1, q_2, q_3) \cdot S(q_2, q_3, q_1) \cdot S(q_3, q_1, q_2) = -1$;

3. *the three simple ratios $S(q_1, q_2, q_3)$, $S(q_2, q_3, q_1)$ and $S(q_3, q_1, q_2)$ commute with each other.*

Proof. Let f be a Möbius transformation mapping $0, 1, \infty$ to q_1, q_2, q_3 respectively. The matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in \mathrm{SL}(2, \mathcal{H})$$

represents the Möbius transformation $S(q) = q^{-1}$, that fixes 1 and swaps 0 with ∞. Setting $g \coloneqq f \circ S$, we get a Möbius transformation that maps $0, 1, \infty$ to q_3, q_2, q_1 respectively and we can thus use g to compute $S(q_3, q_2, q_1)$. We get

$$S(q_3, q_2, q_1) = u_{g,\infty} \cdot u_{g,0}^{-1} = u_{f,S(\infty)} \cdot u_{S,\infty} \cdot u_{S,0}^{-1} \cdot u_{f,S(0)} = u_{f,0} \cdot u_{f,\infty}^{-1},$$

where the last equality follows from the fact that $u_{S,0} = u_{S,\infty} = 1$. This proves the first point. For the second one, we adopt a similar strategy: the two matrices

$$\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$$

of $\mathrm{SL}(2, \mathcal{H})$ represent the Möbius transformations $T_1(q) = (1-q)^{-1}$ and $T_2(q) = 1 - q^{-1}$ respectively. They satisfy

$$\begin{aligned} T_1(0) = 1, \quad T_1(1) = \infty, \quad T_1(\infty) = 0, \\ T_2(0) = \infty, \quad T_2(1) = 0, \quad T_2(\infty) = 1. \end{aligned}$$

We can thus use $g_1 \coloneqq f \circ T_1$ to compute $S(q_2, q_3, q_1)$ and $g_2 \coloneqq f \circ T_2$ to compute $S(q_3, q_1, q_2)$. The cyclic permutation identities follow immediately by evaluating the left isoclinic components of the differentials of g_1 and g_2 using Proposition 2.2.16.

We now move to the last statement. Thanks to the second point, proving that $S(q_1, q_2, q_3)$ commutes with $S(q_2, q_3, q_1)$ is equivalent to showing that

$$u_{f,0} u_{f,1}^{-1} u_{f,\infty} = u_{f,\infty} u_{f,1}^{-1} u_{f,0}. \tag{2.10}$$

If the q_i's are all finite and p_∞ is the pole of f, then by Proposition 2.2.16 we find

$$u_{f,p} = -u_{f,\infty}(p - p_\infty)^{-1}.$$

We can rewrite the left-hand side and the right-hand side of equation (2.10) respectively as

$$u_{f,\infty}(-p_\infty)^{-1}(1 - p_\infty), \quad u_{f,\infty}(1 - p_\infty)(-p_\infty)^{-1}$$

and now the two terms clearly coincide. The limit cases where one of the q_i's is ∞ are verified similarly, keeping in mind that one of $0, 1, \infty$ will now act as the pole of f, and we must use the corresponding case of Proposition 2.2.16. □

While extending the quaternionic cross-ratios at infinity required a specific choice, the definition of the simple ratios presented here is entirely intrinsic. In fact, it is possible to compute the simple ratios explicitly in terms of the three points q_1, q_2, q_3.

Lemma 2.3.13. *Let q_1, q_2, q_3 be three distinct points of $\mathcal{H}$. Then*

$$S(q_1, q_2, \infty) = (q_2 - q_1)^{-1}, \quad S(q_1, \infty, q_3) = 1, \quad S(\infty, q_2, q_3) = (q_2 - q_3),$$

$$S(q_1, q_2, q_3) = (q_2 - q_3)(q_2 - q_1)^{-1}$$

Proof. The proof follows directly by applying Proposition 2.2.16. □

A simple application of the previous lemma enables us to express the cross-ratios of four points q_1, q_2, q_3, q_4 in $\widehat{\mathcal{H}}$ in terms of the simple ratios.

Corollary 2.3.14. *Let q_1, q_2, q_3, q_4 be four distinct points in $\widehat{\mathcal{H}}$. Then*

$$Q(q_1, q_2, q_3, q_4) = S(q_1, q_2, q_3) \cdot S(q_1, q_4, q_3)^{-1}.$$

We conclude this section by proving a useful result.

Lemma 2.3.15. *Let α, β, γ be three commuting, invertible quaternions such that $\alpha\beta\gamma = -1$. Then there exist $x, y, z \in \mathbb{R}^*$ such that $x\alpha, y\beta, z\gamma$ are the simple ratios of three distinct points. More precisely, there exist $q_1, q_2, q_3 \in \widehat{\mathcal{H}}$ such that*

$$S(q_1, q_2, q_3) = x\alpha, \quad S(q_2, q_3, q_1) = y\beta, \quad S(q_3, q_1, q_2) = z\gamma.$$

Proof. We will explicitly find the real numbers $x, y, z \in \mathbb{R}^*$ and construct three points q_1, q_2, q_3 whose simple ratios are $x\alpha, y\beta, z\gamma$. The verification that these points yield the correct parameters is a mechanical computation which we omit. First, assume that α, β, γ are either all real numbers or all non-real quaternions. Since α and γ commute, we can find two real numbers $x, z \in \mathbb{R}^*$ such that

$$z\gamma = (1 - x\alpha)^{-1}.$$

(In the real case this is trivial; in the non-real case it follows from the fact that α and γ have collinear imaginary parts). Setting $y = (xz)^{-1}$, we find that $x\alpha, y\beta, z\gamma$ are the simple ratios of the points

$$q_1 = 0, \quad q_2 = 1, \quad q_3 = 1 - x\alpha.$$

Assume now that exactly one among α, β, γ is a real number. If $\alpha \in \mathbb{R}$, then $\beta, \gamma \notin \mathbb{R}$ and $\alpha\beta\gamma = -1$ implies $\gamma = -\beta^{-1}\alpha^{-1}$. Setting

$$x = \alpha^{-1}, \quad y = \alpha, \quad z = 1,$$

we find that $x\alpha, y\beta, z\gamma$ are the simple ratios of the points

$$q_1 = \gamma^{-1}, \quad q_2 = \infty, \quad q_3 = 0.$$

The cases where $\beta \in \mathbb{R}$ or $\gamma \in \mathbb{R}$ are handled analogously by permuting the points. We have exhausted all possible cases:

Lastly, notice that the case where exactly two of α, β, γ are real cannot occur: the product of two real numbers and a non-real quaternion is non-real, which contradicts the hypothesis $\alpha\beta\gamma = -1$. □

2.3.2 Relations among cross-ratios

A priori, given four pairwise distinct points $q_1, q_2, q_3, q_4 \in \widehat{\mathcal{H}}$ we have 24 possible quaternionic cross-ratios corresponding to all the possible permutations of $\{1, 2, 3, 4\}$. However, by applying (2.6), (2.7) one sees that

$$Q(q_1, q_4, q_3, q_2) = Q(q_1, q_2, q_3, q_4)^{-1}$$

and hence we may restrict our investigation to the 12 cross-ratios that are obtained from even permutations. For every pair $i, k \in \{1, 2, 3, 4\}$ of distinct indices we write

$$Q_{ik} = Q(q_i, q_j, q_k, q_l)$$

where j, l are determined by the requirement that (i, j, k, l) is an even permutation of $(1, 2, 3, 4)$. We use this notation whenever the four points q_1, q_2, q_3, q_4 are fixed or clear from the context.

The reason for adopting the notation Q_{ik} is that q_i and q_k lie respectively in the two interior and the two exterior factors in the generic formula $Q_{ik} = (q_j - q_k)(q_j - q_i)^{-1}(q_l - q_i)(q_l - q_k)^{-1}$ that holds when none of the points is ∞. The original cross-ratio is $Q_{13} = Q(q_1, q_2, q_3, q_4)$. This section is mainly devoted to understanding the relations between the 12 distinct cross-ratios Q_{ik} as $i, k \in \{1, 2, 3, 4\}$ vary.

Remark 2.3.16. If the q_i's commute (for example, if they all lie in $\widehat{\mathbb{C}}$), we have

$$\begin{aligned} Q &:= Q_{13} = Q_{31} = Q_{42} = Q_{24}, \\ Q' &:= Q_{23} = Q_{32} = Q_{41} = Q_{14}, \\ Q'' &:= Q_{12} = Q_{21} = Q_{43} = Q_{34}. \end{aligned}$$

Furthermore, by straightforward computations one finds

$$Q \cdot Q' \cdot Q'' = -1, \quad Q' = (1-Q)^{-1}, \quad Q'' = 1 - Q^{-1}. \tag{2.11}$$

If we drop the commutativity assumption, the above relations may not be satisfied. For example, we have

$$Q(0,1,i,j) = \frac{1}{2}(1-i-j-k), \quad Q(j,i,1,0) = \frac{1}{2}(1-i+j+k).$$

If two cross-ratios share the third index, like Q_{31} and Q_{41}, we may express one in terms of the other similarly as in the familiar complex case, and we recover the relations (2.11) of Remark 2.3.16.

Proposition 2.3.17. *We have*

$$Q_{jk} = (1-Q_{ik})^{-1}, \quad Q_{lk} = 1 - Q_{ik}^{-1}, \quad Q_{ik}Q_{jk}Q_{lk} = -1 \tag{2.12}$$

for every even permutation i,j,k,l of $1,2,3,4$.

Proof. It suffices to prove the first equality with $(i,j,k,l) = (1,2,3,4)$, and then the second follows by iteration and the third one is a direct algebraic consequence of the first two. If $q_a = \infty$ for some a the equality easily follows from the definitions. For instance

$$\begin{gathered} Q(q_2,q_4,\infty,q_1)^{-1} = (q_1-q_2)^{-1}(q_4-q_2) = (q_1-q_2)^{-1}(q_4-q_1+q_1-q_2) = \\ (q_1-q_2)^{-1}(q_4-q_1)+1 = 1 - Q(q_1,q_2,\infty,q_4). \end{gathered}$$

We then may suppose $q_a \neq \infty$. In Remark 2.3.4, we have seen that the Möbius transformation

$$f(q) = (q_2-q_3)(q_2-q_1)^{-1}(q-q_1)(q-q_3)^{-1}$$

maps q_1,q_2,q_3,q_4 to $0,1,\infty,Q_{13}$ respectively. We now compute $Q(f(q_2),f(q_4),f(q_3),f(q_1))$ in two different ways. Firstly, Proposition 2.3.6 yields

$$Q(f(q_2),f(q_4),f(q_3),f(q_1)) = u_{f,q_3} Q_{23} u_{f,q_3}^{-1} = Q_{23},$$

where the last equality follows from the fact that $u_{f,q_3} = 1$ (see the matrix representation for f given in Remark 2.3.4, and use the fact that q_3 is the pole of f). On the other hand, by directly applying f to the q_a's we get

$$Q(f(q_2),f(q_4),f(q_3),f(q_1)) = Q(1,Q_{13},\infty,0) = (1-Q_{13})^{-1}.$$

The same applies for Q_{43}. □

We shall refer to the equations in (2.12) as *Thurston's relations.*

The main subtlety of the cross-ratios Q_{ik} is that there is no general formula that allows to calculate, say, Q_{31} directly from Q_{42}. Some knowledge of the values of $q_1, \dots, q_4$ is also required.

Proposition 2.3.18. *For every even permutation i, j, k, l of $1, 2, 3, 4$ we have*

$$\begin{aligned} Q_{ki} &= S(q_i, q_j, q_k)^{-1} Q_{ik} S(q_i, q_j, q_k) = S(q_i, q_l, q_k)^{-1} Q_{ik} S(q_i, q_l, q_k), \\ Q_{jl} &= S(q_l, q_j, q_k)^{-1} Q_{ik} S(q_l, q_j, q_k) = S(q_l, q_i, q_k)^{-1} Q_{ik} S(q_l, q_i, q_k), \\ Q_{lj} &= S(q_k, q_l, q_j)^{-1} Q_{ik} S(q_k, q_l, q_j) = S(q_k, q_i, q_j)^{-1} Q_{ik} S(q_k, q_i, q_j), \end{aligned}$$

where $S(q_i, q_j, q_k)$ is the simple ratio of the ordered triple (q_i, q_j, q_k). In particular, the four cross-ratios $Q_{ik}, Q_{ki}, Q_{jl}, Q_{lj}$ all lie in the same conjugacy class.

Proof. Without loss of generality, we may suppose $(i, j, k, l) = (1, 2, 3, 4)$. The first relation follows directly from Corollary 2.3.14 and from the first point of Lemma 2.3.12. We now move to the second relation. Thanks to Proposition 2.3.17, we can write

$$Q_{24} = (1 - Q_{34})^{-1}, \quad Q_{43} = 1 - Q_{13}^{-1}. \tag{2.13}$$

Consider the even permutation $(4, 1, 2, 3)$ of $1, 2, 3, 4$. By applying the first relation, we can relate Q_{34} and Q_{43} as follows:

$$Q_{34} = S(q_4, q_1, q_3)^{-1} Q_{43} S(q_4, q_1, q_3) = S(q_4, q_2, q_3)^{-1} Q_{43} S(q_4, q_2, q_3). \tag{2.14}$$

To ease the notation, we set $a = S(q_4, q_1, q_3)^{-1}$ and $b = S(q_4, q_2, q_3)^{-1}$. By combining (2.13) and (2.14) we obtain

$$Q_{24} = (1 - Q_{34})^{-1} = (1 - a Q_{43} a^{-1})^{-1} = a(1 - Q_{43})^{-1} a^{-1} = a(1 - 1 + Q_{13}^{-1})^{-1} a^{-1} = a Q_{13} a^{-1},$$

and similarly $Q_{24} = b Q_{13} b^{-1}$. This proves the second relation, and the last one can be proved in a similar way. □

Chapter 3

Ideal tetrahedra in hyperbolic 5-space

We represent 5-dimensional hyperbolic space $\mathbb{H}^5$ via the conformal half-space model $\mathcal{H} \times (0, +\infty)$. Recall that the metric tensor at a point $(q, t) \in \mathcal{H} \times (0, +\infty)$ is equal to the Euclidean one rescaled by $1/t^2$.

The sphere at infinity $\partial\mathbb{H}^5$ is identified with the compactification $\widehat{\mathcal{H}}$ of the quaternions. As we have seen in Proposition 1.2.15, the group $\mathrm{Isom}(\mathbb{H}^5)$ of isometries of $\mathbb{H}^5$ is isomorphic to the Möbius group $\mathrm{Möb}(\widehat{\mathbb{R}}^4)$. By Theorem 2.2.14, the subgroup of orientation-preserving Möbius transformations $\mathrm{Möb}^+(\widehat{\mathbb{R}}^4)$ is isomorphic to $\mathrm{PSL}(2, \mathcal{H})$, therefore we obtain the following isomorphism

$$\mathrm{Isom}^+(\mathbb{H}^5) \cong \mathrm{PSL}(2, \mathcal{H}). \tag{3.1}$$

The main focus of this chapter is to study the action of $\mathrm{PSL}(2, \mathcal{H})$ on ideal tetrahedra in $\mathbb{H}^5$.

In the first section, we employ the identification (3.1) to give a (partial) classification of orientation-preserving isometries of $\mathbb{H}^5$. The results are mostly based on the works of Foreman [For04] and Gongopadhyay [GLM26], and a more thorough classification can be found in those works.

We then move to study the action of $\mathrm{PSL}(2, \mathcal{H})$ on oriented ideal tetrahedra and pairs of adjacent, oriented ideal tetrahedra in $\mathbb{H}^5$. This is done by introducing *quaternionic moduli* assigned to the oriented edges of an oriented ideal tetrahedron, which are defined as specific quaternionic cross-ratios of its ideal vertices. We remark that the specific ordering of the vertices in the cross-ratio is completely determined by the chosen oriented edge. Similarly to the complex moduli in dimension three, the quaternionic moduli uniquely detect the isometry class of an ideal tetrahedron. However, due to the non-commutativity of the quaternions, they are not isometry invariants, but rather transform under conjugation.

3.1 Isometries of hyperbolic 5-space

As we have reviewed in Section 1.1.2, the groups of orientation-preserving isometries of $\mathbb{H}^2$ and $\mathbb{H}^3$ are respectively isomorphic to $\mathrm{PSL}(2,\mathbb{R})$ and $\mathrm{PSL}(2,\mathbb{C})$. Given an isometry in these dimensions, it is possible to classify its behavior (elliptic, parabolic, or hyperbolic) through explicit algebraic conditions on the trace of a matrix representing it (see Theorem 1.2.21 and Theorem 1.2.22).

While developing a similar classification of the isometries in $\mathrm{Isom}^+(\mathbb{H}^5) \cong \mathrm{PSL}(2,\mathcal{H})$ is possible (see, for instance, [For04, GLM26]), it is beyond the scope of this thesis. Instead, we present a direct classification criterion for the specific case of orientation-preserving isometries induced by upper-triangular matrices, which is a known result in the literature.

Consider an orientation-preserving isometry f of $\mathbb{H}^5$. As recalled in Section 1.2.4, Brouwer's theorem guarantees a fixed point in $\overline{\mathbb{H}}^5$. If this point lies in $\mathbb{H}^5$, f is elliptic; since we are in odd dimension, Lemma 1.2.19 guarantees that f also has at least one fixed point on the boundary. Therefore, in all cases, f has at least one fixed point $q_0 \in \widehat{\mathcal{H}}$. Up to conjugation, we can assume $q_0 = \infty$. In this case, f is represented by an upper-triangular matrix

$$A = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \mathrm{SL}(2,\mathcal{H}), \tag{3.2}$$

and we can easily give a classification for the action of f based on the coefficients of A.

Remark 3.1.1. Since the Dieudonné determinant of A equals 1, we have $|d| = |a|^{-1}$. In particular, $|a| = |d|$ if and only if a and d are both unit quaternions.

Remark 3.1.2 (Notation)**.** In the previous section, we denoted by T_A the quaternionic Möbius transformation induced by a matrix $A \in \mathrm{SL}(2,\mathcal{H})$. In the remainder of this work, when clear from the context, to ease the notation we will often use A to denote both the matrix and its induced quaternionic Möbius transformation.

Theorem 3.1.3. *Let f be a non-trivial orientation-preserving isometry of $\mathbb{H}^5$ represented by an upper-triangular matrix $A \in \mathrm{SL}(2,\mathcal{H})$, as in (3.2). Then the isometry f is:*

1. *hyperbolic if and only if $|a| \neq 1$;*
2. *parabolic if and only if a and d are conjugate, and either $a \in \mathbb{R}$ or $\bar{a}b \neq bd$;*
3. *elliptic otherwise.*

Proof. By construction, A fixes the point $\infty \in \widehat{\mathcal{H}}$. First, we need to check under which conditions there exists another point $q_0 \in \widehat{\mathcal{H}}$ satisfying $Aq_0 = q_0$. By definition, A acts on $\widehat{\mathcal{H}}$ as the quaternionic Möbius transformation

$$q \mapsto aqd^{-1} + bd^{-1},$$

hence $q_0 \in \widehat{\mathcal{H}}$ is a fixed point for A if and only if

$$aq_0 - q_0 d = -b.$$

This happens precisely when the quaternion $-b$ lies in the image of the linear map $T\colon \mathcal{H} \to \mathcal{H}$ defined by $T(q) = aq - qd$. We study the subspace $K = \ker(T)$ of $\mathcal{H}$. Notice that K contains a non-zero quaternion q_0 if and only if

$$d = q_0^{-1} a q_0,$$

and this happens precisely when a and d are conjugate.

Suppose first that $|a| \neq 1$: by Remark 3.1.1, a and d do not have the same norm and thus they cannot be conjugate. Consequently, the kernel K is trivial, T is an isomorphism and there exists a unique point $q_0 \in \mathcal{H}$ satisfying $T(q_0) = -b$. In this case A fixes the points q_0 and ∞, thus the induced isometry f preserves the entire geodesic l joining them in $\mathbb{H}^5$. Since the global dilation factor of A on the boundary is $|a|^2 \neq 1$, f translates along l without fixing any interior points (see Remark 1.2.14). This proves that f is hyperbolic.

We will henceforth assume $|a| = |d| = 1$. Again, if $\mathrm{Re}(a) \neq \mathrm{Re}(d)$, then a and d cannot be conjugate (because conjugate quaternions have the same real parts). By the same argument as before, T is an isomorphism and there exists a unique fixed point $q_0 \in \mathcal{H}$. Since the dilation factor is $|a|^2 = 1$, the isometry f fixes the entire geodesic connecting q_0 and ∞ pointwise, meaning f is elliptic.

Lastly, suppose that $\mathrm{Re}(a) = \mathrm{Re}(d)$, so that a and d are conjugate. In this case, the kernel K is non-trivial, meaning T is not surjective. We distinguish two cases depending on whether a is real or not.

If $a \in \mathbb{R}$, since a and d are conjugate unit quaternions, we must have $a = d = \pm 1$. As a result, the linear map T identically vanishes, meaning $K = \mathcal{H}$ and $\mathrm{Im}(T) = \{0\}$. Since f is a non-trivial isometry, we necessarily have $b \neq 0$. Therefore, $-b \notin \mathrm{Im}(T)$. This implies that ∞ is the unique fixed point, and f acts as a parabolic isometry (in particular, f is a translation).

When $a \notin \mathbb{R}$, by Corollary 1.5.1, the kernel K has dimension 2. We verify that the image of T coincides with the linear subspace

$$H = \{q \in \mathcal{H} : \bar{a} q = q d\}.$$

Indeed, since a and d are not real, H is a plane. The intersection with K is trivial: if $q \in H \cap K$, then $aq = \bar{a}q$, which is only possible if $q = 0$ since $a \notin \mathbb{R}$. We then check that $\mathrm{Im}(T) \subset H$: if $x = aq_0 - q_0 d$ for some quaternion $q_0 \in \mathcal{H}$, then

$$\bar{a}x = \bar{a}(aq_0 - q_0 d) = q_0 - \bar{a} q_0 d, \quad xd = (aq_0 - q_0 d)d = aq_0 d - q_0 d^2.$$

Since d is a unit quaternion, $d^2 = 2\,\mathrm{Re}(d) d - 1$ and the difference $\bar{a}x - xd$ becomes

$$q_0(1 + d^2) - (\bar{a} + a) q_0 d = 2\,\mathrm{Re}(d) q_0 d - 2\,\mathrm{Re}(a) q_0 d = 0,$$

where the last equality follows from the fact that a and d have the same real part.

By dimensional arguments, we must have $\mathrm{Im}(T) = H$. At this point, f admits a fixed point in $\mathcal{H}$ if and only if $-b \in \mathrm{Im}(T) = H$, which translates to $\bar{a}b = bd$. If this condition holds, there is an entire plane of finite fixed points and the isometry fixes a geodesic pointwise (hence it is elliptic). Otherwise, ∞ is the unique fixed point, making f parabolic. □

Along the proof of Theorem 3.1.3, we have discovered the following result.

Corollary 3.1.4. *Let f be a non-trivial, orientation-preserving elliptic isometry of $\mathbb{H}^5$. Then the fixed-point set of f on the boundary at infinity $\partial\mathbb{H}^5 \cong \widehat{\mathcal{H}}$ is either a pair of distinct points, or a 2-dimensional sphere.*

Example 3.1.5. Choose any angle $\theta \in (0, \pi)$ and consider the two matrices

$$A_1 = \begin{pmatrix} e^{\theta i} & 1+k \\ 0 & e^{\theta j} \end{pmatrix}, \quad A_2 = \begin{pmatrix} e^{\theta i} & 1-k \\ 0 & e^{\theta j} \end{pmatrix}$$

in $\mathrm{SL}(2, \mathcal{H})$. Both matrices share the same diagonal entries, which are conjugate unit quaternions. Consequently, the linear part of both transformations is a simple rotation of angle θ around the invariant plane

$$P_A = \{q \in \mathcal{H} : iq = qj\}. \tag{3.3}$$

A direct check shows that the quaternions $1-k$ and $i+j$ form a basis of P_A. Similarly, the orthogonal complement $P_A^{\perp} = \{q \in \mathcal{H} : iq = -qj\}$ coincides with $\mathrm{Span}(1+k, i-j)$.

We denote by $b_1 := 1+k$ and $b_2 := 1-k$ the upper-right entries of A_1 and A_2 respectively. We compute the expression $e^{-\theta i}b_m - b_m e^{\theta j}$ for both $m = 1, 2$:

$$e^{-\theta i}b_1 - b_1 e^{\theta j} = -\sin\theta(i-j) + \sin\theta(i-j) = 0,$$

$$e^{-\theta i}b_2 - b_2 e^{\theta j} = -\sin\theta(i+j) - \sin\theta(i+j) = -2\sin\theta(i+j).$$

Thanks to the proof of Theorem 3.1.3, the matrix A_1 thus represents an elliptic isometry that fixes a copy of $\mathbb{H}^3$ in $\mathbb{H}^5$. Since $\sin\theta \neq 0$, the second expression is non-zero and A_2 represents a parabolic isometry. Parabolic isometries of this kind, which involve a non-trivial rotational part, are known as *screw-parabolic* isometries.

Remark 3.1.6. The two matrices A_1, A_2 of Example 3.1.5 do not commute. Indeed,

$$\begin{pmatrix} e^{\theta i} & b_1 \\ 0 & e^{\theta j} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} e^{\theta i} & b_2 \\ 0 & e^{\theta j} \end{pmatrix}$$

commute if and only if

$$e^{\theta i}(b_2 - b_1) = (b_2 - b_1)e^{\theta j},$$

which is equivalent to asking that $b_2 - b_1 \in P_A$ (the plane defined in (3.3)). But $b_2 - b_1 = -2k \notin P_A$. However, we can easily construct an orientation-preserving parabolic

isometry represented by a matrix $A_3 \in \mathrm{SL}(2,\mathcal{H})$ that commutes with A_1. This is done by choosing a quaternion b_3 such that $b_3 - b_1 \in P_A$ while $b_3 \notin P_A^{\perp}$. A direct verification shows that $b_3 = 1+i+j+k$ satisfies both requirements, yielding a parabolic matrix

$$A_3 = \begin{pmatrix} e^{\theta i} & 1+i+j+k \\ 0 & e^{\theta j} \end{pmatrix}$$

that commutes with A_1. This phenomenon does not occur in dimensions 2 and 3, where non-trivial parabolic isometries and elliptic isometries cannot commute.

Remark 3.1.7. Every matrix $A \in \mathrm{SL}(2,\mathcal{H})$ can be uniquely written as $A = A_0 + A_1 j$, where A_0 and A_1 are 2×2 complex matrices. Similarly to what we did in Section 2.2, we can associate to A the complex matrix

$$\psi(A) = \begin{pmatrix} A_0 & A_1 \\ -\bar{A}_1 & \bar{A}_0 \end{pmatrix} \in \mathrm{SL}(4,\mathbb{C}).$$

The classification of the isometry induced by A on $\mathbb{H}^5$ can be deduced directly from the complex matrix $\psi(A)$. One can verify that each fixed point of A on $\widehat{\mathcal{H}}$ corresponds to either a conjugate pair of complex eigenvalues or to a real eigenvalue with algebraic multiplicity at least 2 for the matrix $\psi(A)$ (see [Rod14]). Consequently, the isometry induced by A is:

- hyperbolic if and only if all the eigenvalues of $\psi(A)$ have norm different from 1 (notice that this implies that $\psi(A)$ is diagonalizable over $\mathbb{C}$);
- elliptic if and only if $\psi(A)$ is diagonalizable over $\mathbb{C}$ and all its eigenvalues have norm 1;
- parabolic if and only if $\psi(A)$ is not diagonalizable over $\mathbb{C}$ and all its eigenvalues have norm 1.

This complex representation is widely used in the literature: for instance, Foreman [For04] and Gongopadhyay [GLM26] provided a full classification of the isometries directly in terms of the coefficients of the characteristic polynomial of $\psi(A)$.

3.1.1 Geometric description of the isometries

In Theorem 3.1.3, we have proved that we can determine whether an orientation-preserving isometry of $\mathbb{H}^5$ represented by a matrix

$$A = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \mathrm{SL}(2,\mathcal{H})$$

is hyperbolic, parabolic or elliptic by directly studying the coefficients of A. We now investigate in more detail the geometric behavior of these maps on the boundary $\partial\mathbb{H}^5 = \widehat{\mathcal{H}}$. Notice that, since A fixes ∞, it leaves $\mathcal{H}$ invariant, acting on it as an affine similarity by Proposition 1.5.7.

Elliptic and hyperbolic isometries both fix at least two points on the boundary at infinity $\partial\mathbb{H}^5 = \widehat{\mathcal{H}}$. Since A already fixes ∞, up to conjugating by a translation we can assume that the other fixed point is $0 \in \mathcal{H}$. This means that A is a diagonal matrix and its action reduces to $q \mapsto aqd^{-1}$, which is a linear similarity on $\mathcal{H}$. We distinguish two cases depending on the moduli of the diagonal entries.

The hyperbolic case By Theorem 3.1.3, the matrix A represents a hyperbolic isometry if and only if $|a| \neq 1$ (which also implies $|d| \neq 1$ by Remark 3.1.1). Since $|a| \neq |d|$, A scales the norm of every quaternion by a strictly positive factor $\lambda = |a||d|^{-1} \neq 1$. Based on the analysis of Section 1.5.2, we distinguish three cases:

- if exactly one of a and d is real, $A|_{\mathcal{H}}$ is the composition of a dilation with an isoclinic rotation. As a vector space, $\mathcal{H}$ decomposes as the direct sum of two orthogonal, invariant planes. On each of them, A acts as a planar similarity with the same scaling factor λ and with rotation angles that are equal up to a sign;

- if both a and d are non-real, $A|_{\mathcal{H}}$ is the composition of a dilation with either a double or a simple rotation. The space again decomposes into two orthogonal, invariant planes; on each of them, the map acts as a 2-dimensional similarity, but with rotation angles having different absolute values. In particular, in the case of a simple rotation, the angle on one of the two planes is zero, meaning the action on that specific plane is a pure dilation;

- lastly, if both a and d are real, the restriction to $\mathcal{H}$ is simply a dilation.

Sometimes, in the literature, the first two classes of maps are called *loxodromic*, while only the isometries in the last class are called hyperbolic.

The elliptic case By Theorem 3.1.3, the matrix A represents an elliptic isometry if and only if $|a| = 1$ (which also implies $|d| = 1$). In this case, A acts on $\mathcal{H}$ as a Euclidean, orientation-preserving linear isometry (see Section 1.5.2). There are two possibilities:

- if a and d are not conjugate, the restriction to $\mathcal{H}$ is either an isoclinic rotation or a double rotation. In both cases, the unique fixed points at infinity are 0 and ∞;

- if a and d are conjugate, the restriction to $\mathcal{H}$ is a simple rotation: in particular, the vector space

$$K = \{q \in \mathcal{H} : aq = qd\} \tag{3.4}$$

  is a plane by Theorem 1.5.1, and A fixes K pointwise. Hence, the fixed-point set on $\widehat{\mathcal{H}}$ is a 2-sphere $(K \cup \{\infty\})$, meaning that the isometry fixes a totally geodesic $\mathbb{H}^3$ inside $\mathbb{H}^5$.

The parabolic case We now move to parabolic isometries. In this case, $b \neq 0$, so A is no longer a diagonal matrix. We furthermore suppose that a and d are conjugate unit quaternions (otherwise A is either elliptic or hyperbolic). The restriction to $\mathcal{H}$ is the Euclidean isometry

$$q \mapsto (aq + b)d^{-1} = aqd^{-1} + bd^{-1}. \tag{3.5}$$

We know by Theorem 3.1.3 that A represents a parabolic isometry of $\mathbb{H}^5$ in exactly two cases, depending on its linear part. We thus get two classes of parabolic isometries:

- *translations*, which occur when $a = d = \pm 1$. In this case, the linear part is trivial;
- *screw-parabolic*, which occur when $a, d \notin \mathbb{R}$ and $\bar{a}b \neq bd$. In this case, the linear part $q \mapsto aqd^{-1}$ is a simple rotation around the plane K defined in (3.4). Notice that A translates the plane K by the vector bd^{-1}, and the condition $\bar{a}b \neq bd$ holds if and only if bd^{-1} is not orthogonal to K. This ensures that the translation along the rotation plane is non-trivial, yielding a screw motion with no fixed points.

The following result shows that the boundary action of every screw-parabolic isometry is a rotation around a uniquely determined affine plane, followed by a translation strictly parallel to it.

Lemma 3.1.8. *Let $f \in \mathrm{Isom}^+(\mathbb{H}^5)$ be a screw-parabolic isometry represented by the upper-triangular matrix*

$$A = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \mathrm{SL}(2, \mathcal{H}),$$

and let K be the plane defined in (3.4). There exists a unique vector $y_0 \in K^\perp$ such that:

1. *the affine plane $y_0 + K$ is A-invariant;*
2. *we have $\pi_{K^\perp}(bd^{-1}) = y_0 - ay_0d^{-1}$.*

In particular, A acts as a rotation around the affine plane $y_0 + K$ composed with a translation strictly parallel to K.

Proof. Let $T : \mathcal{H} \to \mathcal{H}$ denote the linear map

$$T(x) = x - axd^{-1}.$$

Since a and d are both non real, the kernel of T is precisely the plane K, and its image is $K^\perp$. As a consequence, there exists a unique vector $y_0 \in K^\perp$ such that

$$\pi_{K^\perp}(bd^{-1}) = T(y_0) = y_0 - ay_0d^{-1},$$

proving (2). To prove (1), notice that for every $k \in K$ we have

$$f(y_0 + k) = ay_0d^{-1} + k + \pi_K(bd^{-1}) + \pi_{K^\perp}(bd^{-1}) = \underbrace{k + \pi_K(bd^{-1})}_{\in K} + y_0 \in y_0 + K.$$

The proof is complete. □

3.2 Ideal tetrahedra in hyperbolic 5-space

As we have recalled in Section 1.4, if T is an ideal tetrahedron in $\mathbb{H}^3$, we can naturally assign a complex modulus to each of its edges. These complex moduli actually coincide with the complex cross-ratios of the ideal vertices of T in $\widehat{\mathbb{C}}$, ordered appropriately (see Remark 1.4.15), and we know that one of them determines the other five. They are orientation-preserving isometry invariants, and they uniquely determine T up to isometry.

Here we want to carry out a similar assignment for *oriented* edges of oriented ideal tetrahedra in $\mathbb{H}^5$. This is achieved by assigning to each oriented edge of such a tetrahedron T a quaternionic cross-ratio of the ideal vertices of T in $\widehat{\mathcal{H}}$, ordered in an appropriate way. Almost all the results of this section follow from the properties of quaternionic cross-ratios introduced in the previous chapter.

3.2.1 Quaternionic moduli of tetrahedra

Let T be an oriented ideal tetrahedron in $\mathbb{H}^5$. By identifying the boundary at infinity of $\mathbb{H}^5$ with the compactification of the quaternions, the ideal vertices of T are elements q_1, q_2, q_3, q_4 of $\widehat{\mathcal{H}}$. We assume that the ideal vertices are ordered according to the orientation of T.

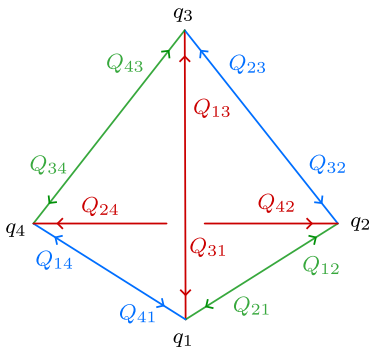


Figure 3.1: An oriented ideal tetrahedron T with its quaternionic cross-ratios $Q_{ik}(T)$. Pairs of opposite edges are drawn with the same colour, as their cross-ratios lie in the same similarity class.

We assign a quaternion to each oriented edge of T as follows. For every pair of indices $i \neq k \in \{1, 2, 3, 4\}$, let e_{ik} be the oriented edge from q_i to q_k. We assign to e_{ik} the quaternion

$$Q_{ik}(T) \coloneqq Q(q_i, q_j, q_k, q_l),$$

where $j, l \in \{1, 2, 3, 4\}$ are chosen so that the permutation $i, j, k, l \mapsto 1, 2, 3, 4$ is even. See Figure 3.1. We will sometimes call $Q_{ik}(T)$ the *quaternionic modulus* of T assigned to the edge e.

Based on the analysis of the previous chapter, we know that the cross-ratios $Q_{ik}(T)$

satisfy the relations established in Proposition 2.3.17 and Proposition 2.3.18. We summarize them in the following statement.

Lemma 3.2.1. *Let T be an oriented ideal tetrahedron in $\mathbb{H}^5$ with ideal vertices q_1, q_2, q_3, q_4 as above. For every even permutation i, j, k, l of $1, 2, 3, 4$ we have:*

1. *the quaternionic moduli assigned to the three edges entering the ideal vertex q_k satisfy the following relations, which we call* Thurston relations*:*

$$Q_{jk}(T) = (1 - Q_{ik}(T))^{-1}, \quad Q_{lk}(T) = 1 - Q_{ik}(T)^{-1}, \quad Q_{ik}(T)Q_{jk}(T)Q_{lk}(T) = -1;$$

2. *the four quaternionic moduli $Q_{ik}(T), Q_{ki}(T), Q_{jl}(T)$, and $Q_{lj}(T)$, corresponding to a pair of opposite edges with their different orientations, all lie in the same similarity class; more precisely:*

$$\begin{aligned} Q_{ki}(T) &= S(q_i, q_j, q_k)^{-1} Q_{ik}(T) S(q_i, q_j, q_k) = S(q_i, q_l, q_k)^{-1} Q_{ik}(T) S(q_i, q_l, q_k), \\ Q_{jl}(T) &= S(q_l, q_j, q_k)^{-1} Q_{ik}(T) S(q_l, q_j, q_k) = S(q_l, q_i, q_k)^{-1} Q_{ik}(T) S(q_l, q_i, q_k), \\ Q_{lj}(T) &= S(q_k, q_l, q_j)^{-1} Q_{ik}(T) S(q_k, q_l, q_j) = S(q_k, q_i, q_j)^{-1} Q_{ik}(T) S(q_k, q_i, q_j), \end{aligned}$$

where we recall that $S(q_i, q_j, q_k)$ is the simple ratio (see Definition 2.3.10).

As seen in Remark 1.4.15, complex cross-ratios assigned to the edges of an ideal tetrahedron in $\mathbb{H}^3$ are invariant under orientation-preserving isometries. Due to the lack of commutativity, this is no longer true for their quaternionic counterparts in $\mathbb{H}^5$. Nevertheless, applying an orientation-preserving isometry to an oriented ideal tetrahedron in $\mathbb{H}^5$ changes its quaternionic moduli by conjugation. This is a direct consequence of Proposition 2.3.6, which we restate below, adapted to our current setting.

Proposition 3.2.2. *Let T be an oriented ideal tetrahedron in $\mathbb{H}^5$ with ideal vertices q_1, q_2, q_3, q_4, as above. Let $f \in \mathrm{Isom}^+(\mathbb{H}^5)$ be an orientation-preserving isometry. Then*

$$Q_{ik}(f(T)) = u_{f,q_k} \cdot Q_{ik}(T) \cdot u_{f,q_k}^{-1},$$

where u_{f,q_k} is the left isoclinic component of the differential of f at the point q_k.

The following results are obtained by a direct translation of Remark 2.3.4 and Theorem 2.3.7.

Corollary 3.2.3. *Let T be an oriented ideal tetrahedron in $\mathbb{H}^5$ with ideal vertices q_1, q_2, q_3, q_4, ordered according to the orientation. There exists an orientation-preserving isometry $f \in \mathrm{Isom}^+(\mathbb{H}^5)$ mapping q_1, q_2, q_3, q_4 to $0, 1, \infty, Q_{13}(T)$, respectively.*

Remark 3.2.4. Let T and f be as in Corollary 3.2.3. Since the conjugacy class of $Q_{13}(T)$ in $\mathcal{H}$ contains a unique complex number $z \in \mathbb{C}$ with strictly positive imaginary part, we can post-compose f with a conjugation to obtain an ideal tetrahedron T' with vertices $0, 1, \infty, z$. The new tetrahedron T' lies in the standard copy of $\mathbb{H}^3 \subset \mathbb{H}^5$ (induced by the inclusion $\mathbb{C} \subset \mathcal{H}$). For every pair of indices $i \neq k \in \{1, 2, 3, 4\}$, the quaternionic modulus $Q_{ik}(T')$ of the new tetrahedron T' coincides with the complex modulus $z_{ik}(T')$. By Proposition 3.2.2, each quaternionic modulus $Q_{ik}(T)$ is hence conjugate to the corresponding complex modulus $z_{ik}(T')$.

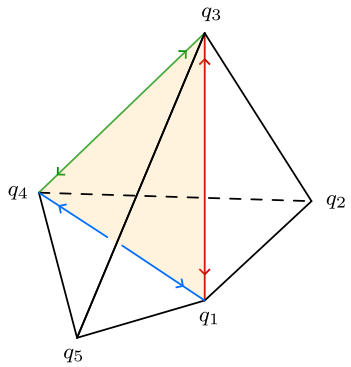


Figure 3.2: Two tetrahedra T_1, T_2 adjacent along the face q_1, q_3, q_4.

Corollary 3.2.5. *Let T and T' be two oriented ideal tetrahedra in $\mathbb{H}^5$, with vertices q_1, q_2, q_3, q_4 and q'_1, q'_2, q'_3, q'_4 respectively, ordered according to the orientations. The following are equivalent:*

1. *there exists an orientation-preserving isometry $f \in \mathrm{Isom}^+(\mathbb{H}^5)$ such that $f(q_i) = q'_i$ for every $i = 1, 2, 3, 4$;*

2. *there exists an invertible quaternion $a \in \mathcal{H}^*$ such that $Q_{13}(T') = aQ_{13}(T)a^{-1}$.*

Remark 3.2.6. Let T and T' be as in Corollary 3.2.5. If $Q_{13}(T') = aQ_{13}(T)a^{-1}$ for some $a \in \mathcal{H}^*$, then by applying Lemma 3.2.1 we know that the cross-ratios $Q_{ik}(T)$ and $Q_{ik}(T')$ are conjugate for every $i \neq k \in \{1, 2, 3, 4\}$.

Unlike in the complex case, the isometry of Corollary 3.2.5 is not unique. For example, any isometry f of the form $q \mapsto aqa^{-1}$ with $a \in \mathbb{C}^*$ fixes $0, 1, \infty$ and $z \in \mathbb{C}$, because a and z commute. This shows that prescribing the image of an orientation-preserving isometry on four points in $\partial\mathbb{H}^5$ is not enough to determine it completely. We therefore extend our investigation to pairs of adjacent tetrahedra, whose ideal vertices provide a configuration of five points.

3.2.2 Pairs of adjacent tetrahedra

Throughout the rest of this chapter, let T_1 and T_2 be two ideal tetrahedra in $\mathbb{H}^5$ sharing a face F. Let $q_1, q_3, q_4 \in \widehat{\mathcal{H}}$ be the ideal vertices of F, and let $q_2, q_5 \in \widehat{\mathcal{H}}$ be the vertices opposite to F in T_1 and T_2 respectively, as illustrated in Figure 3.2. We orient T_1 and T_2 such that the induced orientations on the common face F are opposite. This is achieved by ordering the ideal vertices of T_1 as q_1, q_2, q_3, q_4, and those of T_2 as q_1, q_4, q_3, q_5. From now on, this coherent orientation convention will be implicitly assumed whenever we say that T_1 and T_2 are two oriented adjacent ideal tetrahedra.

We want to investigate the relations between the quaternionic moduli assigned to the oriented common edges of T_1 and T_2. We have seen that the quaternionic modulus $Q_{ik}(T_1)$ of an edge is conjugate to the quaternionic modulus $Q_{ki}(T_1)$ assigned to the

same edge with opposite orientation. Similarly, $Q_{ik}(T_2)$ is conjugate to $Q_{ki}(T_2)$. It turns out that the conjugation can be performed simultaneously for the quaternionic moduli assigned to common edges, as stated in the following result.

Lemma 3.2.7. *Let T_1 and T_2 be two oriented adjacent ideal tetrahedra in $\mathbb{H}^5$ as in Figure 3.2. For $s \in \{1,2\}$, the following relations hold*

$$\begin{aligned} Q_{31}(T_s) &= S(q_3,q_4,q_1)Q_{13}(T_s)S(q_3,q_4,q_1)^{-1},\\ Q_{14}(T_s) &= S(q_1,q_3,q_4)Q_{41}(T_s)S(q_1,q_3,q_4)^{-1},\\ Q_{43}(T_s) &= S(q_4,q_1,q_3)Q_{34}(T_s)S(q_4,q_1,q_3)^{-1}. \end{aligned}$$

Proof. We first apply the second point of Lemma 3.2.1 with $(i,j,k,l) = (1,2,3,4)$ to the quaternionic moduli of T_1. This yields

$$Q_{31}(T_1) = S(q_1,q_4,q_3)^{-1}Q_{13}(T_1)S(q_1,q_4,q_3).$$

Recall that the simple ratio satisfies the identity $S(q_1,q_4,q_3)^{-1} = S(q_3,q_4,q_1)$. Substituting this into the previous equation yields the first formula for T_1. Similarly, applying the same result with $(i,j,k,l) = (1,4,3,5)$ to the quaternionic moduli of T_2 yields exactly the same conjugating element:

$$Q_{31}(T_2) = S(q_1,q_4,q_3)^{-1}Q_{13}(T_2)S(q_1,q_4,q_3),$$

which proves the first formula. The relations for the remaining common edges follow analogously. □

Remark 3.2.8. The simple ratios appearing in the previous lemma commute and their product equals -1 (see Lemma 2.3.12). Moreover, we point out that they depend only on the vertices of the common face.

We are mainly interested in pairs of oriented adjacent ideal tetrahedra that do not lie in the same hyperbolic subspace of dimension three. This property can be detected by comparing cross-ratios assigned to common edges.

Lemma 3.2.9. *Let T_1 and T_2 be two oriented adjacent ideal tetrahedra in $\mathbb{H}^5$ as above. The following are equivalent:*

1. *T_1 and T_2 lie in the same three-dimensional hyperbolic subspace;*
2. *the quaternionic moduli $Q_{13}(T_1)$ and $Q_{13}(T_2)$ commute;*
3. *the quaternionic moduli assigned to the common edges commute; that is, for every $i,k \in \{1,3,4\}$, $Q_{ik}(T_1)$ and $Q_{ik}(T_2)$ commute.*

Proof. Let f be the orientation-preserving isometry of Corollary 3.2.3, which maps q_1,q_2,q_3,q_4 to $0,1,\infty,Q_{13}(T_1)$ respectively. Under this map, the tetrahedron $f(T_1)$ lies in the 3-dimensional subspace

$$\mathrm{Span}(1,Q_{13}(T_1)) \times \mathbb{R}_+.$$

Since T_1 and T_2 share the face with vertices q_1, q_3, q_4, they lie in the same 3-dimensional subspace if and only if the fifth vertex, $f(q_5)$, lies in $\mathrm{Span}(1, Q_{13}(T_1))$. This holds precisely when $f(q_5)$ commutes with $Q_{13}(T_1)$. By definition, we can express the mapped vertex $f(q_5)$ as a product of cross-ratios:

$$f(q_5) = Q(q_1, q_2, q_3, q_5) = Q(q_1, q_2, q_3, q_4)Q(q_1, q_4, q_3, q_5) = Q_{13}(T_1)Q_{13}(T_2),$$

where the intermediate equality is a direct check. Therefore, the product $Q_{13}(T_1)Q_{13}(T_2)$ commutes with $Q_{13}(T_1)$ if and only if $Q_{13}(T_1)$ and $Q_{13}(T_2)$ commute. This establishes the equivalence between (1) and (2). Finally, the equivalence of (2) and (3) follows directly from Lemma 3.2.1. □

If two oriented adjacent ideal tetrahedra T_1, T_2 are not contained in the same copy of $\mathbb{H}^3$, we say that they are in *general position*.

Corollary 3.2.10. *Every element of* $\mathrm{Isom}^+(\mathbb{H}^5)$ *is uniquely determined by its values on five the ideal vertices of two oriented adjacent ideal tetrahedra in general position.*

Proof. By the previous lemma (Lemma 3.2.9), up to the action of $\mathrm{PSL}(2, \mathcal{H})$, we can assume that the five ideal vertices are $0, 1, \infty, p, q$ where p and q do not commute. If f is an orientation-preserving isometry of $\mathbb{H}^5$ that fixes all five points, then f has the form $x \mapsto axa^{-1}$, where a is a quaternion that commutes with both p and q. Proposition 1.5.4 now implies that a is a real number, so f is the identity. The proof is complete. □

Theorem 3.2.11. *Let T_1, T_2 be two oriented adjacent ideal tetrahedra as in Figure 3.2, and consider another pair T_1', T_2' with vertices q_1', q_2', q_3', q_4' and q_1', q_4', q_3', q_5' respectively. The following are equivalent:*

1. *there exists an orientation-preserving isometry $f \in \mathrm{Isom}^+(\mathbb{H}^5)$ mapping q_i to q_i' for every $i = 1, \dots, 5$;*
2. *the pair of quaternionic moduli $(Q_{13}(T_1), Q_{13}(T_2))$ is simultaneously conjugate to the pair $(Q_{13}(T_1'), Q_{13}(T_2'))$;*
3. *for every $i, k \in \{1, 3, 4\}$, the pair of quaternionic moduli $(Q_{ik}(T_1), Q_{ik}(T_2))$ is simultaneously conjugate to the pair $(Q_{ik}(T_1'), Q_{ik}(T_2'))$.*

Moreover, if both T_1, T_2 and T_1', T_2' are in general position, then the isometry f is unique.

Proof. Let $f \in \mathrm{Isom}^+(\mathbb{H}^5)$ be an orientation-preserving isometry mapping $q_1, \dots, q_5$ to $q_1', \dots, q_5'$ respectively. By Proposition 2.3.6, for every pair of indices $i \neq k \in \{1, 3, 4\}$ we have

$$Q_{ik}(T_1') = u_{f,q_k} Q_{ik}(T_1) u_{f,q_k}^{-1}, \quad Q_{ik}(T_2') = u_{f,q_k} Q_{ik}(T_2) u_{f,q_k}^{-1}.$$

This shows that (1) implies (2) and (3).

We now assume that there exists $a \in \mathcal{H}^*$ that simultaneously conjugates the two quaternionic moduli $Q_{13}(T_1), Q_{13}(T_2)$ to $Q_{13}(T_1'), Q_{13}(T_2')$. By the very definition of the quaternionic cross-ratio, we get

$$Q(q_1', q_2', q_3', q_5') = Q_{13}(T_1')Q_{13}(T_2') = aQ_{13}(T_1)Q_{13}(T_2)a^{-1} = aQ(q_1, q_2, q_3, q_5)a^{-1}.$$

Theorem 2.3.7 now shows that (2) implies (1). The equivalence between (2) and (3) is a consequence of Lemma 3.2.1 and Lemma 2.3.12. Finally, if both T_1, T_2 and T_1', T_2' are in general position, then f is uniquely determined by the images of the five points $q_1, \dots, q_5$ by Corollary 3.2.10. □

Normal form Recall that, by Definition 2.1.10, a pair of non-commuting quaternions (p, q) is in normal form if

$$p \in \mathbb{C}^+, \quad \langle q, j\rangle > 0, \quad \langle q, k\rangle = 0.$$

Consider two oriented adjacent ideal tetrahedra T_1, T_2 with vertices $0, 1, \infty, z$ and $0, z, \infty, w$ respectively. Suppose that $z \in \mathbb{C}^+$, and assume moreover that T_1 and T_2 are in general position. This implies that $w \in \mathcal{H}$ is a non-complex quaternion.

Under these assumptions, the quaternionic moduli assigned to the edge from 0 to ∞ in T_1 and T_2 are respectively given by

$$Q(0, 1, \infty, z) = z, \quad Q(0, z, \infty, w) = z^{-1}w.$$

Definition 3.2.12. The pair of adjacent ideal tetrahedra (T_1, T_2) as above is in *normal form* if the pair of non-commuting quaternions $(z, z^{-1}w)$ is in normal form.

Lemma 3.2.13. *Consider two oriented adjacent ideal tetrahedra T_1, T_2 in $\mathbb{H}^5$ that are in normal form. Then the the pair (T_1, T_2) is orientation-preservingly isometric to a pair of adjacent tetrahedra in normal form.*

Proof. Let q_1, q_2, q_3, q_4 be the ideal vertices of T_1, and q_1, q_4, q_3, q_5 be the ideal vertices of T_2, as in Figure 3.2. By Lemma 3.2.9, the quaternionic moduli $Q_{13}(T_1)$ and $Q_{13}(T_2)$ do not commute. As shown in Lemma 2.1.11, this implies the existence of a quaternion $a \in \mathcal{H}^*$ that conjugates the pair $(Q_{13}(T_1), Q_{13}(T_2))$ to a pair of non-commuting quaternions (p, q) in normal form.

Consider now the two adjacent, ideal tetrahedra T_1' and T_2' with vertices $0, 1, \infty, p$ and $0, 1, \infty, pq$ respectively. We look at the quaternionic moduli assigned to the oriented edge connecting 0 and ∞ in the two tetrahedra. For T_1', we simply get $Q(0, 1, \infty, p) = p$, while for T_2' we get

$$Q(0, p, \infty, pq) = p^{-1}pq = q,$$

which is precisely the pair (p, q). In particular, the pair T_1', T_2' is in normal form.

Since, by construction, $(Q_{13}(T_1), Q_{13}(T_2))$ and (p, q) are simultaneously conjugate pairs, Theorem 3.2.11 implies that there exists an orientation-preserving isometry $f \in \mathrm{Isom}^+(\mathbb{H}^5)$ such that

$$f(q_1) = 0, \quad f(q_2) = 1, \quad f(q_3) = \infty, \quad f(q_4) = p, \quad f(q_5) = pq.$$

The proof is complete. □

3.2.3 A key lemma

Consider an oriented, ideal triangle F in $\mathbb{H}^5$ with vertices q_1, q_3, q_4. Suppose that for every distinct $i, k \in \{1, 3, 4\}$ we are given a non-real quaternion q_{ik}, one for each of the six oriented edges of F. We want to understand under which conditions the parameters q_{ik} arise as the quaternionic cross-ratios of an ideal tetrahedron T having F as a face.

Lemma 3.2.14. *Suppose that the quaternions q_{ik} satisfy the following conditions:*

1. *the Thurston relations, i.e.*

$$q_{43} = 1 - q_{13}^{-1}, \quad q_{34} = (1 - q_{14})^{-1}, \quad q_{31} = 1 - q_{41}^{-1}; \tag{3.6}$$

2. *they are conjugate by the simple ratios of the vertices of F. More precisely, for $\alpha = S(q_3, q_4, q_1)$, $\beta = S(q_1, q_3, q_4)$ and $\gamma = S(q_4, q_1, q_3)$, we have*

$$q_{31} = \alpha q_{13} \alpha^{-1}, \quad q_{14} = \beta q_{41} \beta^{-1}, \quad q_{43} = \gamma q_{34} \gamma^{-1}. \tag{3.7}$$

Then there exists, and is unique, a point $q_2 \in \widehat{\mathcal{H}}$ such that the ideal tetrahedron T with vertices q_1, q_2, q_3, q_4 satisfies $Q_{ik}(T) = q_{ik}$.

Proof. We start by defining q_2 in such a way that

$$q_{13} = Q(q_1, q_2, q_3, q_4)$$

This can always be done explicitly by looking at the definition of the cross-ratio. For example, if the q_i's are all finite then

$$q_2 = (q_{13} \cdot \alpha^{-1} - 1)^{-1}(q_{13} \cdot \alpha^{-1} \cdot q_1 - q_3)$$

because $\alpha = S(q_3, q_4, q_1)$. We call T the ideal tetrahedron with vertices q_1, q_2, q_3, q_4. It remains to check that $Q_{ik}(T) = q_{ik}$ also for the other indices, and this follows by combining hypotheses (3.6) and (3.7) with the internal relations for the cross-ratios of the tetrahedron T (see Lemma 3.2.1 and Lemma 3.2.7). □

Suppose now that we are given two sets of non-real quaternions, $q_{ik}^{(1)}$ and $q_{ik}^{(2)}$, assigned to the oriented edges of F. If, for example, the $q_{ik}^{(1)}$ satisfy the hypotheses of Lemma 3.2.14, we can constructs a single ideal tetrahedron having F as a face and the $q_{ik}^{(1)}$ as (some) quaternionic moduli. However, since the conjugating simple ratios depend entirely on the base triangle, we can apply this construction simultaneously to build a pair of adjacent tetrahedra sharing the face F. We only need to ensure that the $q_{ik}^{(2)}$ satisfy the orientation-reversed Thurston relations to account for the fact that the two tetrahedra must induce opposite orientations to the shared face F.

Corollary 3.2.15. *If the parameters $q_{ik}^{(1)}$ satisfy the conditions of Lemma 3.2.14, and the parameters $q_{ik}^{(2)}$ satisfy the same conjugation relations (3.7) alongside the orientation-reversed Thurston relations*

$$q_{43}^{(2)} = (1 - q_{13}^{(2)})^{-1}, \quad q_{34}^{(2)} = 1 - (q_{14}^{(2)})^{-1}, \quad q_{31}^{(2)} = (1 - q_{41}^{(2)})^{-1}, \tag{3.8}$$

then they uniquely determine two adjacent ideal tetrahedra T_1 and T_2 sharing the face F. More precisely, there exist two unique points $q_2, q_5 \in \widehat{\mathcal{H}}$ such that the oriented adjacent ideal tetrahedra T_1, T_2 with vertices q_1, q_2, q_3, q_4 and q_1, q_4, q_3, q_5 satisfy

$$Q_{ik}(T_1) = q_{ik}^{(1)} \quad \text{and} \quad Q_{ik}(T_2) = q_{ik}^{(2)}$$

for every distinct $i, k \in \{1, 3, 4\}$.

3.3 Quaternionic angles

In Section 1.4 of Chapter 1, we defined the complex moduli assigned to the edges of an oriented, ideal tetrahedron T in $\mathbb{H}^3$ as follows:

- suppose that T has ideal vertices $x_1, x_2, x_3, x_4 \in \widehat{\mathbb{C}}$, ordered such that they induce the negative orientation on T;
- there exists a unique orientation-preserving isometry in $\mathrm{PSL}(2, \mathbb{C})$ mapping x_1, x_2, x_3 to $0, 1, \infty$;
- the complex modulus assigned to the edge e joining x_1 and x_3 is the image of the fourth point x_4, which is a (uniquely determined) complex number $z \in \mathbb{C}$ with $\mathrm{Im}(z) > 0$.

If we truncate the mapped tetrahedron T' with vertices $0, 1, \infty, z$ with a small enough horosphere centered at ∞, we get a Euclidean triangle Δ, and the complex modulus z is equal to the complex angle of Δ in the corner corresponding to e

While it is possible to define the *quaternionic angles* of a triangle in $\mathcal{H}$, we cannot employ them to give a definition for the quaternionic moduli in a way that is as direct as in the 3-dimensional case. The main reason for this is that the group $\mathrm{PSL}(2, \mathcal{H})$ does not act freely on ordered triples of points, but rather on 5-tuples (see Corollary 3.2.10).

Still, the quaternionic angles play an important role in our work. Recall from Section 2.3 that extending the definition of the cross-ratio

$$Q(q_1, q_2, q_3, q_4)$$

of four points in $\mathcal{H}$ to the case $q_3 = \infty$ required making a choice, as only the similarity class of the limit as q_3 goes to infinity is well-defined. This choice can be explained using the quaternionic angles.

Definition 3.3.1. Let $\Delta \subset \mathcal{H}$ be an oriented triangle with vertices A, B and C such that (A, B, C) induces the given orientation on Δ. We define the *quaternionic angles* of Δ at each vertex as

$$q_A(\Delta) = (B-A)^{-1}(C-A), \quad q_B(\Delta) = (C-B)^{-1}(A-B), \quad q_C(\Delta) = (A-C)^{-1}(B-C).$$

When it is clear from the context, we will drop the dependency on Δ, and just write q_A, q_B and q_C in place of $q_A(\Delta)$, $q_B(\Delta)$ and $q_C(\Delta)$.

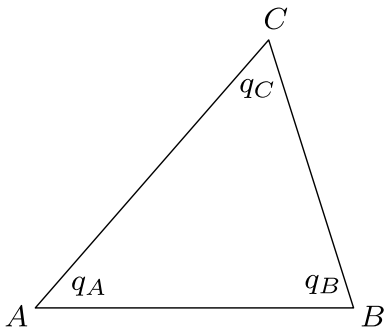


Figure 3.3: An oriented triangle $\Delta \subset \mathcal{H}$ with its quaternionic angles.

Lemma 3.3.2. *The quaternionic angles defined above commute with each other and satisfy the relations*

$$q_A q_B q_C = -1, \quad q_B = (1-q_A)^{-1}, \quad q_C = (1-q_B)^{-1} = 1 - q_A^{-1}. \tag{3.9}$$

Proof. The equalities

$$q_B^{-1} = (A-B)^{-1}(C-B) = 1 + (A-B)^{-1}(C-A) = 1 - q_A,$$
$$q_C = (A-C)^{-1}(B-C) = 1 + (A-C)^{-1}(B-A) = 1 - q_A^{-1}$$

easily imply the last two relations. The first one then follows trivially. □

Recall that the group of orientation-preserving similarities $\mathrm{Sim}^+(\mathcal{H})$ of $\mathcal{H}$ is isomorphic to the subgroup of $\mathrm{PSL}(2, \mathcal{H})$ consisting of (classes of) upper-triangular matrices (see Proposition 1.5.7). This correspondence arises by assigning to each matrix

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \mathrm{SL}(2, \mathcal{H}) \tag{3.10}$$

the similarity $q \mapsto aqd^{-1} + bd^{-1}$.

If $\Delta \subset \mathcal{H}$ is a triangle with vertices A, B, C and $f \in \mathrm{Sim}^+(\mathcal{H})$ is the orientation-preserving similarity represented by the matrix (3.10), then the triangle $f(\Delta)$ has quaternionic angles

$$q_{f(A)} = dq_A d^{-1}, \quad q_{f(B)} = dq_B d^{-1}, \quad q_{f(C)} = dq_C d^{-1}. \tag{3.11}$$

In particular, the quaternionic angles are invariant under translations, left multiplications by an invertible quaternion and dilations.

Lemma 3.3.3. *Two oriented triangles in $\mathcal{H}$ are related by an orientation-preserving similarity if and only if their quaternionic angles are simultaneously conjugate.*

Proof. Let A, B, C and A', B', C' be the vertices of two triangles Δ and Δ' respectively.

We already know that if there exists a similarity $f \in \mathrm{Sim}^+(\mathcal{H})$ such that

$$f(A) = A', \quad f(B) = B', \quad f(C) = C'$$

then the quaternionic angles of Δ and Δ' are simultaneously conjugate. Conversely, suppose there exists $d \in \mathcal{H}^*$ such that we have

$$dq_A d^{-1} = q_{A'}, \quad dq_B d^{-1} = q_{B'}, \quad dq_C d^{-1} = q_{C'}. \tag{3.12}$$

We claim that a similarity sending A, B, C to A', B', C' respectively is represented by the matrix

$$M = \begin{pmatrix} (B'-A')d(B-A)^{-1} & A'd - (B'-A')d(B-A)^{-1}A \\ 0 & d \end{pmatrix} \in \mathrm{GL}(2,\mathcal{H}).$$

Indeed, for every $q \in \mathcal{H}$ we can write

$$M(q) = (B'-A')d(B-A)^{-1}(q-A)d^{-1} + A' \tag{3.13}$$

and we obtain $M(A) = A'$ and $M(B) = B'$ very easily, while the fact that $M(C) = C'$ follows directly from the condition $dq_C d^{-1} = q_{C'}$. Notice, however, that the matrix M could possibly have Dieudonné determinant different from 1. This is not an issue: by replacing d with λd for a non zero real number λ, equation (3.12) still holds, while M and λM define the same element of $\mathrm{PGL}(2,\mathcal{H}) = \mathrm{PSL}(2,\mathcal{H})$ and thus the same similarity of $\mathcal{H}$. The proof is complete. □

Consider an oriented ideal tetrahedron T in $\mathbb{H}^5$ with ideal vertices $q_1, q_2, \infty, q_4 \in \widehat{\mathcal{H}}$, ordered following the orientation. Every small enough horosphere centered at ∞ intersects T in a triangle Δ, and each corner of Δ corresponds to one of the three oriented edges from q_i to ∞. By projecting the horizontal horosphere onto $\mathcal{H}$, the triangle Δ inherits a Euclidean structure, well-defined up to similarities, and we can identify its vertices with q_1, q_2, q_4. Importantly, choosing a different horosphere centered at ∞ affects the Euclidean triangle Δ only by a dilation. Consequently, the quaternionic angles of Δ are unambiguously defined and do not depend on the height of the chosen horosphere.

A direct check yields the following result.

Proposition 3.3.4. *For every $i = 1, 2, 4$, the quaternionic modulus of T assigned to the oriented edge from q_i to ∞ coincides with the quaternionic angle of Δ at the vertex q_i.*

In virtue of Proposition 3.3.4, most of the properties of quaternionic angles can be deduced from those of the quaternionic cross-ratios, and vice-versa. For instance, the 2-dimensional translation of Lemma 3.2.9 yields the following criterion.

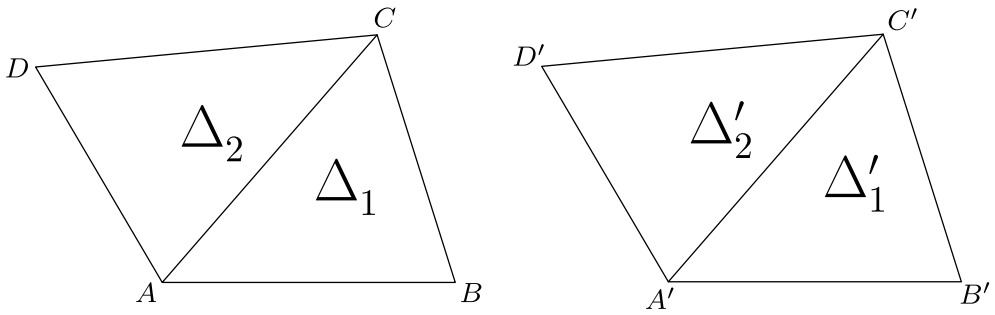


Figure 3.4: Two pairs of adjacent triangles in $\mathcal{H}$.

Corollary 3.3.5. *Two adjacent triangles in $\mathcal{H}$ are coplanar if and only if all their quaternionic angles commute with each other.*

Furthermore, we can classify pairs of adjacent, oriented triangles up to orientation-preserving similarity by combining our findings with Theorem 3.2.11 and Corollary 3.2.10.

Corollary 3.3.6. *Suppose Δ_1, Δ_2 are adjacent triangles in $\mathcal{H}$ with vertices A, B, C, D, and Δ_1', Δ_2' are adjacent triangles in $\mathcal{H}$ with vertices A', B', C', D' (see Figure 3.4). There exists an orientation-preserving similarity sending A, B, C, D to A', B', C', D' respectively if and only if there exists $d \in \mathcal{H}^*$ that simultaneously conjugates all the quaternionic angles of Δ_1, Δ_2 to the corresponding quaternionic angles of Δ_1', Δ_2', meaning*

$$(q_{A'}(\Delta_1'), q_{B'}(\Delta_1'), q_{C'}(\Delta_1')) = d(q_A(\Delta_1), q_B(\Delta_1), q_C(\Delta_1))d^{-1}$$
$$(q_{A'}(\Delta_2'), q_{C'}(\Delta_2'), q_{D'}(\Delta_2')) = d(q_A(\Delta_2), q_C(\Delta_2), q_D(\Delta_2))d^{-1}.$$

Moreover, when such a similarity exists, if Δ_1, Δ_2 (and, equivalently, Δ_1', Δ_2') are not coplanar, then the similarity is unique.

Remark 3.3.7. Let Δ and Δ' be two oriented triangles in $\mathcal{H}$ with vertices A, B, C and A', B', C' respectively. Suppose we know that there exists a quaternion $d \in \mathcal{H}^*$ such that $dq_A(\Delta)d^{-1} = q_{A'}(\Delta')$. Because of the algebraic relations (3.9), it immediately follows that d also conjugates the quaternionic angles of Δ at B and C to those of Δ' at B' and C'. Therefore, to check if the quaternionic angles of two triangles are simultaneously conjugate, it suffices to verify the condition for just one pair of corresponding vertices.

Remark 3.3.8. When the similarity of Corollary 3.3.6 is unique, its explicit form can be easily written down. In fact, every orientation-preserving similarity mapping A, B, C, D to A', B', C', D' respectively is represented by the $\mathrm{GL}(2, \mathcal{H})$ matrix

$$\begin{pmatrix} (B'-A')d(B-A)^{-1} & A'd - (B'-A')d(B-A)^{-1}A \\ 0 & d \end{pmatrix}, \tag{3.14}$$

where $d \in \mathcal{H}^*$ is any quaternion realizing the simultaneous conjugation of the quaternionic angles. By Corollary 2.1.8, the quaternion d is unique up to a real factor, and d and λd determine the same element of $\mathrm{PGL}(2, \mathcal{H}) = \mathrm{PSL}(2, \mathcal{H})$ for every $\lambda \in \mathbb{R}^*$.

Chapter 4

Quaternionic decorations

Let M be an oriented, complete, non-compact hyperbolic 3-manifold with finite volume. Suppose that M admits an ideal triangulation, and let $|\mathcal{T}|$ be the geometric realization of $\mathcal{T}$. We recall that, by definition, M is homeomorphic to $|\mathcal{T}| \setminus \{\text{vertices}\}$. Moreover, $\mathcal{T}$ lifts to an ideal triangulation $\widetilde{\mathcal{T}}$ of $\widetilde{M}$, and again $\widetilde{M}$ is homeomorphic to $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$, where $|\widetilde{\mathcal{T}}|$ is the geometric realization of $\widetilde{\mathcal{T}}$.

In [Thu02], Thurston proved that, under suitable conditions, the problem of finding explicit hyperbolic structures on M translates into solving a system of equations. As we reviewed in Section 1.4,we assign a complex parameter z_i with $\mathrm{Im}(z_i) > 0$ to each topological tetrahedron T_i. This parameter explicitly prescribes the isometry class of its geometric realization in $\mathbb{H}^3$. Then for every edge of $|\mathcal{T}| \setminus \{\text{vertices}\}$ we have a *consistency equation* that guarantees that the tetrahedra glue together consistently around it. Every solution to the consistency equations yields a hyperbolic structure on M, thus a developing pair (D, ρ) up to equivalence. We recall that $\rho\colon \pi_1(M) \to \mathrm{Isom}^+(\mathbb{H}^3)$ is a group morphism (the *holonomy*), while $D\colon \widetilde{M} \to \mathbb{H}^3$ is a local isometry (the *developing map*) which is $\pi_1(M)$-equivariant. In Section 1.4.2, we constructed D step-by-step on $\widetilde{\mathcal{T}}$.

The ultimate goal of this thesis, which is presented in this chapter, is to develop an analogous construction in dimension 5. As discussed in the introduction, the motivation is the following: from the 3-manifold M we can construct a 5-dimensional, complete hyperbolic manifold M' with *infinite* volume, which is diffeomorphic to $M \times \mathbb{R}^2$ and in which M is geodesically embedded as $M \times \{0\}$. If (D_0, ρ_0) denotes a developing pair associated to the complete hyperbolic structure on M, the induced complete structure on the bigger manifold M' has holonomy

$$\rho_0'\colon \pi_1(M') \cong \pi_1(M) \xrightarrow{\rho_0} \mathrm{Isom}^+(\mathbb{H}^3) \hookrightarrow \mathrm{Isom}^+(\mathbb{H}^5).$$

By Mostow-Prasad rigidity, the complete structure on M is unique. Since M' has infinite volume, this result no longer applies and we can look for deformations of the induced complete structure of M'. A natural way to do this is to construct new developing pairs (D', ρ') for M'.

Our first step in this direction is the definition of 5-dimensional developing maps directly on $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$. We specifically restrict our attention to those maps whose image is not entirely contained in a 3-dimensional hyperbolic subspace. Employing the quaternionic formalism introduced in the previous chapter, in the first section we formalize these objects as *generic quaternionic developing maps*. Thanks to the results of the previous chapter, we will see that every quaternionic developing map naturally assigns to every oriented edge of the triangulation a tuple of quaternions—arising from the quaternionic moduli of the images of the ideal tetrahedra winding around it—well-defined up to simultaneous conjugation. These tuples are subject to some compatibility conditions.

Motivated by these observations, in the second section we introduce *quaternionic decorations* of $\mathcal{T}$. More precisely, a quaternionic decoration assigns:

- complex moduli to each tetrahedron of $\mathcal{T}$, to fix its isometry class;
- a generic tuple of quaternions to every oriented edge of $|\mathcal{T}| \setminus \{\text{vertices}\}$ (considered up to simultaneous conjugation) to prescribe the relative position of the adjacent tetrahedra glued around it.

The assigned parameters are required to satisfy two families of compatibility conditions, namely the *Edge relations* and the *Face relations*. We will then prove the main result of this thesis, which establishes a canonical bijection between quaternionic decorations on $\mathcal{T}$ and equivalence classes of generic quaternionic developing maps for $(M, \mathcal{T})$.

Throughout the first two sections, we also study the restrictions of the holonomy induced by a quaternionic developing map to the cusp stabilizers (the *peripheral holonomy*). We present explicit conditions to detect when the images of these restrictions are parabolic or discrete subgroups of $\mathrm{Isom}^+(\mathbb{H}^5) \cong \mathrm{PSL}(2, \mathcal{H})$. Lastly, we show that every quaternionic decoration induces a labeling of the 2-dimensional triangulation of the boundary tori with quaternionic angles, and we use these local parameters to explicitly compute the peripheral holonomy.

Notation In what follows, M will be a complete, oriented, non-compact hyperbolic 3-manifold with finite volume, and $\mathcal{T}$ will denote an ideal triangulation of M. We also identify the group $\mathrm{Isom}^+(\mathbb{H}^5)$ with $\mathrm{PSL}(2, \mathcal{H})$.

Throughout this chapter, we adopt the following conventions:

- "tilded" symbols denote objects in the lifted triangulation $|\widetilde{\mathcal{T}}|$ (e.g., $\widetilde{T}, \tilde{e}, \tilde{v}$), while "non-tilded" symbols refer to their projections in the base triangulation $|\mathcal{T}|$;
- lowercase letters, such as q_{ik}, are used to denote the abstract algebraic parameters assigned to the triangulation by a quaternionic decoration;
- uppercase letters, such as Q_{ik}, are strictly reserved for the actual geometric quaternionic cross-ratios computed from points mapped into $\mathbb{H}^5$.

4.1 Quaternionic developing maps

Definition 4.1.1. A *quaternionic developing map* for $(M, \mathcal{T})$ is a continuous map

$$D\colon |\widetilde{\mathcal{T}}| \to \overline{\mathbb{H}}^5$$

mapping every vertex of $|\widetilde{\mathcal{T}}|$ to an ideal point in $\partial\mathbb{H}^5$, and every ideal simplex in $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$ homeomorphically to the geodesic ideal simplex determined by the images of its vertices, together with a group morphism $\rho\colon \pi_1(M) \to \mathrm{PSL}(2,\mathcal{H})$, called the *holonomy*. The pair (D,ρ) must satisfy

$$D(\gamma \cdot \tilde{x}) = \rho(\gamma) D(\tilde{x}) \tag{4.1}$$

for every $x \in |\widetilde{\mathcal{T}}|$ and for every $\gamma \in \pi_1(M)$.

Let $D\colon |\widetilde{\mathcal{T}}| \to \overline{\mathbb{H}}^5$ be a continuous map sending vertices to points at infinity and every ideal tetrahedron to the geodesic ideal tetrahedron spanned by the images of its vertices. Consider a lift $\tilde{e}$ in $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$ of an oriented edge e having valence k_e in $|\mathcal{T}| \setminus \{\text{vertices}\}$. Let $\widetilde{T}_1, \dots, \widetilde{T}_{k_e}$ be the ideal tetrahedra incident to $\tilde{e}$, following a positive cyclic order induced by the orientation of the edge. For every $i = 1, \dots, k_e$, let $q_i(\tilde{e})$ be the quaternionic cross-ratio of $D(\widetilde{T}_i)$ assigned to $D(\tilde{e})$ in $\mathbb{H}^5$.

We say that the map D is *generic* if the tuple of cross-ratios $(q_1(\tilde{e}), \dots, q_{k_e}(\tilde{e}))$ is generic for every oriented edge $\tilde{e}$, *i.e.* no pair of cyclically adjacent entries commute (see Definition 2.1.3). We will refer to the tuple of quaternions $(q_1(\tilde{e}), \dots, q_{k_e}(\tilde{e}))$ as the *quaternionic cross-ratios around* $D(\tilde{e})$.

Remark 4.1.2. Let $D\colon |\widetilde{\mathcal{T}}| \to \overline{\mathbb{H}}^5$ be a generic, continuous map sending vertices to points at infinity and every ideal tetrahedron to the geodesic ideal tetrahedron spanned by the images of its vertices. Any map $\rho\colon \pi_1(M) \to \mathrm{PSL}(2,\mathcal{H})$ satisfying the holonomy condition (4.1) is a group homomorphism. To see this, consider the vertices $\tilde{v}_1, \dots, \tilde{v}_5$ of two adjacent tetrahedra in $|\widetilde{\mathcal{T}}|$. Since D is generic, the images $D(\tilde{v}_1), \dots, D(\tilde{v}_5)$ are pairwise distinct and form the vertices of two oriented adjacent ideal tetrahedra in $\mathbb{H}^5$ that are in general position. For any $\gamma_1, \gamma_2 \in \pi_1(M)$, the holonomy condition implies:

$$\rho(\gamma_1\gamma_2) D(\tilde{v}_i) = D(\gamma_1\gamma_2 \cdot \tilde{v}_i) = \rho(\gamma_1) D(\gamma_2 \cdot \tilde{v}_i) = \rho(\gamma_1)\rho(\gamma_2) D(\tilde{v}_i)$$

for each $i = 1, \dots, 5$. Since the isometries $\rho(\gamma_1\gamma_2)$ and $\rho(\gamma_1)\rho(\gamma_2)$ coincide on the ideal vertices of two adjacent tetrahedra in general position, they must be identical.

Definition 4.1.3. We say that two quaternionic developing maps (D,ρ) and (D',ρ') are *equivalent* if there exists $A \in \mathrm{PSL}(2,\mathcal{H})$ such that $D' = A \circ D$ and $\rho' = A\rho A^{-1}$.

The following properties of quaternionic developing maps follow directly from the properties of the quaternionic cross-ratio.

Lemma 4.1.4. *Let (D,ρ) be a generic quaternionic developing map for $(M,\mathcal{T})$. The following facts are true:*

(1) if ideal two tetrahedra $\widetilde{T}$ and $\widetilde{T}'$ are lifts of the same ideal tetrahedron of $|\mathcal{T}| \setminus \{\text{vertices}\}$, their images $D(\widetilde{T})$ and $D(\widetilde{T}')$ are orientation-preservingly isometric;

(2) if $\tilde{e}$ and $\tilde{f}$ are two lifts of the same edge, the quaternionic cross-ratios around $D(\tilde{e})$ are simultaneously conjugate to the quaternionic cross-ratios around $D(\tilde{f})$;

(3) let $\tilde{e}_+$ and $\tilde{e}_-$ denote the same edge of $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$ with opposite orientations. The quaternionic cross-ratios around $D(\tilde{e}_+)$ and $D(\tilde{e}_-)$ form two adjoint tuples of quaternions (see Definition 2.1.13). More precisely, if $(q_1, \dots, q_k)$ and $(p_k, \dots, p_1)$ denote the quaternionic cross-ratios around $D(\tilde{e}_+)$ and $D(\tilde{e}_-)$ respectively, ordered in such a way that q_i and p_i refer to the same tetrahedron, then $(q_1, \dots, q_k)$ and $(p_1, \dots, p_k)$ are adjoint. In particular

$$q_1 \cdot \ldots \cdot q_k = 1, \quad p_k \cdot \ldots \cdot p_1 = 1.$$

Proof. If $\widetilde{T}$ and $\widetilde{T}'$ are lifts of the same tetrahedron, then $\widetilde{T}' = \gamma \cdot \widetilde{T}$ for some $\gamma \in \pi_1(M)$. The holonomy condition (4.1) implies that the isometry $\rho(\gamma)$ maps $D(\widetilde{T})$ to $D(\widetilde{T}')$, proving (1). Similarly, if $\tilde{e}$ and $\tilde{f}$ are two lifts of the same edge, there exists an element $\gamma \in \pi_1(M)$ such that $\gamma \cdot \tilde{e} = \tilde{f}$. Now the isometry $\rho(\gamma)$ maps each tetrahedron incident to $D(\tilde{e})$ to a tetrahedron incident to $D(\tilde{f})$, and statement (2) follows from Proposition 2.3.6.

Finally, let $\tilde{e}_+$ and $\tilde{e}_-$ denote the same edge of $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$ with opposite orientations. Up to equivalence, we can assume that $D(\tilde{e}_+)$ is the unique geodesic in $\mathbb{H}^5$ oriented from 0 to ∞. Then the star around the edge $D(\tilde{e}_+)$ consists of k ideal tetrahedra $T_1, \dots, T_k$, where T_i has vertices $0, x_i, \infty, x_{i+1}$ (with indices mod k), see Figure 4.1. We assume moreover that q_i and p_i both refer to the tetrahedron T_i. By definition

$$q_i = Q(0, x_i, \infty, x_{i+1}) = x_i^{-1} x_{i+1}, \quad p_i = Q(\infty, x_{i+1}, 0, x_i) = x_{i+1} x_i^{-1}.$$

This matches precisely the definition of adjoint tuples (Definition 2.1.13), and the product identities $q_1 \cdot \ldots \cdot q_k = 1$ and $p_k \cdot \ldots \cdot p_1 = 1$ follow immediately from Lemma 2.1.14. □

Remark 4.1.5. The reason why we restrict ourselves to generic developing maps is because we want to avoid the case where the image of D is entirely contained in the compactification of a 3-dimensional subspace. This could also be achieved by imposing a weaker condition, for instance that the quaternionic cross-ratios around some edge do not form a commutative k-tuple of quaternions.

Example 4.1.6. Let $\rho_0 \colon \pi_1(M) \to \mathrm{PSL}(2, \mathbb{C})$ be the holonomy of the complete hyperbolic structure on M. Let $\rho_0' \colon \pi_1(M) \to \mathrm{PSL}(2, \mathcal{H})$ be the representation obtained by composing ρ_0 with the standard inclusion $\mathrm{PSL}(2, \mathbb{C}) \hookrightarrow \mathrm{PSL}(2, \mathcal{H})$. Then, ρ_0' is not the holonomy of a generic quaternionic developing map.

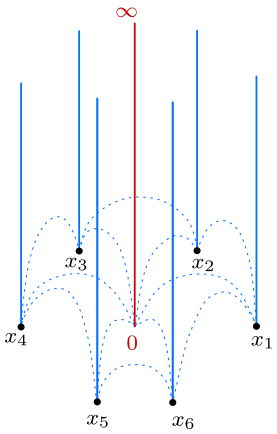


Figure 4.1: The star around the edge $D(\tilde{e}_+)$ in $\mathbb{H}^5$.

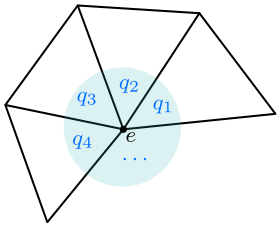


Figure 4.2: A quaternionic lift around the edge e.

Indeed, suppose by contradiction that there exists a generic quaternionic developing map (D, ρ_0') for $(M, \widetilde{\mathcal{T}})$, and consider a vertex $\tilde{v}$ of $|\widetilde{\mathcal{T}}|$. For every element γ in the stabilizer of $\tilde{v}$, by definition we have

$$D(\tilde{v}) = D(\gamma \cdot \tilde{v}) = \rho_0'(\gamma) D(\tilde{v}),$$

so that $D(\tilde{v})$ is a fixed point of $\rho_0'(\gamma)$. By construction, $\rho_0'(\gamma) \in \mathrm{PSL}(2, \mathbb{C})$ is a parabolic isometry, hence its unique fixed point in the whole space $\widehat{\mathcal{H}}$ lies in $\widehat{\mathbb{C}}$. As a consequence, every vertex of $|\widetilde{\mathcal{T}}|$ is mapped into $\widehat{\mathbb{C}}$ and so $D\colon |\widetilde{\mathcal{T}}| \to \overline{\mathbb{H}}^5$ is not generic.

4.1.1 Peripheral holonomy

Assume for simplicity that M has a single cusp, and let (D, ρ) be a quaternionic developing map for $(M, \mathcal{T})$. The boundary of the truncated cusp is a torus L, and its fundamental group $\pi_1(L) \cong \mathbb{Z} \times \mathbb{Z}$ injects into $\pi_1(M)$. We wish to study the restriction $\rho|_{\pi_1(L)}\colon \pi_1(L) \to \mathrm{PSL}(2, \mathcal{H})$, which is sometimes called the *peripheral holonomy*.

Definition 4.1.7. The holonomy ρ is *boundary-parabolic* if for every $\gamma \in \pi_1(L)$, the matrix $\rho(\gamma)$ is either the identity or it represents a parabolic isometry of $\mathbb{H}^5$.

We want to find conditions under which the holonomy is boundary-parabolic, and its restriction to $\pi_1(L)$ is discrete and faithful. As we have seen in Chapter 1, in the 3-dimensional case parabolic isometries fixing ∞ are just Euclidean translations. Therefore, if ρ' is a holonomy of a hyperbolic structure on M, to check whether $\rho'|_{\pi_1(L)}$ is discrete and faithful it is sufficient to verify that some two generators map to translations by linearly independent vectors. When this happens, they generate a 2-dimensional lattice and the hyperbolic structure on the whole manifold M is complete.

In higher dimensions, the situation is more subtle. As we saw in the previous chapter, a parabolic isometry of $\mathbb{H}^5$ fixing ∞ can be screw-parabolic, meaning it involves a non-

trivial linear component. In this case, detecting when such a representation is discrete and faithful requires a more careful analysis.

To investigate this, we make the injection $\pi_1(L) \hookrightarrow \pi_1(M)$ precise. First, recall that L corresponds to a vertex v of $|\mathcal{T}|$, and that $\mathcal{T}$ induces a triangulation $\mathcal{T}_v$ of L obtained by considering the links of those vertices of $\mathcal{T}$ that project to v (see Section 1.4). We identify $M, \widetilde{M}$ and L with the geometric realizations $|\mathcal{T}| \setminus \{\text{vertices}\}$, $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$ and $|\mathcal{T}_v|$ respectively. Fix a lift $\tilde{v}$ of v in $|\widetilde{\mathcal{T}}|$. We choose a basepoint $x_0 \in L$ for both $\pi_1(L)$ and $\pi_1(M)$, and as a basepoint for $\widetilde{M}$ we pick a lift $\tilde{x}_0$ lying in the link of $\tilde{v}$. With this choice, the image of the inclusion $\pi_1(L, x_0) \hookrightarrow \pi_1(M, x_0)$ corresponds exactly to the subgroup of $\pi_1(M, x_0)$ stabilizing the ideal vertex $\tilde{v}$.

Now set $\tilde{v}_3 \coloneqq \tilde{v}$, and consider two adjacent tetrahedra T_1, T_2 of $|\mathcal{T}|$ having vertices v_1, v_2, v_3, v_4 and v_1, v_4, v_3, v_5 respectively, as in Figure 4.3a. Up to post-composing with an orientation-preserving isometry of $\mathbb{H}^5$ (which does not alter the equivalence class of (D, ρ) by definition), we can assume that

$$D(\tilde{v}_1) = 0, \quad D(\tilde{v}_2) = 1, \quad D(\tilde{v}_3) = \infty.$$

By the previous considerations, the holonomy ρ maps the stabilizer $\Gamma \coloneqq \pi_1(L, x_0)$ to an abelian subgroup of $\mathrm{PSL}(2, \mathcal{H})$ fixing ∞. For every $\gamma \in \Gamma$ we choose a matrix representative for $\rho(\gamma)$ in $\mathrm{SL}(2, \mathcal{H})$, which by a slight abuse of notation we still denote by $\rho(\gamma)$:

$$\rho(\gamma) = \begin{pmatrix} a_\gamma & b_\gamma \\ 0 & d_\gamma \end{pmatrix}. \tag{4.2}$$

Remark 4.1.8. In general, there is no guarantee that the chosen matrix lifts yield a group morphism $\Gamma \to \mathrm{SL}(2, \mathcal{H})$. However, replacing the chosen representative of $\rho(\gamma)$ with its negative changes the signs of a_γ, b_γ and d_γ simultaneously. In the next section, these matrix entries appear only in multiplicative combinations, such as the translation vector $A_\gamma = b_\gamma d_\gamma^{-1}$ and the linear map $x \mapsto a_\gamma x d_\gamma^{-1}$, that do not depend on the chosen sign. Therefore, all subsequent results do not depend on the matrix lifts, and do not need the lifts to yield a group morphism.

4.1.2 Boundary-parabolic holonomy

Fix an element $\gamma \in \Gamma$. As we have seen in Theorem 3.1.3, the matrix $\rho(\gamma)$ of (4.2) represents a parabolic isometry of $\mathbb{H}^5$ if and only if:

(Condition 1) a_γ and d_γ are conjugate unit quaternions, and

(Condition 2) either $a_\gamma, d_\gamma \in \mathbb{R}$ and $b_\gamma \neq 0$, or $\bar{a}_\gamma b_\gamma \neq b_\gamma d_\gamma$.

We recall that Condition 1 is satisfied if and only if $|a_\gamma| = |d_\gamma| = 1$ and $\mathrm{Re}(a_\gamma) = \mathrm{Re}(d_\gamma)$, by Theorem 1.5.1.

Lemma 4.1.9. *If there exists $\gamma_0 \in \Gamma$ such that $\rho(\gamma_0)$ is parabolic, then $\rho(\gamma)$ satisfies Condition 1 for every $\gamma \in \Gamma$.*

Proof. Fix an element $\gamma \in \Gamma$, and assume that $\rho(\gamma)$ is not trivial (otherwise there is nothing to prove). We need to show that a_γ and d_γ are conjugate unit quaternions. Since $\rho(\gamma)$ and $\rho(\gamma_0)$ commute, they preserve each other's fixed-point set. In particular, $\rho(\gamma)$ cannot have exactly two fixed points on the boundary at infinity. Indeed, if its fixed-point set were $\{\infty, q\}$, the isometry $\rho(\gamma_0)$ would have to preserve this pair; since $\rho(\gamma_0)$ already fixes ∞, it would be forced to fix q as well, contradicting the fact that $\rho(\gamma_0)$ is parabolic. Thanks to Corollary 3.1.4, $\rho(\gamma)$ is therefore either a parabolic isometry of $\mathbb{H}^5$, or an elliptic isometry of $\mathbb{H}^5$ having a 2-sphere as a fixed-point set on $\partial\mathbb{H}^5$. Both cases arise only if a_γ and d_γ are conjugate unit quaternions, as we wanted. □

We now want to understand Condition 2. Fix a pair of generators γ_1, γ_2 of $\Gamma \cong \mathbb{Z}\times\mathbb{Z}$, and assume that one of them, say γ_1, maps to a parabolic isometry of $\mathbb{H}^5$. By Lemma 4.1.9, $\rho(\gamma)$ satisfies Condition 1 for every $\gamma \in \Gamma$.

If both $\rho(\gamma_1)$ and $\rho(\gamma_2)$ have real diagonal entries, ($a_{\gamma_s} = d_{\gamma_s} = \pm 1$ for $s = 1, 2$), then $\rho(\Gamma)$ consists solely of (possibly trivial) translations. In this case, the geometric behavior of the holonomy is entirely determined by the translation vectors $A_{\gamma_1} = b_{\gamma_1} d_{\gamma_1}^{-1}$ and $A_{\gamma_2} = b_{\gamma_2} d_{\gamma_2}^{-1}$ of the generators.

Lemma 4.1.10. *Suppose that $\rho(\gamma_1)$ and $\rho(\gamma_2)$ both have real diagonal entries, and let A_{γ_1} and A_{γ_2} be their translation vectors. Then the restriction $\rho|_\Gamma$ is injective and ρ is boundary-parabolic if and only if A_{γ_1} and A_{γ_2} are linearly independent over $\mathbb{Z}$. When this is the case, the image $\rho(\Gamma)$ is a discrete subgroup if and only if A_{γ_1} and A_{γ_2} are linearly independent over $\mathbb{R}$.*

Proof. Since the diagonal entries are real, $\rho(\gamma)$ satisfies Condition 1 for every $\gamma \in \Gamma$. A straightforward check shows that for any $\gamma = \gamma_1^n \gamma_2^m$, the translation vector of $\rho(\gamma)$ is simply $A_\gamma = nA_{\gamma_1} + mA_{\gamma_2}$. The isometry $\rho(\gamma)$ satisfies Condition 2 if and only if $b_\gamma \neq 0$, which is equivalent to $A_\gamma \neq 0$. Thus, $\rho|_\Gamma$ is injective and boundary-parabolic if and only if $nA_{\gamma_1} + mA_{\gamma_2} \neq 0$ for all $(n, m) \neq (0, 0)$, which proves the first part of the statement. The condition for discreteness follows from the fact that a lattice of translations in a real vector space is discrete if and only if its generators are linearly independent over $\mathbb{R}$. □

To address the complementary case to Lemma 4.1.10, from now on we assume that at least one of the fixed generators, say γ_1, maps to a screw-parabolic isometry. By hypothesis, a_{γ_1} and d_{γ_1} are conjugate unit quaternions. Moreover, since ρ is a group homomorphism and $\Gamma \cong \mathbb{Z} \times \mathbb{Z}$, the imaginary parts of the a_γ's are all collinear by Proposition 1.5.4, and the same holds for the d_γ's. Consequently, we can write:

$$a_\gamma = \cos\alpha_\gamma + \sin\alpha_\gamma u, \quad d_\gamma = \cos\alpha_\gamma + \sin\alpha_\gamma v \tag{4.3}$$

for some purely imaginary, unit quaternions u, v and for angles $\alpha_\gamma \in [0, 2\pi]$. We fix the notation

$$A_\gamma := \rho(\gamma)0 = b_\gamma d_\gamma^{-1}$$

for the translation vector of $\rho(\gamma)$.

When the diagonal entries of $\rho(\gamma)$ are not real, the linear part of $\rho(\gamma)$ is a simple rotation around the plane

$$K \coloneqq \{x \in \mathcal{H} : ux = xv\}. \tag{4.4}$$

In particular, it acts as a 2-dimensional rotation on its orthogonal plane

$$K^{\perp} = \{x \in \mathcal{H} : ux = -xv\}. \tag{4.5}$$

We denote by $\pi_K \colon \mathcal{H} \to K$ and $\pi_{K^{\perp}} \colon \mathcal{H} \to K^{\perp}$ the orthogonal projections onto K and $K^{\perp}$ respectively. It is easily verified that they are given by

$$\pi_K(x) = \frac{x - uxv}{2}, \quad \pi_{K^{\perp}}(x) = \frac{x + uxv}{2}.$$

Now the condition $\bar{a}_\gamma b_\gamma \neq b_\gamma d_\gamma$ is equivalent to asking that the translation vector $A_\gamma = b_\gamma d_\gamma^{-1}$ is not orthogonal to K, which translates into

$$\pi_K(A_\gamma) \neq 0. \tag{4.6}$$

Lemma 4.1.11. *For each pair of elements $\gamma, \sigma \in \Gamma$, we have*

$$\pi_K(A_{\gamma\sigma}) = \pi_K(A_\gamma) + \pi_K(A_\sigma).$$

Proof. Since ρ is a group homomorphism, a direct check shows that:

$$A_{\gamma\sigma} = A_\gamma + a_\gamma A_\sigma d_\gamma^{-1}. \tag{4.7}$$

If a_γ and d_γ are real, they are equal to ± 1, so $a_\gamma A_\sigma d_\gamma^{-1} = A_\sigma$, and the identity trivially holds. Otherwise, notice that the linear map $x \mapsto a_\gamma x d_\gamma^{-1}$ acts as the identity on K (since a_γ commutes with u and d_γ commutes with v), and preserves its orthogonal complement $K^{\perp}$. In particular, it commutes with the orthogonal projection π_K. Applying π_K to equation (4.7) yields

$$\pi_K(A_{\gamma\sigma}) = \pi_K(A_\gamma) + \pi_K(a_\gamma A_\sigma d_\gamma^{-1}) = \pi_K(A_\gamma) + a_\gamma \pi_K(A_\sigma) d_\gamma^{-1} = \pi_K(A_\gamma) + \pi_K(A_\sigma).$$

The proof is complete. □

Remark 4.1.12. In the case where $\rho(\gamma_2)$ is a translation (*i.e.* $a_{\gamma_2} = d_{\gamma_2} = \pm 1$), commutativity forces its translation vector A_{γ_2} to lie entirely in K. Indeed, a direct check shows

$$A_{\gamma_1\gamma_2} = a_{\gamma_1} A_{\gamma_2} d_{\gamma_1}^{-1} + A_{\gamma_1}, \quad A_{\gamma_2\gamma_1} = A_{\gamma_1} + A_{\gamma_2}$$

and now commutativity yields

$$A_{\gamma_2} = a_{\gamma_1} A_{\gamma_2} d_{\gamma_1}^{-1}.$$

In particular, since $b_{\gamma_2} = A_{\gamma_2}$ in this case, the requirement $b_{\gamma_2} \neq 0$ of Condition 2 is equivalent to $\pi_K(A_{\gamma_2}) \neq 0$.

Lemma 4.1.13. *Suppose that $\rho(\gamma_1)$ is a screw-parabolic isometry of $\mathbb{H}^5$. Then there exists a unique $y_0 \in K^{\perp}$ such that for every $\gamma \in \Gamma$ we have:*

1. *the affine plane $y_0 + K$ is $\rho(\gamma)$-invariant;*
2. *the orthogonal projection of the translation vector A_γ onto $K^\perp$ is given by $\pi_{K^\perp}(A_\gamma) = y_0 - a_\gamma y_0 d_\gamma^{-1}$.*

Proof. Since $\rho(\gamma_1)$ is screw-parabolic, by Lemma 3.1.8 we know that there exists a unique vector $y_0 \in K^\perp$ satisfying conditions (1) and (2) for γ_1. We want to show that this exact same y_0 satisfies properties (1) and (2) for every other $\gamma \in \Gamma$. Let $T_\gamma \colon \mathcal{H} \to \mathcal{H}$ denote the linear map

$$T_\gamma(x) = x - a_\gamma x d_\gamma^{-1},$$

so that our desired condition (2) rewrites as $\pi_{K^\perp}(A_\gamma) = T_\gamma(y_0)$. Suppose first that a_γ and d_γ are both non-real. Since Γ is abelian, the matrices $\rho(\gamma_1)$ and $\rho(\gamma)$ commute, yielding

$$T_{\gamma_1}(\pi_{K^\perp}(A_\gamma)) = T_\gamma(\pi_{K^\perp}(A_{\gamma_1})) = T_\gamma(T_{\gamma_1}(y_0)) = T_{\gamma_1}(T_\gamma(y_0)).$$

Because T_{γ_1} is injective on $K^\perp$ we find

$$\pi_{K^\perp}(A_\gamma) = T_\gamma(y_0),$$

as we wanted. This proves (2).

Condition (1) now follows immediately from (2) by the exact same algebraic check used in Lemma 3.1.8. Indeed, for every $k \in K$, we have

$$\rho(\gamma)(y_0 + k) = a_\gamma y_0 d_\gamma^{-1} + k + \pi_K(A_\gamma) + \pi_{K^\perp}(A_\gamma) = \underbrace{k + \pi_K(A_\gamma)}_{\in K} + y_0 \in y_0 + K,$$

as we wanted. The case where the diagonal entries of $\rho(\gamma)$ are real reduces to a trivial check, as by Remark 4.1.12 $\pi_{K^\perp}(A_\gamma) = 0$. □

Remark 4.1.14. Geometrically, Lemma 4.1.13 shows that if $\rho(\gamma_1)$ is screw-parabolic, then the entire image $\rho(\Gamma)$ acts as a family of (possibly trivial) rotations around the affine plane $y_0 + K$, composed with translations parallel to K. In this setting, the nature of any element $\rho(\gamma)$ is entirely determined by its translational component: it is parabolic if and only if $\pi_K(A_\gamma) \neq 0$; otherwise, it is an elliptic isometry (or the identity) fixing the plane $y_0 + K$ pointwise.

To ease the notation, we define the map $\Psi \colon \Gamma \to K$ as

$$\Psi(\gamma) \coloneqq \pi_K(A_\gamma). \tag{4.8}$$

By Lemma 4.1.11, Ψ is a group morphism.

Theorem 4.1.15. *Suppose that $\rho(\gamma_1)$ is a screw-parabolic isometry of $\mathbb{H}^5$, and let $w_1 = \Psi(\gamma_1)$ and $w_2 = \Psi(\gamma_2)$. Then the restriction $\rho|_\Gamma$ is injective and ρ is boundary-parabolic if and only if w_1 and w_2 are $\mathbb{Z}$-independent in K. When this is the case, the image $\rho(\Gamma)$ is a discrete subgroup if and only if w_1 and w_2 are $\mathbb{R}$-independent in K.*

Proof. We first notice that since $\rho(\gamma_1)$ is screw-parabolic, by Lemma 4.1.9 the isometry $\rho(\gamma)$ satisfies Condition 1 for every $\gamma \in \Gamma$.

Let $\gamma = \gamma_1^n \gamma_2^m$ be an arbitrary element of Γ, for some pair of integers (n, m). Since $\rho(\gamma)$ already satisfies Condition 1, it is parabolic and non-trivial if and only if it also satisfies Condition 2, which is equivalent to $\pi_K(A_\gamma) \neq 0$. By Lemma 4.1.11 and the definition of Ψ, we have

$$\pi_K(A_\gamma) = nw_1 + mw_2.$$

Suppose that w_1 and w_2 are linearly independent over $\mathbb{Z}$. Then for any $(n, m) \neq (0, 0)$ we have $\pi_K(A_\gamma) \neq 0$. This implies that for every non-trivial γ, the isometry $\rho(\gamma)$ satisfies Condition 2, meaning it is parabolic. Consequently, ρ is boundary-parabolic, and since no non-trivial element maps to the identity, the restriction $\rho|_\Gamma$ is injective.

Conversely, if w_1 and w_2 are $\mathbb{Z}$-dependent, there exists a non-trivial pair of integers (n, m) such that $\pi_K(A_\gamma) = 0$. For this specific element $\gamma \neq 1$, the matrix $\rho(\gamma)$ represents either an elliptic isometry or the identity. If it is elliptic, ρ is not boundary-parabolic; if it is the identity, $\rho|_\Gamma$ is not injective. This completes the proof of the first part of the statement.

We move to the condition for discreteness. Suppose we have a sequence $\{\gamma = \gamma_1^{n_k} \gamma_2^{m_k}\}_{k \in \mathbb{N}}$ of distinct elements in Γ, and consider their images $\rho(\gamma_1^{n_k} \gamma_2^{m_k}) \in \mathrm{PSL}(2, \mathcal{H})$. To ease the notation, for each $k \in \mathbb{N}$ we set

$$a_k \coloneqq a_{\gamma_1^{n_k} \gamma_2^{m_k}}, \quad d_k \coloneqq d_{\gamma_1^{n_k} \gamma_2^{m_k}}, \quad A_k \coloneqq A_{\gamma_1^{n_k} \gamma_2^{m_k}}.$$

Since the diagonal elements a_k, d_k lie in S^3, up to subsequences they converge to some unitary quaternions a_∞, d_∞. We now move to the sequence of translation vectors, which we split into their orthogonal components $A_k = \pi_K(A_k) + \pi_{K^\perp}(A_k)$.

By Lemma 4.1.13, there exists a unique quaternion $y_0 \in \mathcal{H}$ such that for every $k \in \mathbb{N}$ we have

$$\pi_{K^\perp}(A_k) = y_0 - a_k y_0 d_k^{-1},$$

which converges. Thus, the convergence of the matrices $\rho(\gamma_1^{n_k} \gamma_2^{m_k})$ depends solely on that of the sequence of vectors

$$\pi_K(A_k) = n_k w_1 + m_k w_2 \in K.$$

Since w_1, w_2 belong to the 2-dimensional real vector space K, the sequence $\{n_k w_1 + m_k w_2\}_k$ accumulates if and only if w_1 and w_2 are linearly dependent over $\mathbb{R}$. This proves that the group $\rho(\Gamma)$ is discrete if and only if w_1 and w_2 are $\mathbb{R}$-independent. □

4.2 Quaternionic decorations on a triangulation

Recall that we have fixed an ideal triangulation $\mathcal{T}$ of an oriented, complete, finite volume hyperbolic 3-manifold M. We want to assign algebraic data to the oriented edges of $|\mathcal{T}| \setminus \{\text{vertices}\}$ in such a way that, if certain conditions are satisfied, then every assignment will provide us with a generic quaternionic developing map for our triangulated 3-manifold.

Before giving the definition, we recall some notation. Suppose that $\mathcal{T}$ consists of n tetrahedra $T_1, \dots, T_n$. Recall that the quotient projection

$$\pi_{\mathcal{T}} \colon \bigsqcup_{i=1}^{n} T_i \to |\mathcal{T}|$$

is injective when restricted to the interior of each tetrahedron. Because of this, as we did in Chapter 1, we still denote by $T_1, \dots, T_n$ the images of the tetrahedra with their vertices removed in $|\mathcal{T}| \setminus \{\text{vertices}\}$. In what follows, when referring to the ideal vertices of a tetrahedron T in $|\mathcal{T}| \setminus \{\text{vertices}\}$, we implicitly refer to the four vertices of the abstract simplex prior to the topological identifications of the manifold. Moreover, we recall that $\pi|_{\mathcal{T}}$ is injective when restricted to the interior of every face and every edge.

Definition 4.2.1. A *quaternionic decoration* of $\mathcal{T}$ is the assignment of:

1. for every tetrahedron T of $\mathcal{T}$ with vertices v_1, v_2, v_3, v_4, 12 complex numbers $z_{ik}(T) \in \mathbb{C}_+$ assigned to the oriented edges of T (where $z_{ik}(T)$ is assigned to the edge connecting v_i to v_k), satisfying the relations

$$z_{ik}(T) = z_{ki}(T) = z_{jl}(T) = z_{lj}(T),$$

$$z_{jk}(T) = (1 - z_{ik}(T))^{-1}, \quad z_{lk}(T) = 1 - z_{ik}(T)^{-1}.$$

   We call $z_{ik}(T)$ the *complex moduli* of T;

2. for every oriented edge e having valence k_e in $|\mathcal{T}| \setminus \{\text{vertices}\}$, a generic k_e-tuple of quaternions defined up to simultaneous conjugation

$$(q_1(e), \dots, q_{k_e}(e)),$$

   where the i-th component $q_i(e)$ corresponds to the i-th tetrahedron incident to e (following a positive cyclic order around the edge). We require that each component $q_i(e)$ is conjugate to the complex modulus along e of the corresponding tetrahedron. We will call every such k_e-tuple a *quaternionic lift* around e (see Figure 4.2) and, when clear from the context, we will denote by $q_{ik}(T)$ the component of the tuple assigned to the edge $v_i v_k$ corresponding to the tetrahedron T.

We furthermore require that the complex angles and the quaternionic lifts satisfy the following compatibility requirements:

(R1) (*Edge relations*) Let e_+ and e_- denote the same edge with the opposite orientations. Let $q_1, \dots, q_k$ and $p_k, p_{k-1} \dots, p_1$ be the two quaternionic lifts around e_+ and e_- respectively, ordered such that the quaternions q_i and p_i refer to the same tetrahedron.

Then the tuples $(q_1, \dots, q_k)$ and $(p_1, \dots, p_k)$ are adjoint (see Definition 2.1.13). In particular we have

$$q_1 \cdot \ldots \cdot q_k = 1, \quad p_k \cdot \ldots \cdot p_1 = 1.$$

This condition is well-posed thanks to Lemma 2.1.14. The reversed order in the second product corresponds to the fact that passing to e_- reverses the cyclic order in which the tetrahedra wind around the edge.

(R2) *(Face relations).* Let T_1, T_2 be two tetrahedra of $\mathcal{T}$, and suppose that a face F of T_1 with vertices v_1, v_3, v_4 is paired with a face F' of T_2 with vertices v_1', v_3', v_4', in this order. This pairing naturally identifies the 6 oriented abstract edges of F with the corresponding 6 oriented abstract edges of F'. For every $i \neq k \in \{1,3,4\}$, let e_{ik} denote the resulting oriented edge in the quotient space $|\mathcal{T}| \setminus \{\text{vertices}\}$ (note that these 6 oriented quotient edges are not necessarily distinct). Then there exist representatives for the quaternionic lifts around every e_{ik} that satisfy the following conditions:

(a) the parameters satisfy the internal Thurston relations

$$q_{43}(T_1) = 1 - q_{13}(T_1)^{-1}, \quad q_{34}(T_1) = (1 - q_{14}(T_1))^{-1}, \quad q_{31}(T_1) = 1 - q_{41}(T_1)^{-1};$$
$$q_{43}(T_2) = (1 - q_{13}(T_2))^{-1}, \quad q_{34}(T_2) = 1 - q_{14}(T_2)^{-1}, \quad q_{31}(T_2) = (1 - q_{41}(T_2))^{-1};$$

(b) there exist three commuting quaternions $\alpha, \beta, \gamma \in \mathcal{H}^*$ (associated to the corners of the face) with $\alpha\beta\gamma = -1$ such that for both $s = 1, 2$:

$$\alpha q_{13}(T_s)\alpha^{-1} = q_{31}(T_s), \quad \beta q_{41}(T_s)\beta^{-1} = q_{14}(T_s), \quad \gamma q_{34}(T_s)\gamma^{-1} = q_{43}(T_s).$$

Remark 4.2.2. Fix a quaternionic decoration of $\mathcal{T}$ and consider a quaternionic lift $(q_1(e), \dots, q_{k_e}(e))$ around an oriented edge e. The Edge relations imply

$$q_1(e) \cdot \ldots \cdot q_{k_e}(e) = 1. \tag{4.9}$$

By definition, each component $q_i(e)$ is conjugate to the corresponding complex modulus z_i. If the tuple were commutative (see Definition 2.1.3), equation (4.9) would trivially imply $z_1 \cdot \ldots \cdot z_{k_e} = 1$. However, since the quaternionic lift is generic, adjacent entries do not commute and this implication fails. As a consequence, the Edge relations can be satisfied even when the product of the underlying complex moduli is not equal to 1.

4.2.1 The correspondence theorem

Given a generic quaternionic developing map (D, ρ), we can easily define a quaternionic decoration of $\mathcal{T}$. The parameters are defined as follows:

(1) for every ideal tetrahedron T with vertices v_1, v_2, v_3, v_4, we fix a lift $\widetilde{T}$ and we look at the quaternionic moduli of $D(\widetilde{T})$ in $\mathbb{H}^5$. Every quaternionic modulus $Q_{ik}(\widetilde{T})$ is conjugate to a unique complex number $z_{ik}(T)$ with strictly positive imaginary part. For every $i \neq k \in \{1,2,3,4\}$, we take the $z_{ik}(T)$ as the complex modulus assigned to the oriented edge from v_i to v_k in T;

(2) for every oriented edge e, fix a lift $\tilde{e}$ and let $(q_1(\tilde{e}), \dots, q_{k_e}(\tilde{e}))$ be the quaternionic moduli around $D(\tilde{e})$. We choose the tuple $(q_1(\tilde{e}), \dots, q_{k_e}(\tilde{e}))$ as the quaternionic lift around e.

Observe that these assignments are well defined, since quaternionic lifts of the edges are defined only up to simultaneous conjugation. We now verify that these assignments satisfy the compatibility conditions (R1) and (R2) from Definition 4.2.1. The Edge relations (R1) are satisfied thanks to point (3) of Lemma 4.1.4. The Face relations (R2) follow from the fact that the quaternionic cross-ratios assigned to the common edges of adjacent tetrahedra are conjugate by simple ratios (via Lemma 3.2.7). By Lemma 2.3.12, these ratios commute, exactly playing the roles of the elements α, β, γ required by the definition. Furthermore, we have the following result.

Lemma 4.2.3. *Equivalent generic quaternionic developing maps induce the same quaternionic decoration of $\mathcal{T}$.*

Proof. Let (D, ρ) and (D', ρ') be two equivalent, generic quaternionic developing maps for $(M, \mathcal{T})$. By definition, there exists $f \in \mathrm{Isom}^+(\mathbb{H}^5)$ such that $D' = f \circ D$. As a consequence, we have:

- for every ideal tetrahedron T, the geodesic ideal tetrahedra $D(\widetilde{T})$ and $D'(\widetilde{T})$ are orientation-preservingly isometric. In particular, their quaternionic moduli are conjugate and so they induce the same complex moduli;
- for every edge e, the isometry f maps the geodesic ideal tetrahedra incident to $D(\tilde{e})$ to those incident to $D'(\tilde{e})$. As a consequence, the tuples of quaternionic cross-ratios around $D(\tilde{e})$ and $D'(\tilde{e})$ are simultaneously conjugate, hence inducing the same quaternionic lift around e.

The proof is complete. □

Remark 4.2.4. The assignment we have described at the beginning of this section is well-defined also for non-generic quaternionic developing maps. However, the induced labeling on the combinatorial pieces of $\mathcal{T}$ does not necessarily yield a quaternionic decoration. In particular, it is not guaranteed that the quaternionic moduli around the edges form generic tuples of quaternions.

Conversely, we intend to show that it is possible to reverse this construction in a canonical way. The following preliminary result is a crucial point in our work, as it is a direct bridge between the algebraic data of quaternionic decorations and the geometric realizations in $\mathbb{H}^5$. The proof follows directly by combining Lemma 2.3.15 with Corollary 3.2.15.

Corollary 4.2.5 (Face realization)**.** *For every distinct $i, k \in \{1, 3, 4\}$, let $q_{ik}^{(1)}$ and $q_{ik}^{(2)}$ be non real quaternions satisfying the requirements of the Face relations in Definition 4.2.1. Then they arise as the quaternionic moduli of two oriented adjacent ideal tetrahedra in $\mathbb{H}^5$.*

Proof. By hypothesis, there exists three commuting quaternions $\alpha, \beta, \gamma \in \mathcal{H}^*$ with $\alpha\beta\gamma = -1$ such that for both $s = 1, 2$:

$$q_{31}^{(s)} = \alpha q_{13}^{(s)} \alpha^{-1}, \quad q_{14}^{(s)} = \beta q_{41}^{(s)} \beta^{-1}, \quad q_{43}^{(s)} = \gamma q_{34}^{(s)} \gamma^{-1}. \tag{4.10}$$

By to Lemma 2.3.15, up to multiplying each of α, β, γ with a real number, we can assume that they arise as the simple ratios of three distinct points $q_1, q_3, q_4 \in \widehat{\mathcal{H}}$. Notice that this operation does not alter the conjugation equalities of (4.10). We have

$$\alpha = S(q_3, q_4, q_1), \quad \beta = S(q_1, q_3, q_4), \quad \gamma = S(q_4, q_1, q_3).$$

Since the quaternions $q_{ik}^{(1)}$, $q_{ik}^{(2)}$ also satisfy the Thurston relations, we find ourselves in the hypothesis of Corollary 3.2.15, which guarantees the existence (and uniqueness) of two other points $q_2, q_5 \in \widehat{\mathcal{H}}$ such that:

- the ideal tetrahedron T_1 in $\mathbb{H}^5$ with ideal vertices q_1, q_2, q_3, q_4 realizes the $q_{ik}^{(1)}$, *i.e.* $Q_{ik}(T_1) = q_{ik}^{(1)}$;
- the ideal tetrahedron T_2 in $\mathbb{H}^5$ with ideal vertices q_1, q_4, q_3, q_5 realizes the $q_{ik}^{(2)}$, *i.e.* $Q_{ik}(T_2) = q_{ik}^{(2)}$.

The proof is complete. □

We now state and prove the main result of this thesis.

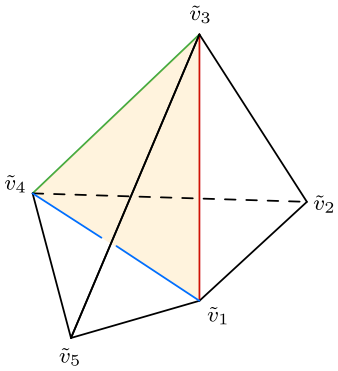
Theorem 4.2.6 (Correspondence Theorem)**.** *Given a quaternionic decoration of $\mathcal{T}$, there exists a generic quaternionic developing map (D, ρ) for $(M, \mathcal{T})$ such that:*

- *the complex moduli of an ideal tetrahedron T are uniquely determined by the isometry class of $D(\widetilde{T})$ in $\mathbb{H}^5$, where $\widetilde{T}$ is any lift of T;*
- *any quaternionic lift around an oriented edge e is simultaneously conjugate to the tuple of quaternionic cross-ratios around $D(\tilde{e})$ in $\mathbb{H}^5$, where $\tilde{e}$ is any lift of e.*

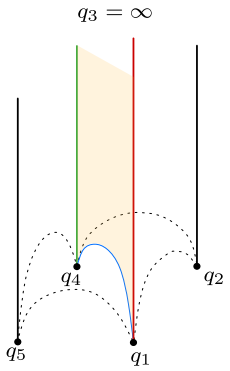
Moreover, (D, ρ) is unique up to equivalence.

Proof. We show the existence of a generic map $D\colon |\widetilde{\mathcal{T}}| \to \overline{\mathbb{H}}^5$ that satisfies the requirements of the statement. We start by defining the map D on the vertices of two adjacent tetrahedra, and then we will show that the Edge relations and the Face relations together (Definition 4.2.1) allow us to extend D to every vertex of $|\widetilde{\mathcal{T}}|$ uniquely. Every time we fix the image in $\partial\mathbb{H}^5$ of the vertices of a lifted tetrahedron $\widetilde{T}$, we continuously extend D to $\widetilde{T} \setminus \{\text{vertices}\}$ mapping every cell homeomorphically to the corresponding unique totally geodesic ideal simplex. We moreover require this extension to be coherent across shared faces.

At each step, we will have to confront the assigned data of the decoration (*i.e.* the quaternionic lifts around edges) with the "geometric" cross-ratios that we can compute by looking at the portion of the triangulation we have mapped in $\mathbb{H}^5$.

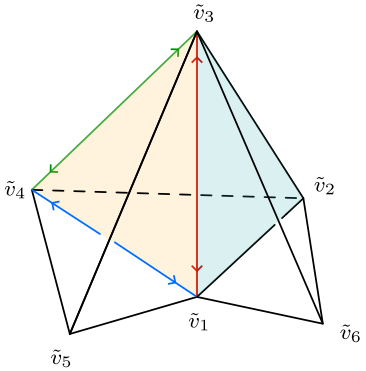


(a) The adjacent tetrahedra $\widetilde{T}_1$ and $\widetilde{T}_2$ of $|\widetilde{\mathcal{T}}|$

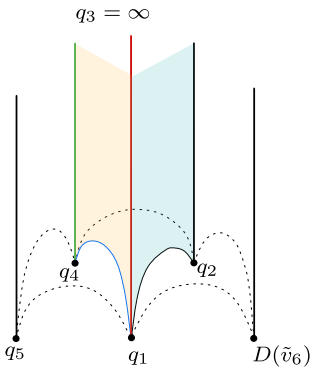


(b) The images of $\widetilde{T}_1$ and $\widetilde{T}_2$ in $\mathbb{H}^5$. Here we have set $q_3 = \infty$

Figure 4.3: The first step in the definition of the developing map (D, ρ).



(a) The adjacent tetrahedra $\widetilde{T}_1$, $\widetilde{T}_2$ and $\widetilde{T}_k$ of $|\widetilde{\mathcal{T}}|$



(b) The images of $\widetilde{T}_1$, $\widetilde{T}_2$ and $\widetilde{T}_k$ in $\mathbb{H}^5$. Here we have set $q_3 = \infty$.

Figure 4.4: The second step in the definition of the developing map (D, ρ)

Step 1: definition of D Let $\widetilde{T}_1$ and $\widetilde{T}_2$ be two adjacent tetrahedra of $|\widetilde{\mathcal{T}}|$ with vertices $\tilde{v}_1, \tilde{v}_2, \tilde{v}_3, \tilde{v}_4$ and $\tilde{v}_1$, $\tilde{v}_4$, $\tilde{v}_3$, $\tilde{v}_5$ respectively (see Figure 4.3a). Let T_1 and T_2 be the corresponding tetrahedra in $\mathcal{T}$. The adjacency in $\widetilde{\mathcal{T}}$ implies that a face of T_1 is paired with a face of T_2 by a simplicial isomorphism identifying the abstract vertices corresponding to $\tilde{v}_1, \tilde{v}_3, \tilde{v}_4$. By applying the Face relations (R2) to this pairing, there exist representatives $q_{ik}(T_1), q_{ik}(T_2)$ for the six paired oriented edges such that they satisfy the internal Thurston relations and are related by conjugation via commuting quaternions α, β, γ. Thanks to Corollary 4.2.5, $q_{ik}(T_1)$ and $q_{ik}(T_2)$ then arise as quaternionic moduli of two ideal hyperbolic tetrahedra in $\mathbb{H}^5$, say with vertices q_1, q_2, q_3, q_4 and q_1, q_3, q_4, q_5 respectively. We set

$$D(\tilde{v}_i) \coloneqq q_i \quad \forall\, i = 1, \dots, 5.$$

With this definition, the geometric cross-ratios match the decoration data. Notice that by making this choice, we have fixed the representative for the entire quaternionic lift around each common edge, thank to Corollary 2.1.9.

We now extend D to a new tetrahedron $\widetilde{T}_k$, adjacent fo $\widetilde{T}_1$ along a different face $\widetilde{F}'$, say with vertices $\tilde{v}_1$, $\tilde{v}_2$, $\tilde{v}_3$ (see Figure 4.4a). Since the positions of $D(\tilde{v}_1)$ $D(\tilde{v}_2)$, $D(\tilde{v}_3)$ were already fixed in the first step, we just need to determine the image of the last vertex, say $D(\tilde{v}_6)$. Consider the common edge $\tilde{e}_{13}$. Having already fixed the image of $\widetilde{T}_1$ and $\widetilde{T}_2$, the quaternionic lift $(q_{13}(T_1), \dots, q_{13}(T_k))$ around e_{13} is rigid and forces the desired cross ratio $Q_{13}(D(\tilde{T}_k))$ to be $q_{13}(T_k)$. We thus use $q_{13}(T_k)$ to define $D(\tilde{v}_6)$ so that

$$Q(D(\tilde{v}_1), D(\tilde{v}_6), D(\tilde{v}_3), D(\tilde{v}_2)) = q_{13}(T_k),$$

see Figure 4.4b. We need to check that the quaternionic moduli of the tetrahedron $D(\widetilde{T}_k)$ are compatible with the decoration data. This is easy for the edges of valence one, but not so straightforward for the edge e_{31}. Again, having already chosen the image of two tetrahedra incident to e_{31}, we have also fixed a representative for its assigned quaternionic lift. Let us call it $(q_{31}(T_k), \dots, q_{31}(T_1))$, where $q_{31}(T_j)$ refers to T_j. We need to prove that

$$q_{31}(T_k) = Q_{31}(D(\widetilde{T}_k)).$$

By the edge relations, we know that the k-tuples

$$(q_{13}(T_1), \dots, q_{13}(T_k)) \quad \text{and} \quad (q_{31}(T_1), \dots, q_{31}(T_k))$$

are adjoint, and so we write

$$q_{13}(T_1) = \alpha_1\alpha_2^{-1}, \quad q_{13}(T_2) = \alpha_2\alpha_3^{-1}, \quad q_{13}(T_k) = \alpha_k\alpha_1^{-1} \tag{4.11}$$

$$q_{31}(T_1) = \alpha_2^{-1}\alpha_1, \quad q_{31}(T_2) = \alpha_3^{-1}\alpha_2, \quad q_{31}(T_k) = \alpha_1^{-1}\alpha_k. \tag{4.12}$$

for some quaternions $\alpha_1, \alpha_2, \alpha_k \in \mathcal{H}^*$. We immediately see that α_2^{-1} conjugates the pair $(q_{13}(T_1), q_{13}(T_2))$ to $(q_{31}(T_1), q_{31}(T_2))$. On the other hand, all the quaternions in (4.11) and (4.12) but $q_{31}(T_k)$ already match their geometric counterparts:

$$q_{13}(T_1) = Q_{13}(D(\widetilde{T}_1)), \quad q_{13}(T_2) = Q_{13}(D(\widetilde{T}_2)), \quad q_{13}(T_k) = Q_{13}(D(\widetilde{T}_k)) \tag{4.13}$$

$$q_{31}(T_1) = Q_{31}(D(\widetilde{T}_1)), \quad q_{31}(T_2) = Q_{31}(D(\widetilde{T}_2)). \tag{4.14}$$

Using Corollary 2.3.14, we can write each cross-ratio in (4.13) and (4.14) as the product of the simple ratios $S_{ijk} \coloneqq S(D(\tilde{v}_i), D(\tilde{v}_j), D(\tilde{v}_k))$. By Lemma 3.2.7, the simple ratio S_{341} also conjugates $(q_{13}(T_1), q_{13}(T_2))$ to $(q_{31}(T_1), q_{31}(T_2))$, and since both pair are generic, up to scaling we must have $S_{341} = \alpha_2$. From this we deduce

$$\alpha_1 = q_{13}(T_1) \cdot \alpha_2 = Q_{13}(D(\widetilde{T}_1)) \cdot S_{341}^{-1} = S_{123},$$

and similarly $\alpha_k = S_{163}$. As a consequence we finally get

$$q_{31}(T_k) = \alpha_1^{-1}\alpha_k = S_{123}^{-1} \cdot S_{163} = S_{321} \cdot S_{361}^{-1},$$

which is precisely the cross-ratio $Q_{31}(D(\widetilde{T}_k))$ of the newly mapped tetrahedron. Thus the geometric cross-ratios match the decoration parameters exactly.

We can iterate this argument by moving face-by-face. We only need to be careful that if at a given point, the face has more than one edge with valence greater or equal to 2, then the lifts around those edges are already defined and we need to repeat the same argument as the one about e_{13} and e_{31} involving simple ratios

Step 2: definition of the holonomy Fix an element $\gamma \in \pi_1(M)$, and consider the two pairs of adjacent, ideal tetrahedra in $\mathbb{H}^5$

$$D(\widetilde{T}_1), D(\widetilde{T}_2) \quad \text{and} \quad D(\gamma \cdot \widetilde{T}_1), D(\gamma \cdot \widetilde{T}_2).$$

By construction, the quaternionic cross-ratios around $D(\tilde{e}_{13})$ and $D(\gamma \cdot \tilde{e}_{13})$ are simultaneously conjugate. Thanks to Theorem 3.2.11, there exists a unique orientation-preserving isometry $\rho(\gamma) \in \mathrm{Isom}^+(\mathbb{H}^5)$ mapping $D(\tilde{v}_i)$ to $D(\gamma \cdot \tilde{v}_i)$ for each $i = 1, \dots, 5$. In particular, $\rho(\gamma)$ satisfies the holonomy condition (4.1) for the first five vertices. We want to show that the relation holds for every vertex of the lifted triangulation.

Let $\widetilde{T}_k$ be another tetrahedron adjacent to $\widetilde{T}_1$ along the face F' with vertices $\tilde{v}_1$, $\tilde{v}_2$, $\tilde{v}_3$, as in Figure 4.4a. By repeating the same argument as above for the pairs

$$D(\widetilde{T}_k), D(\widetilde{T}_1), \quad D(\gamma \cdot \widetilde{T}_k), D(\gamma \cdot \widetilde{T}_1)$$

we find another isometry $\rho'(\gamma)$ mapping $D(\tilde{v}_i)$ to $D(\gamma \cdot \tilde{v}_i)$ for $i = 1, 2, 3, 4, 6$. We want to prove that $\rho(\gamma)$ and $\rho'(\gamma)$ coincide. Since both isometries agree on the vertices of $D(\widetilde{T}_1)$, the composition $g \coloneqq \rho'(\gamma)^{-1}\rho(\gamma)$ fixes q_1, q_2, q_3, q_4. Up to conjugation, we may assume these points are $0, 1, \infty$ and some $z \in \mathbb{C} \setminus \mathbb{R}$, which implies g is represented by a diagonal matrix

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \in \mathrm{SL}(2, \mathbb{C}).$$

We must prove that a is a real number. By construction, both $\rho(\gamma)$ and $\rho'(\gamma)$ map the edge $D(\tilde{e}_{13})$ to $D(\gamma \cdot \tilde{e}_{13})$, and due to Proposition 2.3.6 the left isoclinic components $u_{\rho(\gamma),\infty}$ and $u_{\rho'(\gamma),\infty}$ of their differentials at ∞ both conjugate the quaternionic cross-ratios around $D(\tilde{e}_{13})$ to those around $D(\gamma \cdot \tilde{e}_{13})$. Since these two tuples are generic,

$u_{\rho(\gamma),\infty}$ and $u_{\rho'(\gamma),\infty}$ differ only by the multiplication by a real number. On the other hand, by the chain rule we find

$$a = u_{g,\infty} = u_{\rho'(\gamma)^{-1},\rho(\gamma)(\infty)} \cdot u_{\rho(\gamma),\infty} = u^{-1}_{\rho'(\gamma),\infty} \cdot u_{\rho(\gamma),\infty} \in \mathbb{R}$$

and so we are done. This proves that $\rho(\gamma)$ and $\rho'(\gamma)$ are the same isometry, and that the holonomy condition (4.1) holds also for $\tilde{v}_6$. Since our manifold is connected, we can repeat the same argument to show that $\rho(\gamma)$ satisfies the holonomy condition for every vertex. We furthermore notice that the extension of D to every tetrahedron can be chosen so that the equation (1.23) holds for every $\tilde{x} \in |\widetilde{\mathcal{T}}|$.

This defines a map $\rho\colon \pi_1(M) \to \mathrm{Isom}^+(\mathbb{H}^5)$ that satisfies the holonomy condition: given that $D\colon |\widetilde{\mathcal{T}}| \to \overline{\mathbb{H}}^5$ is generic, Remark 4.1.2 implies that ρ is a group morphism. We have thus constructed the desired quaternionic developing map (D, ρ).

Step 3: uniqueness up to equivalence Notice that the definition of the map D only depends on the following choices:

- the starting pair of adjacent tetrahedra $\widetilde{T}_1, \widetilde{T}_2$ and their lifts;
- the representative $(q_{13}(T_1), \dots, q_{13}(T_k))$ of the quaternionic lift around the edge e_{13};
- the images $D(\widetilde{T}_1), D(\widetilde{T}_2)$.

By changing any of the above, we would have obtained another generic quaternionic developing map (D', ρ') for $(M, \mathcal{T})$ satisfying the required properties. We now look at the two pairs of ideal tetrahedra

$$D(\widetilde{T}_1), D(\widetilde{T}_2), \quad D'(\widetilde{T}_1), D'(\widetilde{T}_2)$$

in $\mathbb{H}^5$. Since both D and D' must be compatible with the fixed quaternionic decoration, we know that the k-tuples of quaternionic lifts around $D(e_{13})$ and $D'(e_{13})$ are simultaneously conjugate. By repeating the same argument as in the construction of the holonomy ρ, we conclude that there exists a new isometry $f \in \mathrm{Isom}^+(\mathbb{H}^5)$ such that $f \circ D = D'$.

Now if we put everything together, for every $\gamma \in \pi_1(M, \tilde{v}_1)$ and for every vertex $\tilde{v}$ we have

$$\rho'(\gamma) f'(D(\tilde{v})) = \rho'(\gamma) D'(\tilde{v}) = D'(\gamma \cdot \tilde{v}) = f(D(\gamma \cdot \tilde{v})) = f(\rho(\gamma)) D(\tilde{v}).$$

Since D is generic, this yields

$$\rho' = f \circ \rho \circ f^{-1},$$

as we wanted. This shows that (D, ρ) and (D', ρ') are equivalent. □

4.2.2 Computing the holonomy

Assume that M has a single cusp, and suppose that we can find a quaternionic decoration on an ideal triangulation $\mathcal{T}$ of M. By Theorem 4.2.6, the decoration induces a quaternionic developing map (D, ρ) for $(M, \mathcal{T})$, unique up to equivalence. We conclude this chapter by computing the peripheral holonomy $\rho\colon \pi_1(L) \to \mathrm{PSL}(2, \mathcal{H})$. The main ideas are a translation of the construction used for the completeness equations in the 3-dimensional case (1.4.3), and most of the proofs are an adaptation of the corresponding arguments in [BP92] and [Mar16].

We briefly recall the setting. We consider two adjacent, oriented tetrahedra $\widetilde{T}_1$, $\widetilde{T}_2$ of $|\widetilde{\mathcal{T}}|$ having ideal vertices $\tilde{v}_1, \tilde{v}_2, \tilde{v}_3, \tilde{v}_4$ and $\tilde{v}_1, \tilde{v}_4, \tilde{v}_3, \tilde{v}_5$ respectively, as in Figure 4.3a. Up to equivalence, we can assume that

$$D(\tilde{v}_1) = 0, \quad D(\tilde{v}_2) = 1, \quad D(\tilde{v}_3) = \infty.$$

The remaining vertices are mapped to some points

$$D(\tilde{v}_4) = p, \quad D(\tilde{v}_5) = q,$$

and we can choose p and q as the unique points such that $D(\tilde{T}_1)$ and $D(\tilde{T}_2)$ are in normal form (see Lemma 3.2.13).

We fix a basepoint $x_0 \in L$ for both $\pi_1(L)$ and $\pi_1(M)$, and we pick a lift $\tilde{x}_0 \in \widetilde{M}$ lying in the link of $\tilde{v}_3$ as a basepoint for $\widetilde{M}$. Without loss of generality, we can choose $\tilde{x}_0$ to lie on the edge connecting $\tilde{v}_1$ to $\tilde{v}_3$.

With this choice, the subgroup $\Gamma = \pi_1(L) \subset \pi_1(M)$ acts on $\widetilde{M}$ by stabilizing $\tilde{v}_3$. Consequently, for every $\gamma \in \Gamma = \pi_1(L)$, its image $\rho(\gamma)$ fixes ∞, and is thus represented by an upper-triangular matrix in $\mathrm{SL}(2, \mathcal{H})$. As we did in Section 4.1.1, we fix a matrix representative for $\rho(\gamma)$ in $\mathrm{SL}(2, \mathcal{H})$ for every $\gamma \in \Gamma$, which we still denote by $\rho(\gamma)$ by a slight abuse of notation:

$$\rho(\gamma) = \begin{pmatrix} a_\gamma & b_\gamma \\ 0 & d_\gamma \end{pmatrix}.$$

We denote by $A_\gamma \coloneqq \rho(\gamma)0 = b_\gamma d_\gamma^{-1}$ the translation vector of $\rho(\gamma)$. We want to show how to determine the diagonal entries a_γ and d_γ, up to a sign, from the decoration data.

The induced 2-dimensional decoration We recall again that L corresponds to the unique vertex v of $|\mathcal{T}|$, and that $\mathcal{T}$ induces a triangulation $\mathcal{T}_v$ of L obtained by considering the links of those vertices of $\mathcal{T}$ that project to v. By choosing a specific lift $\tilde{v}$ of v, this triangulation lifts to a triangulation $\widetilde{\mathcal{T}}_v$ of the universal cover. The geometric realization $|\widetilde{\mathcal{T}}_v|$ naturally lies inside $|\widetilde{\mathcal{T}}| \setminus \{\text{vertices}\}$, as it is obtained by gluing the links of $\tilde{v}$.

We now show that the quaternionic decoration of $\mathcal{T}$ induces a labeling of every corner of $\widetilde{\mathcal{T}}_v$ with a quaternion. We start by fixing $\tilde{v} \coloneqq \tilde{v}_3$.

If $\tilde{e}$ is an edge in $\widetilde{M}$ directed towards $\tilde{v}_3$, its image $D(\tilde{e})$ is a vertical geodesic in $\mathbb{H}^5$ ending at $D(\tilde{v}_3) = \infty$. By truncating $\mathbb{H}^5$ with a small enough horosphere centered at ∞ and passing through our mapped basepoint, we cut the developed ideal tetrahedra

incident to $D(\tilde{e})$ into Euclidean triangles $\Delta_1, \ldots, \Delta_k$. The vertices of the triangles Δ_i correspond exactly to the finite ideal vertices of the developed tetrahedra.

By Proposition 3.3.4, the quaternionic moduli around the edge $D(\tilde{e})$ precisely coincide with the quaternionic angles of the triangles $\Delta_1, \ldots, \Delta_k$ at the common vertex. Thus, the quaternionic decoration directly provides a well-defined labeling on the angles of the lifted 2-dimensional triangulation $\widetilde{\mathcal{T}_v}$ in terms of quaternionic angles. Here lies another crucial difference from the classical case: since the peripheral holonomy $\rho(\pi_1(L))$ acts on $\mathcal{H}$ via orientation-preserving similarities, the quaternionic angles change by simultaneous conjugation. As a consequence, when projecting down to $\mathcal{T}_v$, the labeling is well-defined only up to simultaneous conjugation.

Still, the Edge relations guarantee that the product of the quaternionic angles around any vertex is exactly 1, and this condition does not depend on the conjugacy class. Geometrically, this ensures that the local similarity structures of the triangles glue together smoothly around the vertices.

Description of the diagonal entries We use the quaternionic angles to describe the peripheral holonomy. Let Δ_1, Δ_2 be the Euclidean triangles obtained by truncating $D(\tilde{T}_1)$ and $D(\tilde{T}_2)$ respectively. By our normalization, Δ_1 has vertices $0, 1, p$ and Δ_2 has vertices $0, p, q$.

Consider an element $\gamma \in \pi_1(L)$. By the equivariance of the developing map, the holonomy $\rho(\gamma)$ is an orientation-preserving isometry of $\mathbb{H}^5$ mapping $D(\tilde{v}_i)$ to $D(\gamma \cdot \tilde{v}_i)$ for $i = 1, 2, 3, 4, 5$. Since $\rho(\gamma)$ fixes $D(\tilde{v}_3) = \infty$, its geometric action can be viewed as an orientation-preserving similarity of $\mathcal{H}$ mapping the adjacent triangles Δ_1, Δ_2 to the pair of adjacent triangles having vertices $D(\gamma \cdot \tilde{v}_1), D(\gamma \cdot \tilde{v}_2), D(\gamma \cdot \tilde{v}_4)$ and $D(\gamma \cdot \tilde{v}_1), D(\gamma \cdot \tilde{v}_4), D(\gamma \cdot \tilde{v}_5)$, respectively. Since both pairs of triangles are not coplanar, there exists an invertible quaternion, unique up to a real factor, that simultaneously conjugates the quaternionic angles of the first pair into those of the second pair. By construction, this quaternion is precisely d_γ. Now the isometry $\rho(\gamma)$ is explicitly described in Remark 3.3.8. Recalling that $A_\gamma = \rho(\gamma)(0)$ and setting $B_\gamma \coloneqq D(\gamma \cdot \tilde{v}_2) = \rho(\gamma)(1)$, we get

$$\begin{pmatrix} (B_\gamma - A_\gamma)d_\gamma & A_\gamma d_\gamma \\ 0 & d_\gamma \end{pmatrix}. \tag{4.15}$$

Now we show how to determine $B_\gamma - A_\gamma$. We view every element $\gamma \in \pi_1(L)$ as a path in the lifted, 2-dimensional triangulation, with basepoint in $\tilde{v}_1$. Since the labeling on $\mathcal{T}_v$ is well-defined only up to simultaneous conjugation, it is convenient to fix a landmark edge on the universal cover, where the exact quaternionic angles are known, allowing us to compute $(B_\gamma - A_\gamma)$ by sequentially multiplying the quaternionic angles along the path without any ambiguity.

For this purpose, let $\tilde{\alpha}$ be the oriented edge from $\tilde{v}_1$ to $\tilde{v}_2$ and consider its projection α on the base triangulation. For every loop γ in $\mathcal{T}$ with basepoint v_1, we consider the curve

$$\tilde{\gamma}^\alpha = \tilde{\alpha}^{-1} * \widetilde{\gamma * \alpha},$$

where $\widetilde{\gamma * \alpha}$ is the lift of $\gamma * \alpha$ with starting point $\tilde{v}_1$, see Figure 4.5.

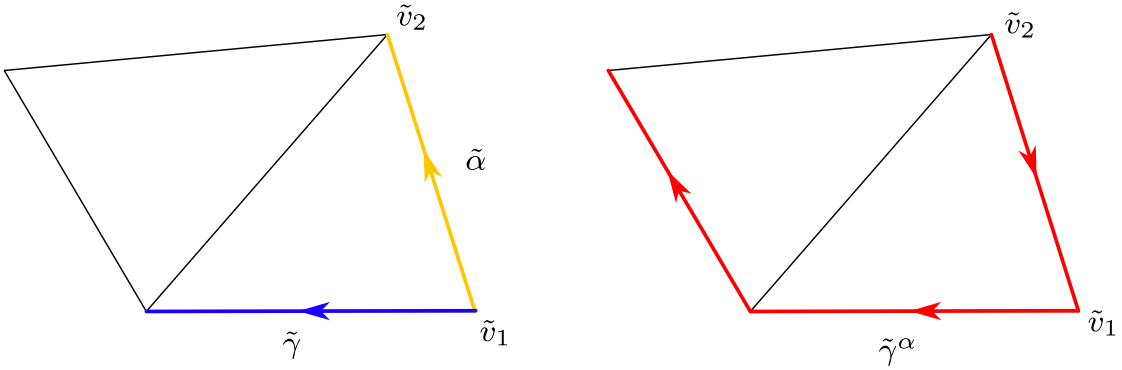


Figure 4.5: A simple example to illustrate the definition of the curve $\tilde{\gamma}^\alpha$.

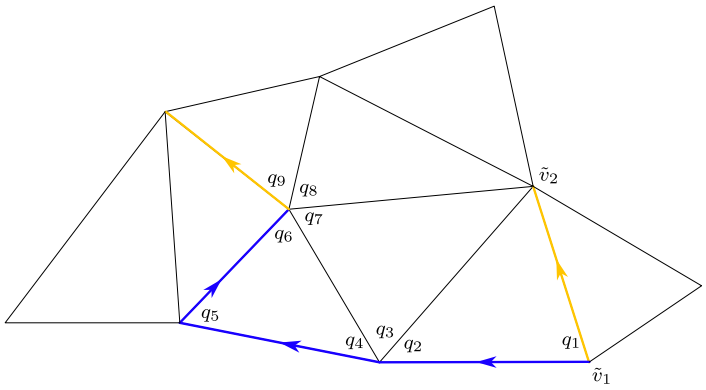


Figure 4.6: If $\tilde{\gamma}$ is the blue curve, we have $\mu(\gamma) = -q_1q_2q_3q_4q_5q_6q_7q_8q_9$.

If $q_1, \dots, q_k$ are the quaternionic angles that $D(\tilde{\gamma}^\alpha)$ encounters on the right (without considering its endpoints), and γ_0 is the number of edges of $\tilde{\gamma}$ - the lift of γ with starting point $\tilde{v}_1$ - we define

$$\mu(\gamma) \coloneqq (-1)^{\gamma_0} q_1 \cdot \ldots \cdot q_k \in \mathcal{H}^*. \tag{4.16}$$

See Figure 4.6 for an example.

Proposition 4.2.7. *The quaternion $\mu(\gamma)$ defined in (4.16) only depends on the homotopy class of γ.*

Proof. It is enough to show that $\mu(\gamma)$ doesn't change if we alter γ with the move described in Figure 4.7.

When performing this move (or its inverse), γ_0 is increased (decreased) by 1 - thus the sign of $\mu(\gamma)$ is changed once - while the product of the quaternionic angles encountered on the right is affected by a factor

$$q \cdot q'' \cdot q' = -1.$$

Summing up, μ is left unchanged. □

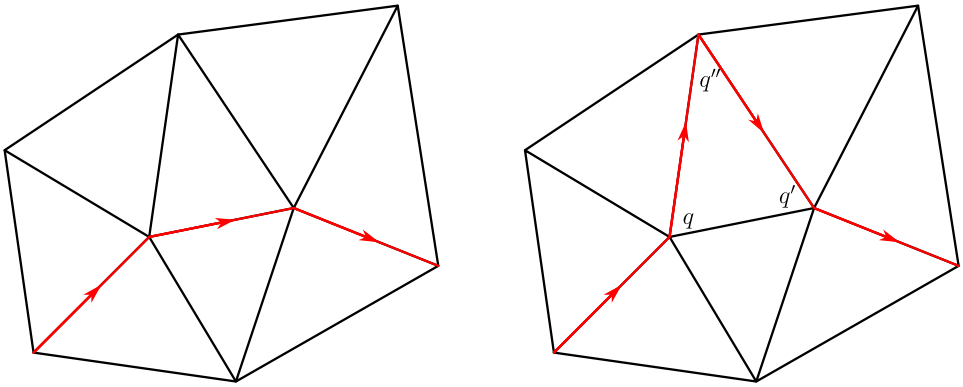


Figure 4.7: This move and its inverse do not change $\mu(\gamma)$.

We have thus defined a map $\mu\colon \pi_1(L) \to \mathcal{H}^*$.

Proposition 4.2.8. *For every $\gamma \in \Gamma$ we have $\mu(\gamma) = B_\gamma - A_\gamma$.*

Proof. Let

$$A_0 \coloneqq 1, \quad A_1 \coloneqq 0, \quad \ldots, \quad A_{n-1} \coloneqq A_\gamma, \quad A_n \coloneqq B_\gamma$$

be the vertices of $\tilde{\gamma}^\alpha$. For every $i = 1, \ldots, n-1$, we define $p_i \in \mathcal{H}^*$ to be the product of the quaternionic angles incident to $D(A_i)$ that $D(\tilde{\gamma}^\alpha)$ encounters on its right (see Figure 4.8).

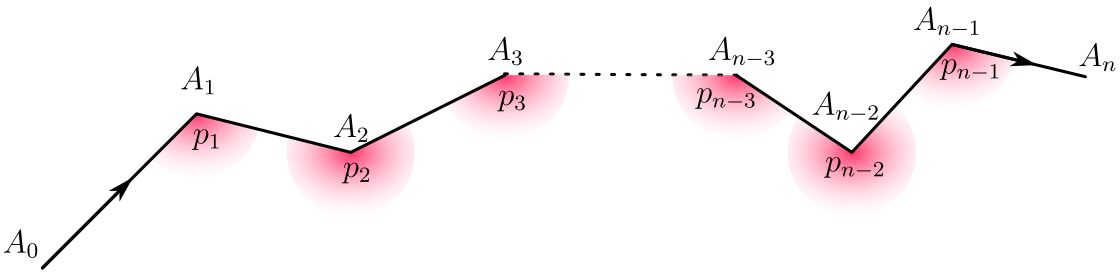


Figure 4.8: The vertices of the curve $\tilde{\gamma}^\alpha$.

By definition, we have

$$p_i = (A_{i-1} - A_i)^{-1}(A_{i+1} - A_i) \implies A_{i+1} - A_i = (A_{i-1} - A_i)p_i.$$

By repeatedly applying the above relation, we find

$$\begin{aligned} B_\gamma - A_\gamma &= A_n - A_{n-1} = -(A_{n-1} - A_{n-2})p_{n-1} \\ &= (-1)^2(A_{n-2} - A_{n-3})p_{n-2}p_{n-1} \\ &= \ldots \\ &= (-1)^{n-1}(A_1 - A_0)p_1 \cdot \ldots \cdot p_{n-1} \\ &= (-1)^n p_1 \cdot \ldots \cdot p_{n-1} \\ &= \mu(\gamma), \end{aligned}$$

where the last equality follows from the fact that $\tilde{\gamma}$ has $n-2$ edges and clearly $(-1)^{n-2} = (-1)^n$. This concludes the proof. □

By combining (4.15) and Proposition 4.2.8, we get

$$\rho(\gamma) = \begin{pmatrix} \mu(\gamma)d_\gamma & A_\gamma d_\gamma \\ 0 & d_\gamma \end{pmatrix}. \tag{4.17}$$

Remark 4.2.9. The map $\mu\colon \pi_1(L) \to \mathcal{H}^*$ is a 1-cocycle, satisfying

$$\mu(\gamma\sigma) = \mu(\gamma)d_\gamma \mu(\sigma) d_\gamma^{-1}$$

for every $\gamma, \sigma \in \pi_1(L)$.

We can now combine the combinatorial form for $\rho|_{\pi_1(L)}\colon \pi_1(L) \to \mathrm{PSL}(2,\mathcal{H})$ of (4.17) with Theorem 4.1.15. We obtain a procedure to detect whether a quaternionic decoration of $(M,\mathcal{T})$ induces a holonomy with good peripheral properties. We write it down for completeness. Start by fixing two generators γ_1, γ_2 for $\pi_1(L) \cong \mathbb{Z}\times\mathbb{Z}$.

Step 1 Check whether $\rho(\gamma_1)$ is a parabolic isometry of $\mathbb{H}^5$. This amounts to proving Condition 1 and Condition 2 from Section 4.1.2. Since d_{γ_1} is well-defined only up to multiplication by a real number, we can assume that both $\mu(\gamma)d_\gamma$ and d_γ are unit quaternions. Condition 1 reduces to proving

$$\mathrm{Re}(\mu(\gamma_1)d_{\gamma_1}) = \mathrm{Re}(d_{\gamma_1}). \tag{4.18}$$

If (4.18) hold, we move to Condition 2 (otherwise, we have found an elliptic isometry and the representation cannot be boundary-parabolic).

Step 2 We test Condition 2 for $\rho(\gamma_1)$. If $d_{\gamma_1} \in \mathbb{R}$, then $\rho(\gamma_1)$ is a translation and we need to check whether $A_{\gamma_1} \neq 0$. If $d_{\gamma_1} \notin \mathbb{R}$, we define the plane K as in (4.4) and we need to verify that

$$\pi_K(A_{\gamma_1}) \neq 0.$$

If this test is successful, $\rho(\gamma_1)$ is a non-trivial parabolic isometry of $\mathbb{H}^5$. By Lemma 4.1.9, this guarantees that for every $\gamma \in \pi_1(L)$, the matrix $\rho(\gamma)$ satisfies Condition 1. We can thus proceed to study the entire peripheral group. Otherwise, $\rho(\gamma_1)$ is either the identity or an elliptic isometry, meaning the holonomy cannot be boundary-parabolic.

Step 3 To determine the global properties of the peripheral holonomy, we evaluate $\rho(\gamma_2)$ and combine it with the information from $\rho(\gamma_1)$ using the results of Section 4.1.1. We distinguish two cases:

- *translation case:* if both $d_{\gamma_1} \in \mathbb{R}$ and $d_{\gamma_2} \in \mathbb{R}$, then the peripheral holonomy consists solely of translations. By Lemma 4.1.10, the restriction $\rho|_{\pi_1(L)}$ is injective and boundary-parabolic if and only if the translation vectors A_{γ_1} and A_{γ_2} are linearly independent over $\mathbb{Z}$. Furthermore, the image $\rho(\pi_1(L))$ is a discrete subgroup if and only if they are linearly independent over $\mathbb{R}$;

- *screw-parabolic case:* if at least one of d_{γ_1} or d_{γ_2} is not real, we assume without loss of generality that $d_{\gamma_1} \notin \mathbb{R}$ (swapping the generators if necessary). Let $v_1 = \pi_K(A_{\gamma_1})$ and $v_2 = \pi_K(A_{\gamma_2})$. By Theorem 4.1.15, the restriction $\rho|_{\pi_1(L)}$ is injective and boundary-parabolic if and only if v_1 and v_2 are linearly independent over $\mathbb{Z}$. Finally, the image is a discrete subgroup if and only if v_1 and v_2 are linearly independent over $\mathbb{R}$.

4.3 Final observations

We recall the setting of the introduction. We consider an oriented, complete, non-compact hyperbolic 3-manifold M with finite volume, and we denote by (D_0, ρ_0) the corresponding developing pair. Recall that the composition

$$\rho_0' \colon \pi_1(M) \xrightarrow{\rho_0} \mathrm{Isom}^+(\mathbb{H}^3) \hookrightarrow \mathrm{Isom}^+(\mathbb{H}^5)$$

is the holonomy of a 5-dimensional, complete hyperbolic manifold $M' \cong M \times \mathbb{R}^2$ with infinite volume. We want to find a continuous path of non-similar representations

$$\rho_t' \colon \pi_1(M) \to \mathrm{Isom}^+(\mathbb{H}^5)$$

with good peripheral properties, such that $\rho_t' \to \rho_0'$ as $t \to 0$.

Let $\mathcal{T}$ be an ideal triangulation of M. In this thesis, we have proved that the representations $\pi_1(M) \to \mathrm{Isom}^+(\mathbb{H}^5)$ that arise as the holonomy of a generic quaternionic developing map for $(M, \mathcal{T})$ are parametrized by quaternionic decorations on the ideal triangulation. Assume now that $\mathcal{T}$ is a geometric ideal triangulation for M, and let $z^{(0)} = (z_1^{(0)}, \dots, z_n^{(0)})$ be a solution to both the consistency and the completeness equations. To find the desired deformations, a natural approach is to take a sufficiently small neighborhood U of $z^{(0)}$ in $\mathbb{C}^n$, such that every tuple $z \in U$ still consists of complex numbers with strictly positive imaginary parts. For every $z \neq z^{(0)}$ in U, the consistency equations generally fail; however, we can try to lift z to a valid quaternionic decoration. By this we mean that we can try to compute a quaternionic decoration with complex moduli induced by z.

Solving this problem is highly non-trivial, as the transition from $\mathbb{C}$ to $\mathcal{H}$ introduces significant algebraic challenges. In Thurston's classical framework, both the consistency and completeness equations involve complex, hence commuting, parameters. In our framework, the parameters are non-commutative and involve simultaneous conjugations. These two factors make explicitly solving the Edge and Face relations a hard task. Furthermore, as we pointed out in Remark 1.4.21, the hypothesis that M admits a geometric ideal triangulation is a delicate one. While M always admits a partially flat ideal triangulation, allowing the triangulation to contain flat tetrahedra is currently difficult in our setting: for instance, it would require the quaternionic lifts to admit some real entries, thus breaking the genericity assumption.

The next obvious step would be to compute a concrete example: quaternionic decorations for the standard triangulation of the figure-eight knot complement are under

computation. We expect to find a continuous family of quaternionic decorations that induces a 1-parameter, continuous family of generic quaternionic developing maps (D_t, ρ'_t), with boundary-parabolic holonomies ρ'_t.

As observed in Example 4.1.6, the holonomy ρ'_0 does not arise as the holonomy of a generic quaternionic developing map. However, we expect ρ'_0 to appear as the limit of the family (D_t, ρ'_t). For $t > 0$, the quaternionic cross-ratios around the mapped edges form generic tuples of quaternions. As $t \to 0$, the tuples flatten: the components orthogonal to $\mathbb{C}$ vanish, and we get a tuple of complex numbers. Geometrically, the image of the developing map D_t flattens into the totally geodesic subspace $\mathbb{H}^3 \subset \mathbb{H}^5$, recovering the original pair (D_0, ρ'_0).

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